

Multimethod Investigation of Wood Degradation in Ice Patch Preservation Environments

Celine Beier

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Supervisors:

Susan Braovac

Margrethe Felter

Julian Post-Melbye

Nanna Bjerregaard Pedersen

Eleni Pavlidou

Abstract

Ice patches in the Norwegian mountains are melting due to climate change and revealing archaeological artifacts that have been preserved in the ice for thousands of years. The recovered wooden objects are well preserved but their degradation has not been studied in detail. This thesis investigates wooden sticks preserved in ice patches using visual examination, color and density measurements, stereomicroscopy, light microscopy, ATR-FTIR spectroscopy and dynamic vapor sorption and compares the results to modern and waterlogged archaeological wood.

The wood preserved in ice patches has a well-preserved interior with similar density and appearance to modern wood. The cells and cell walls remain intact and cellulose and lignin are preserved, while hemicelluloses are degraded. The surfaces of the wood show strong differences from the interior in color and chemical composition, but degradation is limited to the outermost layer and likely caused by environmental factors like sunlight, meltwater, wind and freeze-thaw and drying-wetting cycles. Degradation is localized and depends on the specific environmental conditions.

Both wood preserved in ice patches and wood preserved in waterlogged conditions are considered archaeological wood, but have significantly different material properties and degradation patterns. In waterlogged wood, biological decay can affect cellulose and hemicelluloses, while the lignin structure is typically preserved. In contrast, no biological decay is observed in the ice patch wood and the wood surface shows an opposite degradation pattern with degradation of lignin while cellulose remains preserved.

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1 Introduction

Ice patches, permanent stationary accumulations of snow and ice, in the Norwegian mountains are shrinking due to climate change. These sites have been used for hunting for at least the past 6000 years as reindeer gather on the snow during summer to cool down and avoid insects [1, 2]. As ice patches melt, archaeological artifacts are exposed, including bows, arrows, hunting and horse equipment, skis, textiles, shoes and everyday objects [1]. The finds are generally well preserved and include organic materials that rarely survive. The Glacial Archaeology Programme was established to collect, conserve and study these objects and has already recovered over 4000 artifacts [3].

Wooden artifacts found in ice patches include skis, withies, composite objects like knives and arrows and scaring sticks [1, 4]. Scaring sticks are roughly worked wooden sticks, 1-2 m long, to which a flag made of thin wooden sheets was attached [1]. They were stuck either into the snow or scree near the ice patch to guide reindeer during hunting [2]. The recovered scaring sticks are in good condition and require little conservation treatment [1]. They do not show the typical degradation of waterlogged archaeological wood, but the preservation has not been studied in detail. While wooden artifacts have also been recovered from ice patches in the Yukon, Canada [5, 6, 7] and the Swiss alps [8, 9], the published work on these objects is mainly focused on the archaeological interpretation and conservation treatments and not on the condition of the wood [6, 7, 8, 9].

This thesis investigates the biological and chemical degradation of wood preserved in ice patches in comparison to modern and waterlogged archaeological wood. Samples from scaring sticks are analyzed using color and density measurements as well as stereomicroscopy for surface characterization. Light microscopy is used to investigate the cell structure while Fourier-transform infrared (FTIR) spectroscopy can provide information on the chemical composition. Dynamic vapor sorption (DVS) is used to investigate water sorption behavior.

2 Overview of Wood Structure and Composition

From the outside in, tree trunks consists of the outer bark, inner bark, vascular cambium, sapwood, heartwood and pith. Wood is produced by the vascular cambium through cell division and cells created during the same growth interval form a growth ring. Cells from the beginning of a growth period are called earlywood and cells from later stages are called latewood. Sapwood is the active wood which conducts water from the roots to the leaves while heartwood is the non-conductive inner wood that provides mechanical support. It contains extractives that can cause a darker color compared to sapwood. [10]

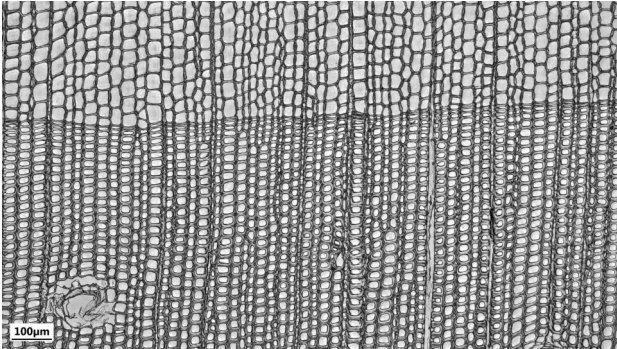
Trees can be divided into two groups, softwood (gymnosperm) and hardwood (angiosperm), which have different cell types and structures. Figure 1 compares softwood and hardwood in the three main anatomical perspectives. The transverse section corresponds to a cross section perpendicular to the vertical axis of the tree trunk. The radial section runs from pith to bark and the tangential section cuts parallel to the growth rings. [10]

In softwood, water transport and mechanical support are provided by tracheids, long vertical cells that in the transverse section appear as square or rectangular cells aligned in radial rows. Some species show a transition from thin-walled tracheids in the earlywood to thicker-walled tracheids in the latewood. Water conduction between tracheids occurs through pits in the tapered ends where tracheids overlap with adjacent cells. Rays contribute to the synthesis, storage and lateral transport of biochemicals and are made of ray parenchyma cells. In the radial section, rays are visible as horizontal bands of rectangular brick-shaped cells while in the tangential section, rays appear as vertical stacks of parenchyma cells. Rays with a single row of cells are called uniserate and are the most common type of ray found in softwood. Some softwood species have ray tracheids, specialized horizontal tracheids at the edges of rays. The intersection of rays with axial tracheids is called the cross-field and contains pits that connect the cells. Some softwood species also have resin canals, which are voids in the wood surrounded by parenchyma cells that produce resin. [10]

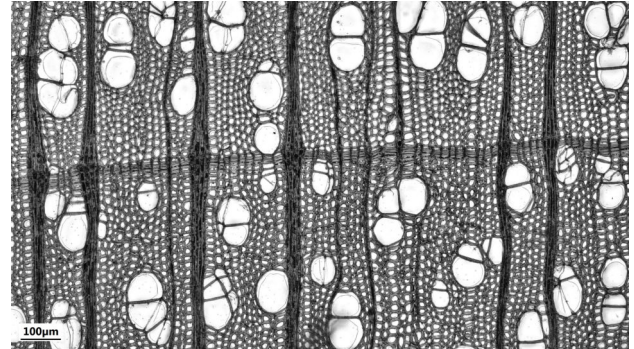
Hardwood has a more complex structure than softwood and contains vessels, fibers, axial parenchyma and rays. Water is conducted through vessels which are connected to by perforation plates. In the transverse section, vessels appear as large openings, while fibers, which provide mechanical support, are visible as small squares. Axial parenchyma cells are similar in size and shape to ray parenchyma cells, but are vertically oriented and exist in a variety of patterns. Hardwood rays can be uniserate or multiserate. [10]

A living plant cell consists of a protoplast and a cell wall. In wood, most cells are dead at maturity and no longer contain a protoplast. The empty space inside the cell wall is called the lumen. The wood cells that are alive at maturity are the parenchyma cells. [10]

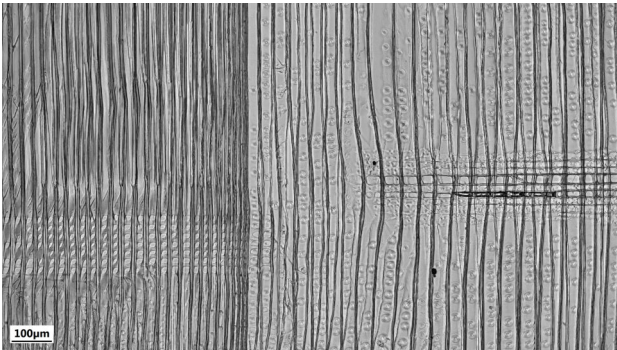
From the outside in, wood cell walls consists of the middle lamella, primary wall and secondary wall. The middle lamella is a layer of cell wall material connecting bordering cells. Adjacent to the middle lamella is the thin primary cell wall with randomly oriented cellulose microfibrils.



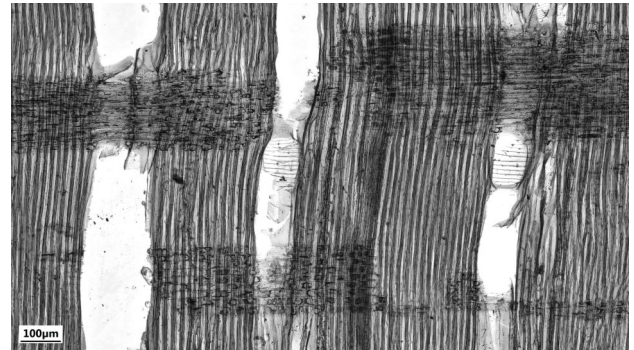
(a) Transverse Section of Softwood



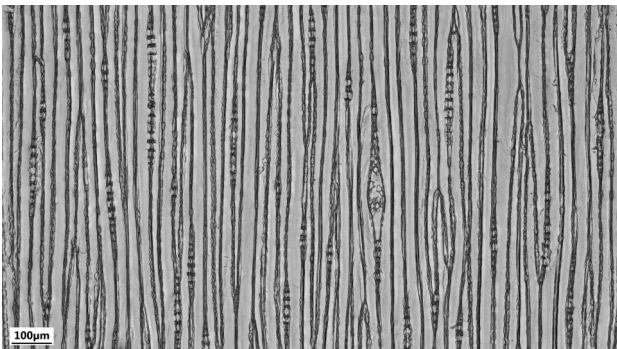
(b) Transverse Section of Hardwood



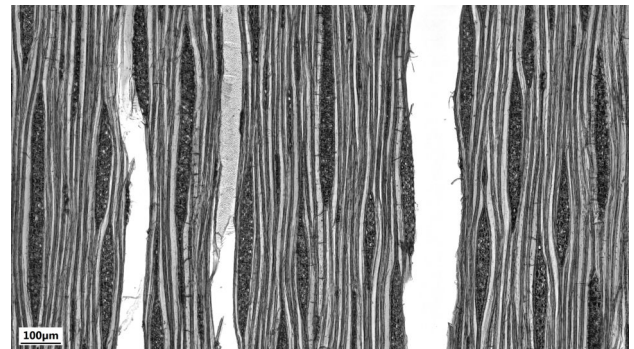
(c) Radial Section of Softwood



(d) Radial Section of Hardwood



(e) Tangential Section of Softwood



(f) Tangential Section of Hardwood

Figure 1: Light microscopy images of transverse (top), radial (middle) and tangential (bottom) sections comparing softwood (pine, left) and hardwood (birch, right) anatomy.

In light microscopy, the middle lamella and primary cell walls of two adjacent cells cannot be distinguished and are called the compound middle lamella. The secondary cell wall consists of three layers. The thin S_1 layer has helical microfibrils with a large microfibril angle between 50° and 70° . The microfibril angle describes the angle between the direction of the cellulose microfibrils and the axial direction of the cell which is defined as 0° . S_2 is the thickest secondary cell wall layer and characterized by a lower lignin content than S_1 and a small microfibril angle of 5° to 30° . Interior of S_2 is the S_3 layer, which has the lowest lignin content of the secondary wall and a high microfibril angle similar to S_1 . [10]

Wood cells mainly consist of cellulose, hemicelluloses and lignin, but also contain extractives and inorganic chemical components. Cellulose is the main component of wood and makes up 40% to 50% of the dry mass. Linked glucose molecules form linear cellulose polymers which combine into microfibrils through hydrogen bonds. Hemicelluloses account for 20% to 30% of the dry wood mass and consist of polymers with different monosaccharide composition, linkage structure and polymer size. In softwood, the main hemicelluloses are glucomannans with smaller amounts of xylans while in hardwood the main hemicelluloses are xylans. Hemicelluloses can form hydrogen bonds, but do not arrange into ordered microfibrils because they are highly branched and connect the cellulose microfibrils with the surrounding lignin matrix. Lignin makes up 26% to 32% of the dry wood mass in softwood and 20% to 28% in hardwood. It consists of three monomers, coniferyl and sinapyl alcohols and trace amounts of p-coumaryl alcohol, which combine to form three-dimensional network polymers. In softwood, lignin is mainly made of coniferyl alcohol which are called guaiacyl units in the lignin polymer while hardwood lignin contains mainly sinapyl and coniferyl alcohol. [10]

3 Site and Sample Context

An ice patch is a perennial snow or ice field that is mostly stagnant and while slow internal deformation can occur, it does not transport mass downslope like glaciers [2]. In glaciers, the constant renewal of ice and strain caused by ice movement limits the preservation of archaeological artifacts older than around 500 years [2]. In contrast, objects preserved in ice patches have been dated to more than 9000 years ago [5].

In an ice patch, archaeological artifacts can be moved vertically between stratigraphic layers during melting periods or laterally through ice deformation. After melting out of the ice, finds can be displaced by wind, water, permafrost processes, animals or humans. These processes generally move objects downslope and most objects are recovered at the lower edges of ice patches. The location of recovery therefore does not necessarily represent the original place of deposition. [2]

The preservation of artifacts depends on the circumstances, location and timing of deposition. Objects that are rapidly included in ice and remain continuously frozen are generally better preserved than objects that underwent repeated melting and freezing cycles. While in the ice, breakage and deformation can occur as snow is compressed and transformed into ice or through deformation of the ice. After melting out, the condition of artifacts is affected by the length of exposure before recovery, weight of seasonal snow and ice, meltwater, moisture, temperature, trampling, scree movement, downslope movement and the specific environmental conditions at the location of the artifact. However, artifacts from ice patches that have partially or completely melted show that wooden objects can survive for decades after melting out of the ice. [2]

The objects investigated in this thesis were recovered from Åndfonne ice patch in Breheimen National Park shown in Figure 2 [4]. The ice patch is located on the north slope of Mount Storivilen at an altitude of 1760-1980 m above sea level. Based on aerial photographs its area can be estimated to be around 233.000 m² in 2010, which makes it one of the largest ice areas in the region. The ice patch is surrounded by a lichen-free zone and high-alpine scree with several nearby ice patches where archaeological artifacts related to hunting and mountain crossings have been found. [4]

First archaeological finds from Åndfonne were reported in the 1980s and rescue work began in 2006 followed by fieldwork from 2010 to 2015. In 2011, a systematic survey of around 11.000 m² along the lower edge of the ice patch was conducted as most artifacts are found in the scree below the ice. In 2015, additional 16.000 m² were surveyed but due to poor preservation few artifacts were collected. In the two surveys 231 scaring sticks, six arrows, a complete bow, two hide objects and reindeer antlers and bones were recovered. Eleven finds have been carbon dated and show dates ranging from the beginning of the Bronze Age (2nd millennium BC) to the Early Medieval Period (8th-9th centuries AD). [4]

The 231 scaring sticks from Åndfonne are in the collection of the Museum of Cultural History in Oslo [4]. Scaring sticks are roughly worked wooden sticks, 1-2 m long, to which a flag made

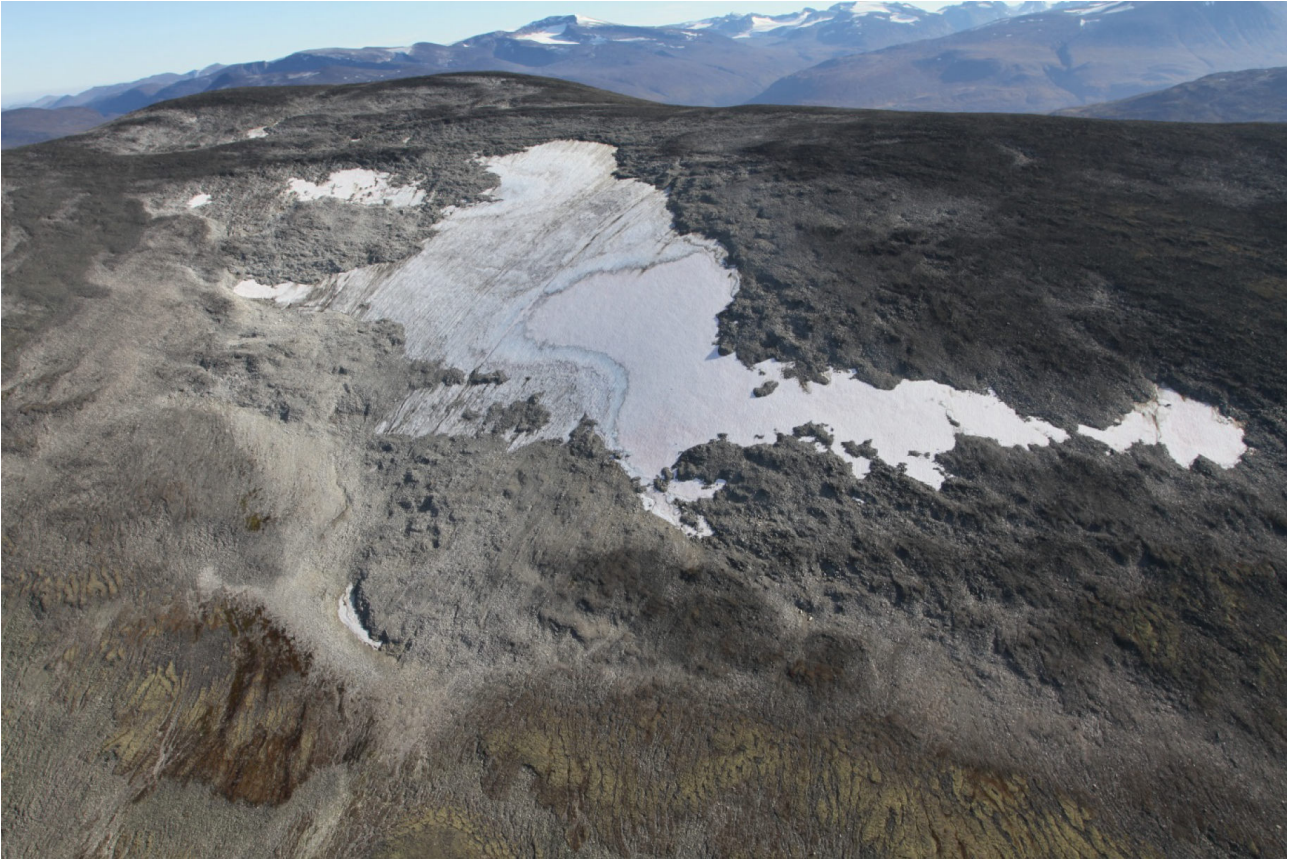


Figure 2: Aerial photograph of the Åndfonne ice patch from northwest in September 2014. Photo: Innlandet County Municipality, Norway.

of thin wooden sheets was attached [1]. They were stuck either into the snow or scree near the ice patch to guide reindeer during hunting [2]. The scaring sticks were wet when recovered but not waterlogged and a pin test confirmed the soundness of wood [1]. The conservation treatment included cleaning with brushes and water and controlled air drying [1].

Some scaring sticks were sampled in 2013 for a study that was not completed, with limited documentation on the sampling strategy or method. The samples slightly differ in size, but are generally around 1 cm^3 . For short stick fragments, one sample was taken from the middle while for longer sticks, two samples from the upper and lower section of the stick were taken. Little information on the specific location of the sticks at recovery was recorded, so no correlations between the preservation and environmental conditions, like exposure to sunlight, meltwater or wind, can be made.

4 Materials and Methods

4.1 Samples

All sampled scaring sticks are first photographed. For sticks from which two samples are taken, below ground (BG) and above ground (AG) are used to distinguish the two ends of the stick. Below ground refers to the sharpened side that was driven into the ground while above ground refers to the end where a flag would have been attached. Since all sticks are recovered lying on the ground, these terms describe the orientation of the stick and not their position at recovery. For analyses of the surface of samples, measurements are taken on the outermost exposed surface. For the interior, a thin layer of the cut surface is removed with a razor blade as changes to the exposed surfaces may have occurred during storage. The exception is the color measurement for which the interior was measured directly on the cut surface without removal of an additional layer.

Based on differences in surface appearance, a subset of samples is selected for further analysis consisting of four groups with three samples each and one additional sample. The analyzed samples are listed in Table 1.

Modern and archaeological pine wood (*Pinus sylvestris*) are used as reference material. The archaeological wood is recovered waterlogged from archaeological excavations and two samples are used, one is stored in water while the other is freeze-dried. Depending on the analysis, one or both archaeological wood samples are included. For all measurements, a thin layer of the sample surface is removed to minimize effects from changes occurring during storage. Cotton linters cellulose powder (Sigma-Aldrich, C6663) is used as a reference for cellulose.

4.2 Color Measurement

Color measurements are taken with a hand-held CM-700d spectrophotometer (Konica Minolta, Japan) using the 3 mm aperture and the specular component excluded (SCE) mode. At each measurement position, three individual measurements are averaged. To account for color variation within the sample, for example for differences between earlywood and latewood, each sample is analyzed at three different positions and the three measurements are averaged. The data is exported using the SpectraMagic NX software (Konica Minolta, Japan) and processed and plotted using Microsoft Excel and Python. The CIE $L^*a^*b^*$ colour space is used, which describes colors on a lightness axis L^* (black - white) and two color axes a^* (green - red) and b^* (blue - yellow).

4.3 Density Measurement

The samples are stored at 26°C and 32% relative humidity (RH) and the density is measured under these conditions. The ice patch wood samples are irregular in shape but are cut into smaller, approximately cuboid pieces for the DVS measurements which also simplifies volume measurement. All pieces contain some surface material, but the amount of surface material varies between pieces. While the surface shows different properties from the interior in some analyses, these differences are limited to a very thin surface layer and the influence on the measured density is expected to be small. The length, width and height of the pieces are determined with a digital caliper with an accuracy of ± 0.03 mm. To account for deviations in the shape, each dimension is measured at both ends and averaged. The weight is measured with a scale with a readability of 0.1 mg and the density can then be calculated as the ratio of mass to volume. Deviations of the cut pieces from a cuboid shape are the main source of uncertainty and a 10% uncertainty is estimated for the density values. Pieces with a visually more noticeable shape differences are assigned a higher uncertainty of 15%.

4.4 Stereomicroscopy

The sample surfaces are investigated using a stereomicroscope (Nikon, Japan). Lighting is positioned at low angles to create a raking light effect and highlight surface features. Images are acquired with a Euromex UHD-4K camera (Euromex, The Netherlands) and the ImageFocus Alpha software (Euromex, The Netherlands).

4.5 Light Microscopy

The sections for light microscopy are cut by hand from the interior and surface of the samples with a razor blade and mounted onto a microscope slide with a drop of glycerol. A cover slip is placed on top of the sample. The sections are examined using a Olympus BX60 light microscope (Olympus Optical, Japan) in transmitted light mode and with parallel- and cross-polarized light. Images are acquired with a Euromex UHD-4K camera (Euromex, The Netherlands) and the ImageFocus Alpha software (Euromex, The Netherlands). Extended depth of field (EDF) imaging is applied to improve focus for uneven sections.

4.6 ATR-FTIR Spectroscopy

For Fourier-transform infrared (FTIR) spectroscopy, small chips are cut from the interior and surface of the samples with a razor blade and analyzed on a Nicolet iS50 spectrometer (Thermo Scientific, USA) with a diamond attenuated total reflection (ATR) crystal and DTGS detector. A clamp is used to apply the same pressure to all samples and ensure good contact between the ATR crystal and samples. The spectra are recorded with 32 scans at 4 cm^{-1} resolution over the

mid-infrared region between 4000 cm^{-1} and 500 cm^{-1} . The effects of anatomical section and alignment of the sample on the ATR crystal are investigated but for all other measurements, these parameters are kept constant. Radial sections are used and the longitudinal axis of the wood is aligned parallel to a small visible line along the center of the ATR crystal. To account for variation within the sample, each wood chip is analyzed at three different positions and the three measurements are averaged with the OMNIC software (Thermo Scientific, USA), which is also used to apply a baseline correction. The spectra are normalized and plotted using Python. The intensities of all spectra are normalized to the 1370 cm^{-1} band which is assigned to C–H stretching in CH_3 mainly originating from cellulose [11]. This band has a comparatively high intensity, central position and strong stability during degradation and is therefore often used as a normalization band [12].

4.7 DVS Measurement

The samples selected for dynamic vapor sorption (DVS) measurements are cut into pieces of approximately 100 mg and from each sample, two pieces are investigated. The pieces have different amounts of surface material, but as the surface differences are limited to a very thin layer, the influence on the DVS measurement is expected to be small. Before the experiment, the samples are stored at 26°C and 32% RH. The DVS measurements are performed on a Vsorp Enhanced analyzer (ProUmid, Germany). During the experiment, the temperature is kept constant at 25°C . The DVS measurement starts with conditioning the samples at 0% RH for 72 h. The RH is then set to increase in steps of 5% from 0% to 90%. Each step is maintained until the wood reaches its equilibrium moisture content or for a maximum duration of 100 h. The equilibrium conditions are $dm/dt < 0.02\%$ per 120 min or that the sample mass fluctuates within the resolution range of the balance. The measurement is stopped at 90% RH before equilibrium is reached due to an instrument malfunction.

5 Results and Discussion

5.1 Visual Description

The scaring sticks vary considerably in appearance, which is illustrated in Figure 3. While some sticks are preserved as intact pieces, others are recovered in fragments and in some cases only a small part of the original stick survives. The shape ranges from straight to considerably bent. The original orientation can be identified through the sharpened lower end where the stick was driven into the ground or the upper section which is shaped to allow for a flag to be attached. While some sticks are uniform in appearance, others have visible differences between fragments or between sections that suggest different parts of the same stick may have undergone different degradation processes.

One of the most apparent differences between sticks is their surface appearance. The surface color ranges from dark brown to bleached light white and many surfaces have a slightly shiny appearance. Patchy discolorations and small attached deposits are present on some sticks. The surface texture varies between fibrous surfaces with loose protruding fibers and comparatively smooth sticks. Most sticks have small cracks along the longitudinal direction and some show separation along the growth rings. One distinct sample is C57838/4 F.169 (Figure 3h) with surface features that might be related to insect activity [10].

These surface features appear to be restricted to the outer layer and do not seem to affect the inner wood. The interior wood shows no visual differences to modern wood, it has a light brown color, often visible growth rings and there are no significant differences between the sticks.

As analyzing every sample in detail is beyond the scope of this thesis, a representative subset is selected based on surface appearance. Two groups are based on surface color, which for most sticks appears to correlate with surface texture. A dark *Brown Surface* as shown in Figures 4a and 4b is typically associated with a smooth surface. In contrast, sticks with a light *White Surface* often have loose fibers attached to the surface. Examples are shown in Figures 4c and 4d. Besides color, two surface features can be observed across the sticks: large diffuse dark discolorations of the surface (Figures 4e and 4f) and small black deposits strongly attached to the surface (Figures 4g and 4h), which correspond to the *Surface with Discolorations* and *Surface with Deposits* categories. The sticks with these surface features show some variation in surface color, however the discolorations appear mainly on light grey or white sticks while the deposits predominantly occur on brown sticks with some surface fibers. These four categories represent the most commonly observed surfaces. However, some sticks do not fit into these groups, for example white surfaces without loose fibers or brown sticks with surface fibers. For each surface category, three samples are selected which are listed in Table 1.



(a)



(b)



(c)



(d)



(e)



(f)



(g)



(h)

Figure 3: Photographs of scaring sticks illustrating the variation in appearance. The samples from the sticks in (a), (b), (e) and (f) are selected for further analysis while the sticks in (c), (d), (g) and (h) are shown for comparison. (a) C57838/4 F.238, (b) C57838/4 F.135-2, (c) C57838/4 F.219, (d) C57838/3 F.165, (e) C57838/4 F.183, (f) C57838/4 F.205, (g) C57838/4 F.117, (h) C57838/4 F.169. Photo: Eva Brænd, Museum of Cultural History, Norway.



(a) Brown Surface



(b) Brown Surface



(c) White Surface



(d) White Surface



(e) Surface with Discolorations



(f) Surface with Discolorations



(g) Surface with Deposits



(h) Surface with Deposits

Figure 4: Photographs of selected scarring sticks illustrating the surface appearance categories defined in Table 1. (a) C57838/4 F.230, (b) C57838/4 F.174 AG, (c) C57838/4 F.183 AG, (d) C57838/3 F.253 BG, (e) C57838/4 F.172 AG, (f) C57838/4 F.135-2 BG, (g) C57838/4 F.175 AG, (h) C57838/4 F.260 AG. Photo: Eva Brænd, Museum of Cultural History, Norway.

Table 1: Categories of ice patch wood samples based on surface appearance and samples selected for further investigation.

Surface Group	Surface Appearance	Samples
Brown Surface	Dark brown color, smooth surface	C57838/4 F.205 AG C57838/4 F.230 C57838/4 F.174 AG
White Surface	Light grey to white color, loose surface fibers	C57838/4 F.238 C57838/4 F.183 BG C57838/3 F.253 BG
Surface with Discolorations	Diffuse dark grey to black discolorations, grey color, mostly smooth surface with some loose surface fibers	C57838/4 F.172 AG C57838/4 F.136 BG C57838/4 F.135-2 BG
Surface with Deposits	Small black adhering deposits, brown color, some loose surface fibers	C57838/4 F.175 AG C57838/4 F.260 AG C57838/4 F.167 BG
Additional Sample	Dark brown, smooth and light grey, fibrous surfaces on the same sample	C57838/4 F.133 AG

5.2 Color Measurement

The color of the ice patch wood samples is measured and compared to modern and freeze-dried archaeological pine. The results on the CIE $L^*a^*b^*$ colour scale are shown in Figure 5 for the interior and Figure 6 for the surface.

For modern pine, the color can vary between sapwood and heartwood and depend on geographical origin and sampling height within the tree [13]. As only one modern pine sample is used this variation is not represented. The measured color of the modern pine sample agrees with the literature value within one standard deviation [13].

The freeze-dried archaeological pine has a similar lightness to modern pine but lower values on the a^* and b^* axes which corresponds to less red and less yellow in the color.

For the ice patch wood samples the color measurements of the interior are very close to each other in all color coordinates. This is in agreement with the visual observations showing that the interior of ice patch wood is similar between all samples. The L^* value is lower than that of modern and freeze-dried archaeological pine, but a^* and b^* show similar values. As the color of the ice patch wood interior is measured directly on the cut surfaces, changes that may have occurred during storage cannot be distinguished. For example, UV exposure can darken wood and change its color which could contribute to the lower L^* value [14]. For all further investigations of the ice patch wood interior, a thin surface layer is removed before measurement.

The color measurements of the ice patch wood sample surfaces show much larger variation in all color coordinates. To illustrate this, Figures 7 and 8 show averaged L^* , a^* and b^* values as bar diagrams including the standard deviation calculated from the three measurements on each sample. The *Brown Surface* samples generally show low lightness values and particularly C57838/4 F.230 and C57838/4 F.174 AG have significantly lower lightness than all other samples, while the *White Surface* samples show comparatively high lightness values. For the *Surface with Discolorations* samples, C57838/4 F.172 AG shows a relatively high lightness, while the other two samples (C57838/4 F.136 BG and C57838/4 F.135-2 BG) have intermediate values similar to the *Surface with Deposits* samples, which have little variation in lightness. Compared to modern pine, the surfaces of ice patch wood samples show lower a^* and b^* values indicating less red and less yellow coloration, which is consistent with the color differences observed. In contrast to the lightness, no significant differences between the surface groups can be observed in the a^* and b^* values.

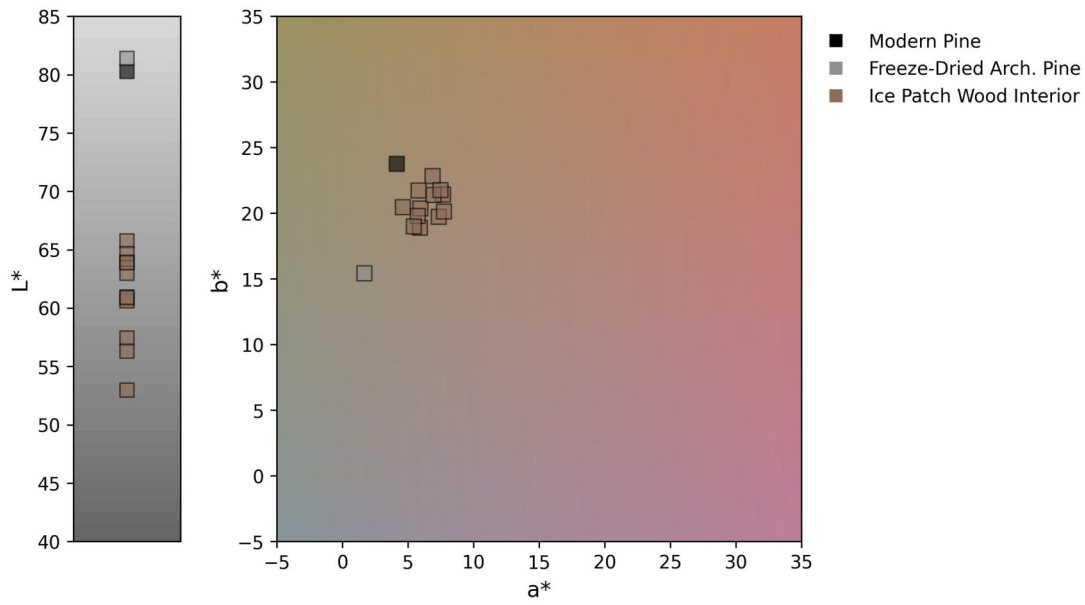


Figure 5: CIE L*a*b* color measurements comparing the interior of ice patch wood samples with modern and freeze-dried archaeological pine.

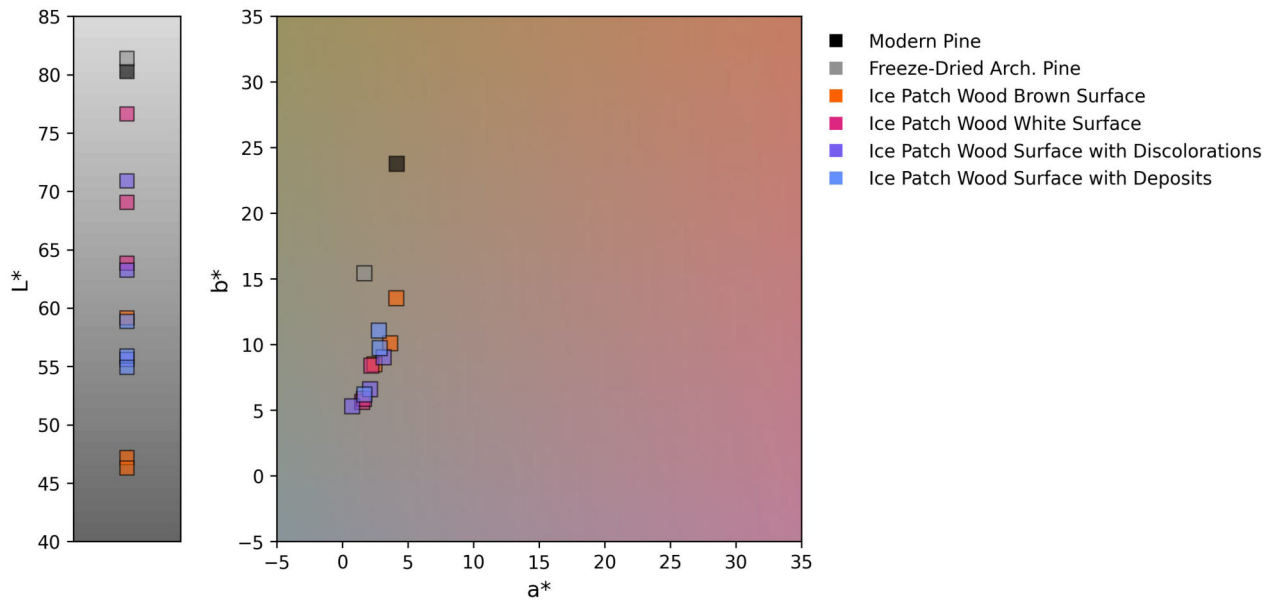


Figure 6: CIE L*a*b* color measurements comparing the surface of ice patch wood samples from the *Brown Surface*, *White Surface*, *Surface with Discolorations* and *Surface with Deposits* categories with modern and freeze-dried archaeological pine.

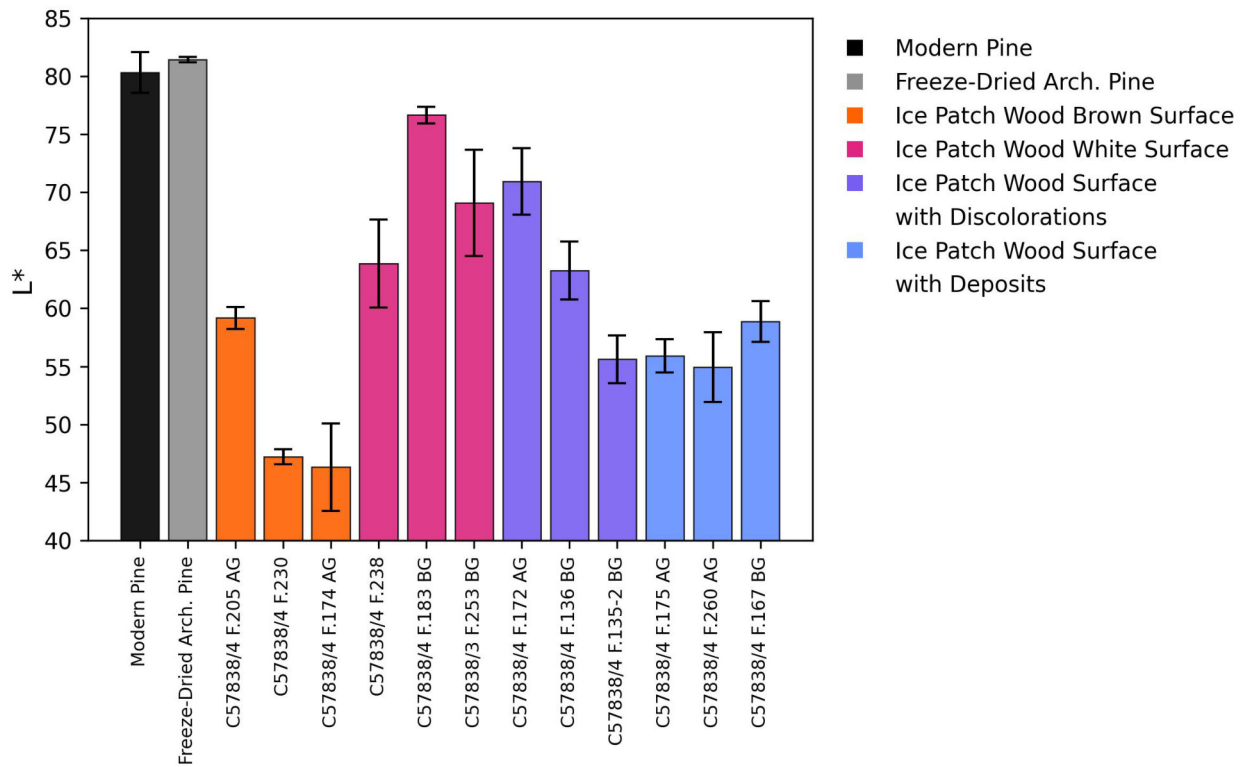


Figure 7: CIE L^* averaged color measurement comparing the surface of ice patch wood samples from the *Brown Surface*, *White Surface*, *Surface with Discolorations* and *Surface with Deposits* categories with modern and freeze-dried archaeological pine.

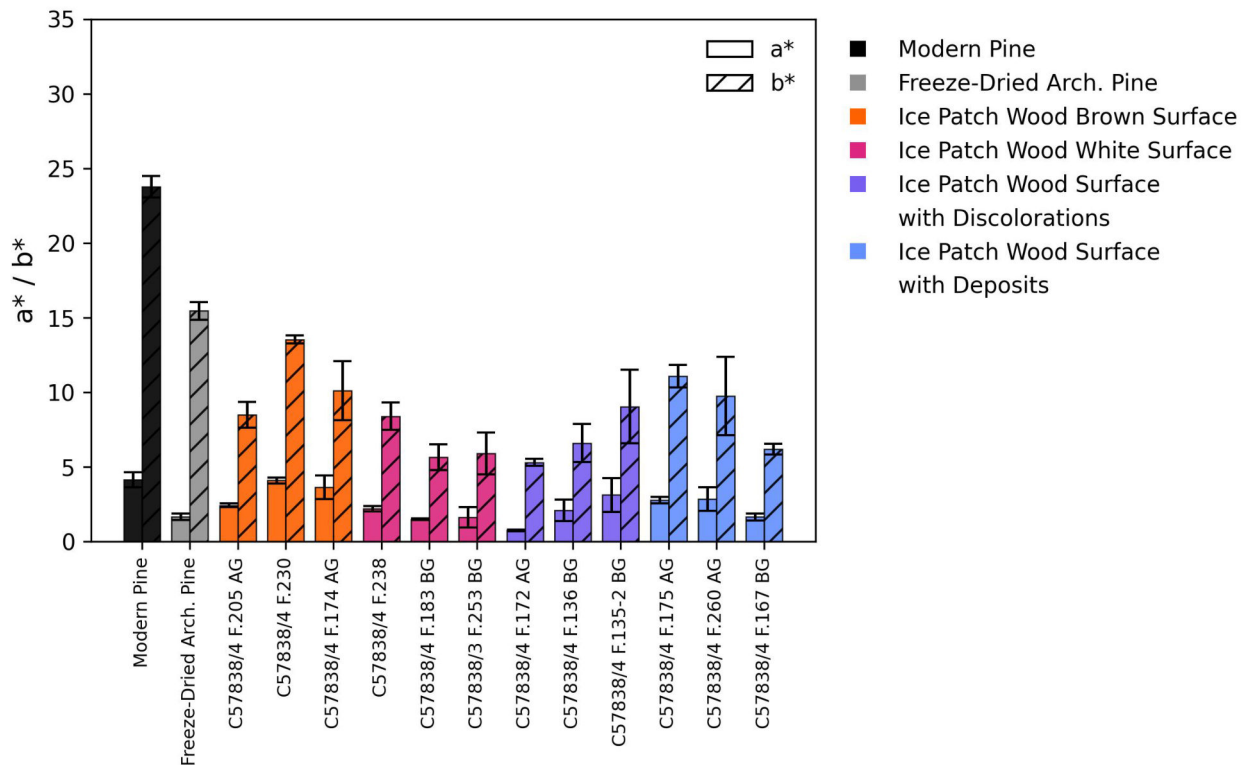


Figure 8: CIE a^* and b^* averaged color measurement comparing the surface of ice patch wood samples from the *Brown Surface*, *White Surface*, *Surface with Discolorations* and *Surface with Deposits* categories with modern and freeze-dried archaeological pine.

Table 2: Measured densities of ice patch wood samples and modern and freeze-dried archaeological pine at 26°C and 32% RH. Literature values for pine with at a moisture content of 12% are shown for comparison.

Sample	Density (g/cm^3)
Modern Pine	0.51 ± 0.05
Archaeological Pine	0.19 ± 0.02
C57838/4 F.205 AG	0.42 ± 0.04
C57838/4 F.230	0.36 ± 0.04
C57838/4 F.238	0.39 ± 0.06
C57838/4 F.172 AG	0.34 ± 0.05
C57838/4 F.175 AG	0.44 ± 0.04
Literature <i>Pinus sylvestris</i> [16]	0.330 ... 0.510 ... 0.890
Literature <i>Pinus sylvestris</i> (Norway) [17]	0.448 ± 0.041

5.3 Density Measurement

The density of the ice patch wood samples is measured to assess the condition of the wood. The calculated densities at 26°C and 32% RH for the ice patch wood samples and modern and freeze-dried archaeological pine are listed in Table 2. In literature, the density of wood is often given at a standard moisture content of 12% which can usually be achieved by conditioning wood at 20°C and 65% RH [15]. For *Pinus sylvestris* the range of density at a moisture content of 12% is 0.330 - 0.890 g/cm^3 with a mean of 0.510 g/cm^3 [16]. For *Pinus sylvestris* from Norway the density at a moisture content of 12% is (0.448 ± 0.041) g/cm^3 [17]. The wood samples are stored and measured at 26°C and 32% RH, which corresponds to an equilibrium moisture content of $\sim 6\%$ [10]. Since this moisture content is lower than the 12% moisture content of the literature values, the measured density is expected to be lower than the literature values.

The measured density of modern pine agrees well with literature values, it matches the mean literature value for pine and the 1σ interval overlaps with the density range for Norwegian pine.

The freeze-dried archaeological pine has a significantly lower density and the 3σ interval does not overlap with the literature range. Low densities are common for archaeological wood as a result of degradation processes [18]. In literature, the density of archaeological wood is often given as the basic density, which is calculated with the sample mass after drying related to the volume before drying [19]. The basic density of the investigated archaeological pine has been previously determined to be 0.132-0.179 g/cm^3 [20]. The difference of the measured value to the basic density is due to the archaeological wood sample already being freeze-dried, so the fully saturated volume is not determined and it was also not further dried before density measurement.

The densities of the ice patch wood samples fall within the literature range for *Pinus sylvestris*. With the exception of C57838/4 F.230 and C57838/4 F.172 AG, the values agree within 1σ with the value for Norwegian pine. No significant difference in density between the ice patch wood samples and modern pine can be found.

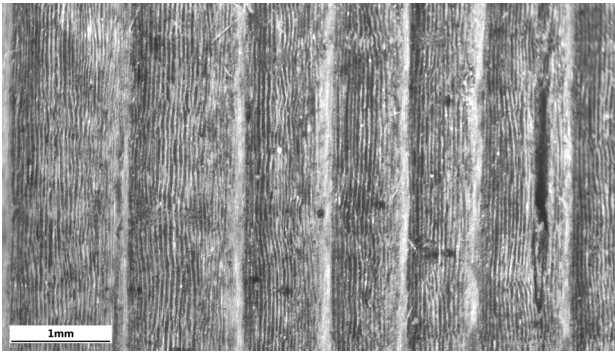
5.4 Stereomicroscopy

Stereomicroscopy imaging is used to investigate the surfaces of the ice patch wood samples and representative results are shown in Figure 9. Some surface features that are visible by eye become more apparent in the stereomicroscopy images, for example the shiny appearance and longitudinal cracks of most surfaces. Small cracks or splitting along the growth rings which are observed on many sticks could be formed during drying due to differential shrinkage where the outer parts of the wood dry and shrink before the inner material has started to dry and shrink [10] or in freeze-thaw cycles [2].

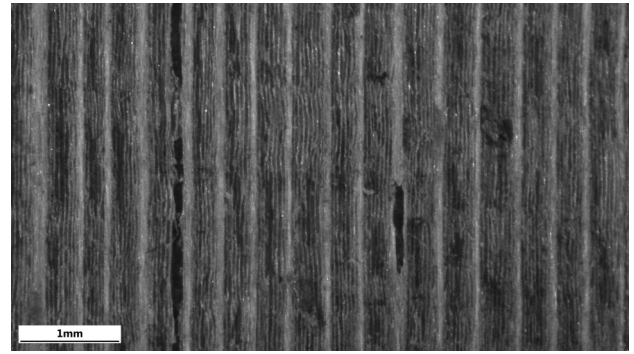
Most sticks display on at least one surface a pattern where the earlywood is eroded and lower in profile while the latewood is preserved and protruding. The resulting surface profile is clearly visible in the stereomicroscopy images of *Brown Surface* samples in Figures 9a and 9b. This type of erosion is common on wood surfaces exposed to water or wind carrying small particles like ice or sand which abrade the thinner-walled earlywood more strongly than the thick-walled latewood [2, 21, 22].

Stereomicroscopy also allows a closer investigation of the surface fibers on some samples. Figures 9c and 9d show that these are wood fibers separating from the surface. This might be similar to observations from wooden huts in Antarctica where defibration can occur under the influence of salt, water and wind [22]. On ice patches, degradation through exposure to water and wind is possible and could contribute to the separation of fibers.

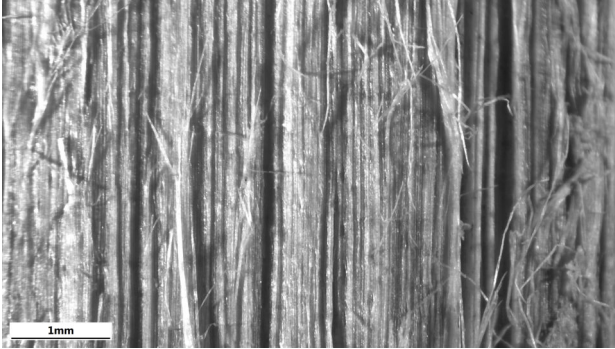
The *Surface with Discolorations* samples in Figures 9e and 9f show small black spots on the surface in the darker areas which could be connected to the observed discolorations. The deposits on the *Surface with Deposits* samples in Figures 9g and 9h appear as raised rounded dark material attached to the wood surface. The deposits vary in size with some individual particles forming larger aggregates.



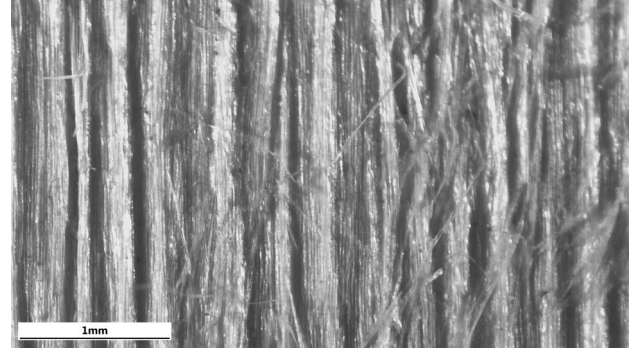
(a) Ice Patch Wood Brown Surface



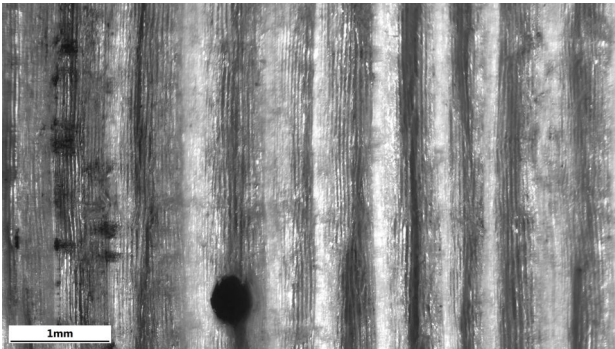
(b) Ice Patch Wood Brown Surface



(c) Ice Patch Wood White Surface



(d) Ice Patch Wood White Surface



(e) Ice Patch Wood Surface with Discolorations



(f) Ice Patch Wood Surface with Discolorations



(g) Ice Patch Wood Surface with Deposits



(h) Ice Patch Wood Surface with Deposits

Figure 9: Representative stereomicroscopy images of the surfaces of ice patch wood samples from the *Brown Surface*, *White Surface*, *Surface with Discolorations* and *Surface with Deposits* categories. (a) C57838/4 F.230, (b) C57838/4 F.205 AG, (c) C57838/4 F.183 BG, (d) C57838/3 F.253 BG, (e) C57838/4 F.172 AG, (f) C57838/4 F.136 BG, (g) C57838/4 F.175 AG, (h) C57838/4 F.260 AG.

5.5 Light Microscopy

Light microscopy is used to investigate several ice patch wood samples in all three anatomical sections. The samples are relatively hard which makes it difficult to cut thin sections by hand and the resulting sections and light microscopy images have varying quality. The features discussed in this chapter are consistently observed across the ice patch wood samples, but only the clearest images are shown. A more complete overview of all light microscopy images can be found in Appendix A.

5.5.1 Comparison with Reference Samples

Figure 10 compares transverse sections of the interior from ice patch wood samples with modern and archaeological pine. The images in the left column are taken at different magnifications and show both earlywood and latewood. For most samples, the earlywood cells are more easily torn during sectioning than the smaller and more compact cells in the latewood. In the right column all images are taken at the same magnification and the differences in cell size between samples are due the position of the section in the growth ring.

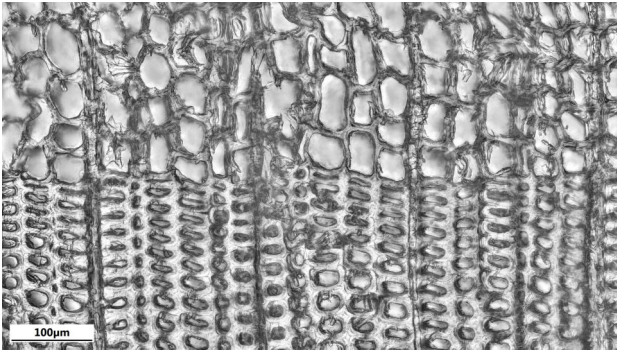
The images of the modern pine show the expected cellular structure of softwood. The tracheid lumina are empty and surrounded by intact cell walls in which the secondary cell walls and the compound middle lamellae are visible. The sections also show bordered pits between tracheids.

In contrast, the images of water-stored archaeological pine show strong degradation of all wood cells. The observed pattern is consistent with degradation by erosion bacteria which enter the cell lumen through rays and pits and initially erode the cellulose-rich S_2 layer while the S_3 layer is decayed at later stages [23]. Erosion bacteria leave behind residual material which can be seen in the images as granular material filling the lumina and cell wall spaces [23, 24]. The lignin-rich middle lamella is mostly unaffected by erosion bacteria [23, 24]. All cells in the section are heavily degraded and no cells remain intact or partially preserved which indicates very advanced degradation of the archaeological wood [24].

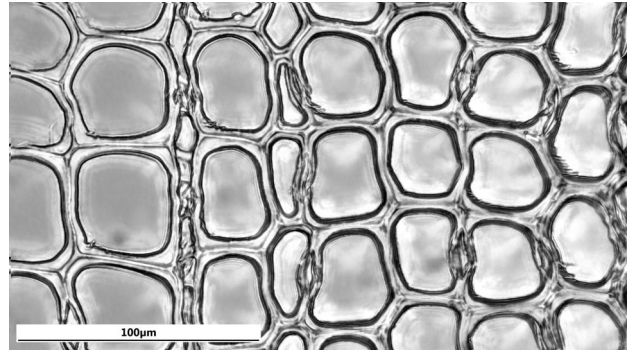
Figures 10e and 10f show the result of freeze-drying highly degraded waterlogged archaeological pine. Degraded wood cells can maintain their structure as long as the wood is waterlogged but when water is removed by freeze-drying the cell walls shrink as the supporting cell wall layers have deteriorated [23]. This shrinkage is visible in the images. The structure of the freeze-dried archaeological pine also explains its low density as the bacterial degradation results in a reduced amount of remaining material per volume.

In comparison with the reference samples the transverse sections of the ice patch wood samples are similar to modern pine. The tracheids are intact and the secondary cell walls and compound lamellae are visible. The lumina are preserved and empty. No evidence for biological degradation can be observed.

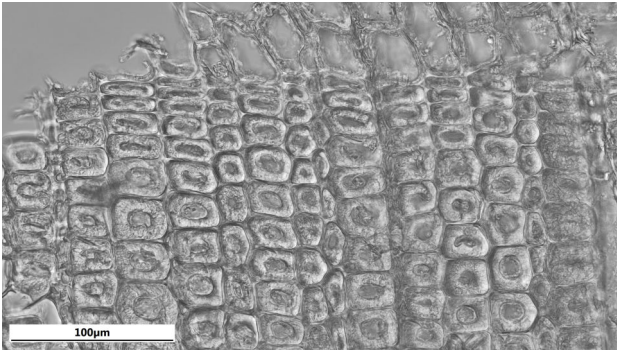
For wood frozen in ice, the absence of biological decay can be explained by the low temperatures



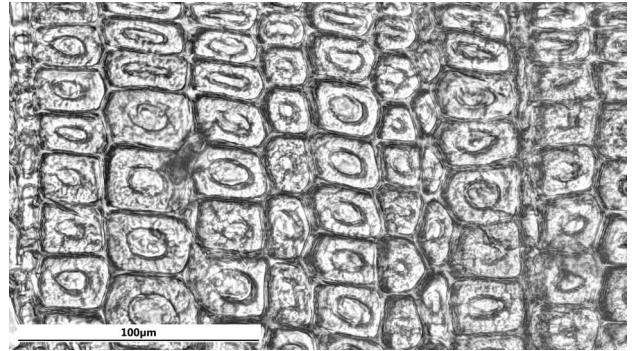
(a) Modern Pine



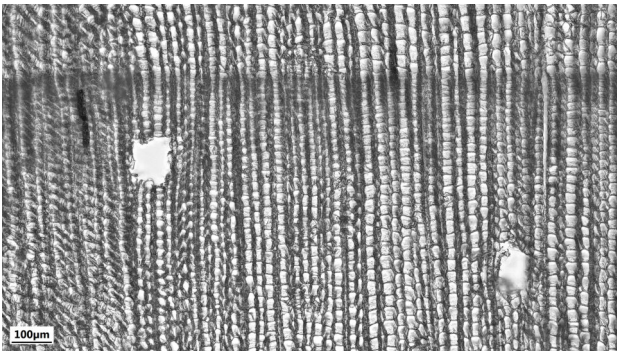
(b) Modern Pine



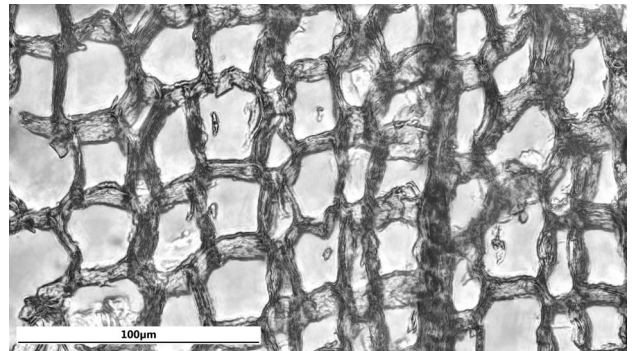
(c) Water-Stored Arch. Pine



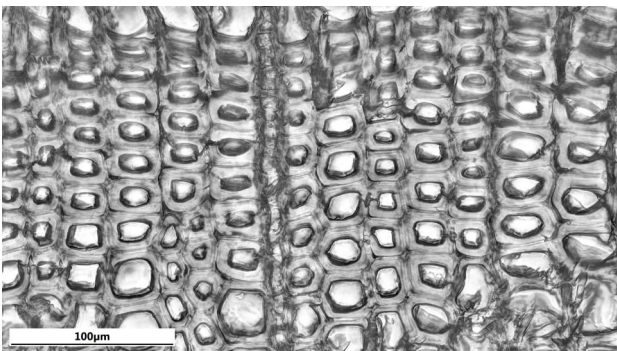
(d) Water-Stored Arch. Pine



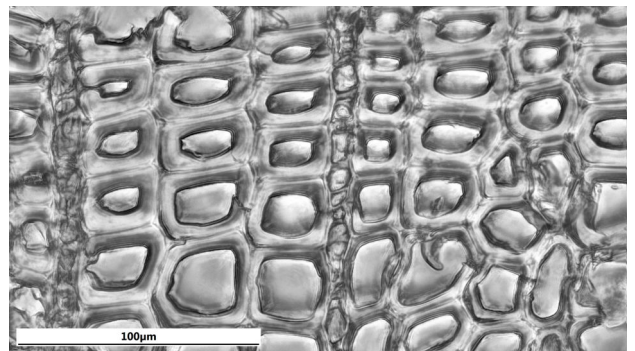
(e) Freeze-Dried Arch. Pine



(f) Freeze-Dried Arch. Pine

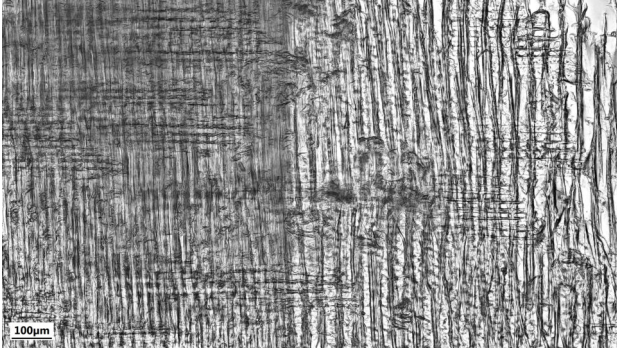


(g) Ice Patch Wood Interior

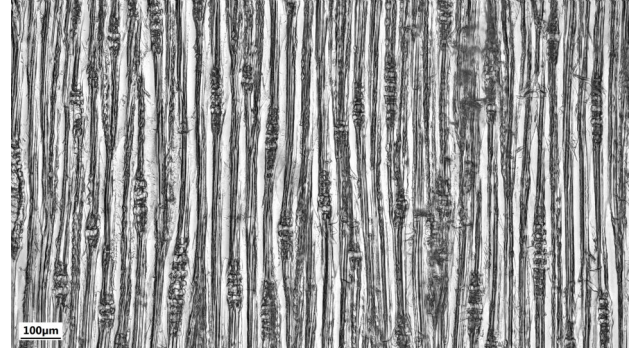


(h) Ice Patch Wood Interior

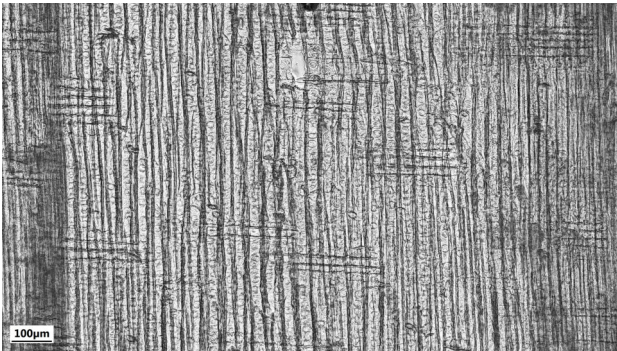
Figure 10: Light microscopy images of transverse sections comparing the interior of ice patch wood samples with modern and archaeological pine. (g) C57838/4 F.230, (h) C57838/4 F.230.



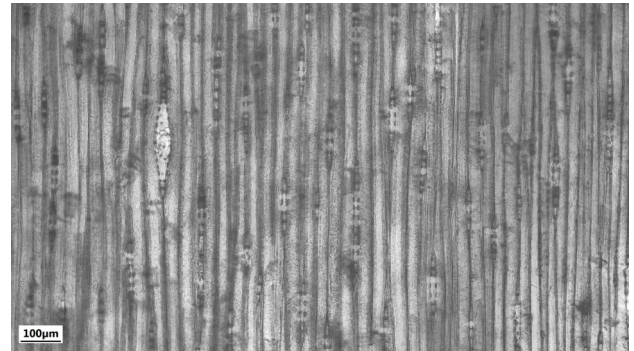
(a) Modern Pine



(b) Modern Pine



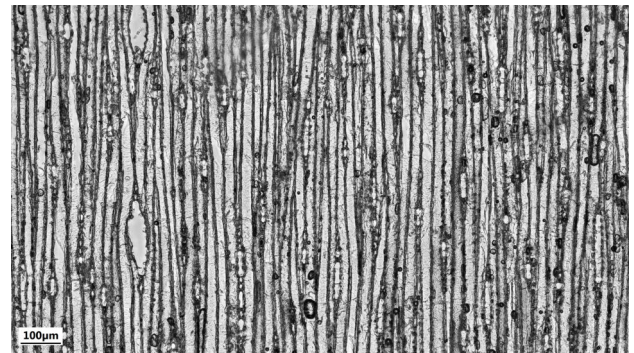
(c) Water-Stored Arch. Pine



(d) Water-Stored Arch. Pine



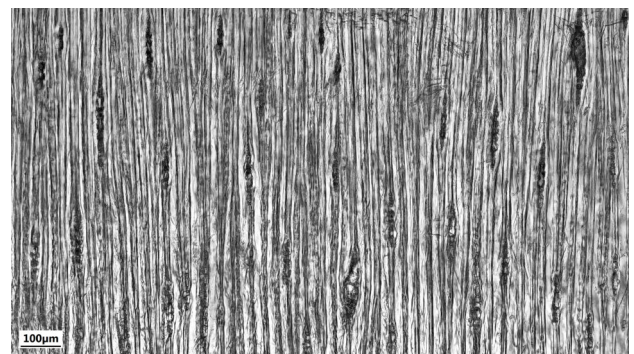
(e) Freeze-Dried Arch. Pine



(f) Freeze-Dried Arch. Pine



(g) Ice Patch Wood Interior



(h) Ice Patch Wood Interior

Figure 11: Light microscopy images of radial (left) and tangential (right) sections comparing the interior of ice patch wood samples with modern and archaeological pine. (g) C57838/4 F.205 AG, (h) C57838/4 F.238.

and lack of oxygen and liquid water which limit biological activity [25]. Studies of wood preserved in soil that undergoes yearly freeze-thaw cycles find that the wood can show loss of cell wall structure by soft rot after only a few years [25, 26]. The lack of biological decay in the interior could indicate that the sticks have been continuously frozen since they were first incorporated into the ice patch. If the sticks had thawed and frozen each year, some biological decay might be expected. A study comparing the extent of ice patches in Jotunheimen National Park, Norway, with the ages of preserved archaeological objects also suggests that finds remain continuously frozen [27]: The preserved artifacts date to periods when the ice patches are contracted and in a growing phase. When ice patches are reduced in size, during summer reindeer are concentrated on small ice areas high in the mountains. A following growth of the ice patches covers artifacts in snow shortly after deposition. In contrast, ice patches have few archaeological finds from colder time periods, like the Little Ice Age, when ice patches are larger. During these periods, reindeer could also use ice areas at lower altitudes which later melted and artifacts are not preserved. [27]

Figure 11 compares the radial and tangential sections of the interior from ice patch wood samples with the reference samples and confirms the observations made for the transverse sections.

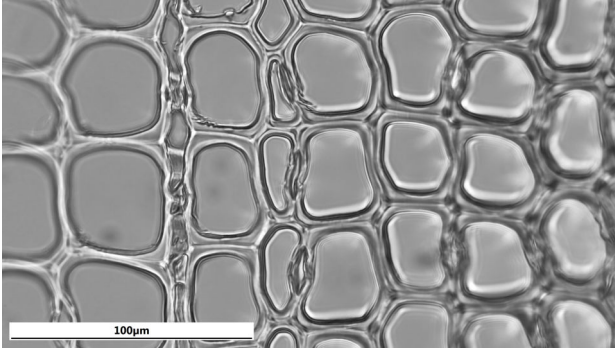
5.5.2 Polarized Light Microscopy

Polarized light microscopy can be used to show the presence and location of crystalline cellulose through birefringence [24, 28]. Figure 12 compares the transverse sections of ice patch wood samples with modern and archaeological pine under parallel- and cross-polarized light. The cross-polarized images are less sharp because the extended depth of field processing, which was used for most parallel-polarized light images, could not be applied.

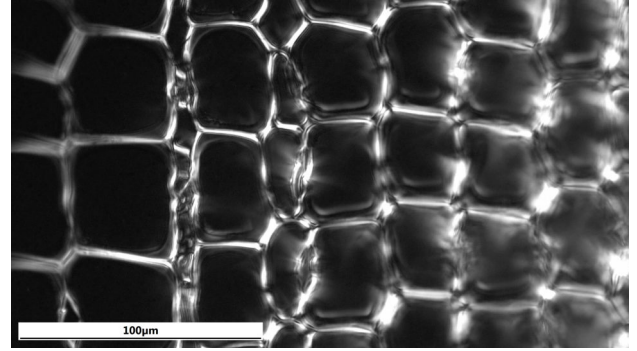
In modern pine, crystalline cellulose is present in all cell wall layers, but in different amounts and with different microfibril angles [10]. While S_3 has the highest relative cellulose content, S_2 is the thickest cell wall layer and has the largest absolute amount of cellulose that can show birefringence [10]. In the polarized light image of modern pine the empty lumina appear dark and all cell wall parts visible under parallel-polarized light are also present in the cross-polarized images, with differences resulting from variations in cellulose content and orientations [29, 30].

In the cross-polarized light image of water-stored archaeological pine, weak light can be observed only around the cell borders. Bacterial decay degrades the cellulose and the residual material left behind in the lumina and the spaces of the cell wall no longer contains crystalline cellulose, so these areas appear dark. As erosion bacteria degrade the S_2 layer before the S_3 layer, birefringence of S_3 can still be observed in less advanced stages of decay [24]. For this sample, birefringence of the S_3 layer is absent, which indicates advanced decay.

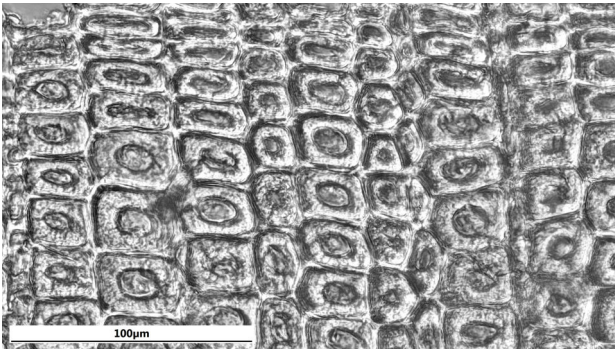
Under cross-polarized light, freeze-dried archaeological pine has a similar appearance to water-stored archaeological pine. Only the remaining cell borders are visible while the empty cell



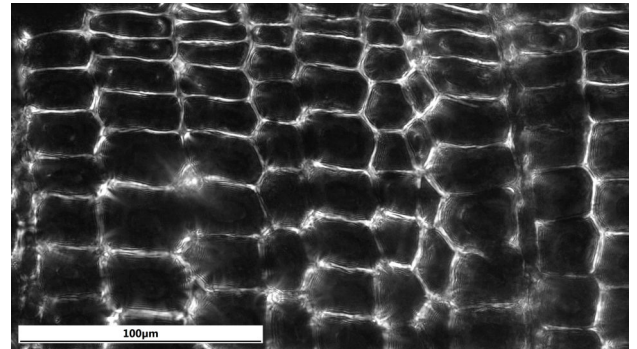
(a) Modern Pine



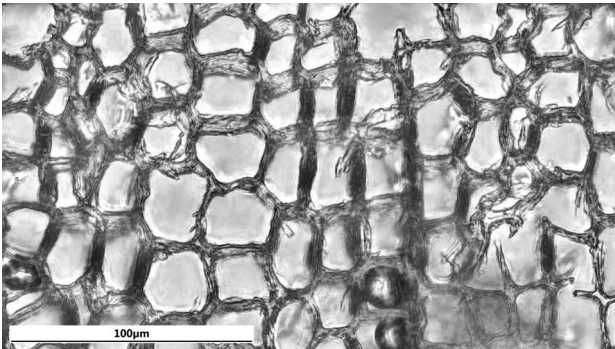
(b) Modern Pine



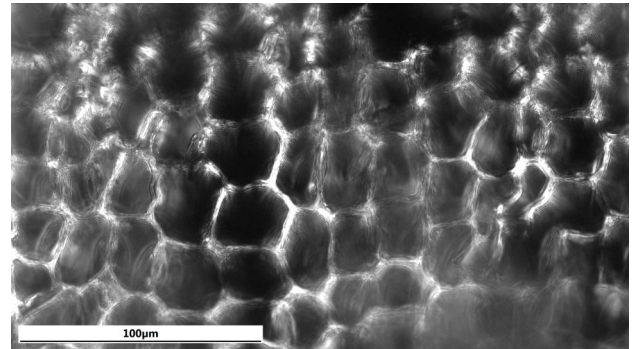
(c) Water-Stored Arch. Pine



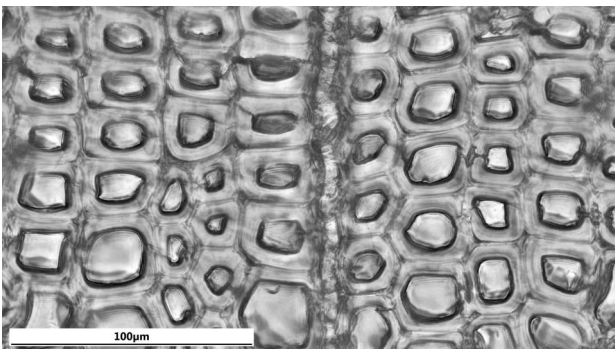
(d) Water-Stored Arch. Pine



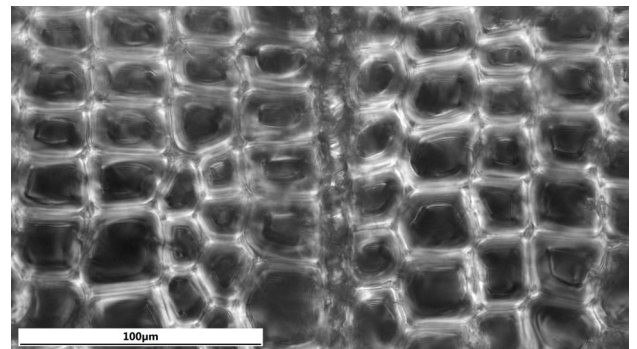
(e) Freeze-Dried Arch. Pine



(f) Freeze-Dried Arch. Pine



(g) Ice Patch Wood Interior



(h) Ice Patch Wood Interior

Figure 12: Light microscopy images of transverse sections comparing the interior of ice patch wood samples with modern and archaeological pine under parallel-polarized light (left) and cross-polarized light (right). (g) C57838/4 F.230, (h) C57838/4 F.230.

interiors appear dark.

For the ice patch wood samples the intact cell walls observed under parallel-polarized light are also visible under cross-polarized light. The birefringence of the cellulose-rich S_3 layer on the lumen side of the cell wall is particularly visible in some cells. Overall, the cross-polarized image is similar to modern pine, with small differences likely due to section and image quality. This supports the observations made under parallel-polarized light that the tracheids maintain their cell wall structure and composition.

5.5.3 Genus-Level Identification

Wood identification is based on the identification key in [31] for Northern European tree species. Figures 13a and 13b show tangential sections of ice patch wood samples. The wood has no vessels, indicating that it is a softwood. Spiral thickening is absent while radial resin canals are present. The rays are uniseriate and less than 15 cells high. The transverse sections in Figures 13c and 13d show an abrupt transition between earlywood and latewood as well as axial resin canals in the latewood. In the radial sections in Figures 13e and 13f, the axial tracheid pits are uniseriate. Figures 13g and 13h show ray tracheids with dentate tracheid walls and fenestriform cross-field pits in both the earlywood and latewood. [31]

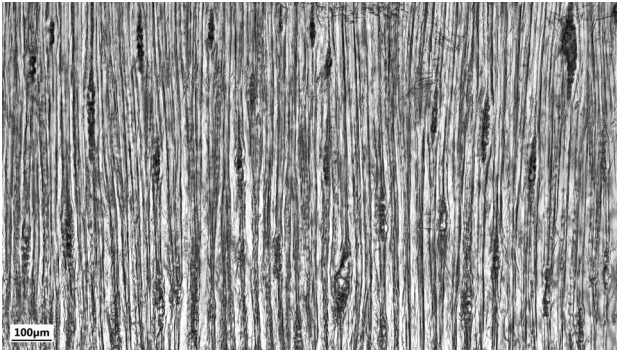
This combination of anatomical features identifies the genus as *Pinus*. The identification key includes both *Pinus sylvestris* and *Pinus mugo* which cannot be differentiated based on their anatomy. However as *Pinus sylvestris* is the only species native to Norway, it can be considered the most likely species [31].

5.5.4 Surface Features

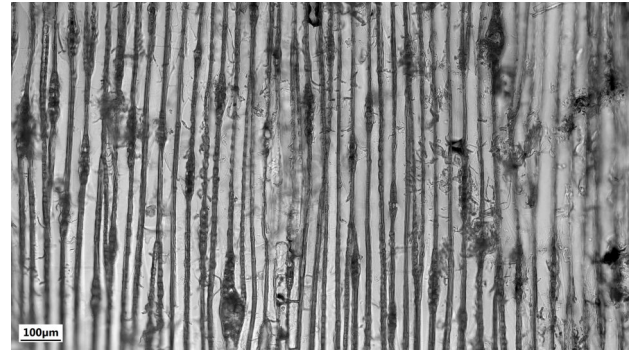
Light microscopy is used to examine sections from the surface of ice patch wood samples and several samples show notable surface features. It is important to note that only a small surface area is examined, which may not be representative of the whole sample or stick. All observed features appear to be present only on the surface and do not extend into the interior.

Figures 14a - 14d show thin, branching filaments on *Brown surface*, *White Surface* and *Surface with Discolorations* samples. One possible explanation is that these are hyphae from wood-staining fungi, which can cause discoloration without degrading the wood [23, 32]. This could explain the surface discolorations on some sticks and potentially contribute to the dark brown color of others.

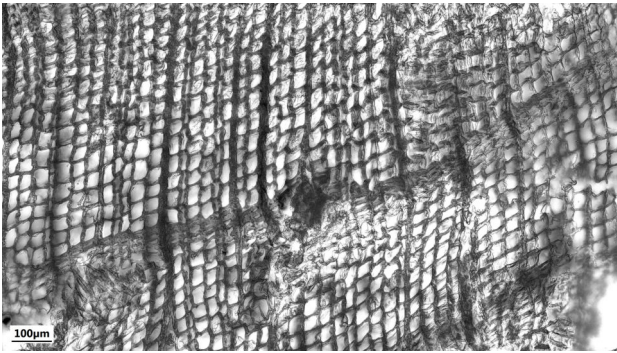
In addition, black spots with different sizes, shown in Figures 14e - 14h, are observed on some samples but cannot be further identified. These spots are found on *Brown Surface* and *Surface with Deposits* samples and may be related to the surface deposits. The larger spots in Figure 14g are similar in size to the smaller deposits shown in Figures 9g and 9h.



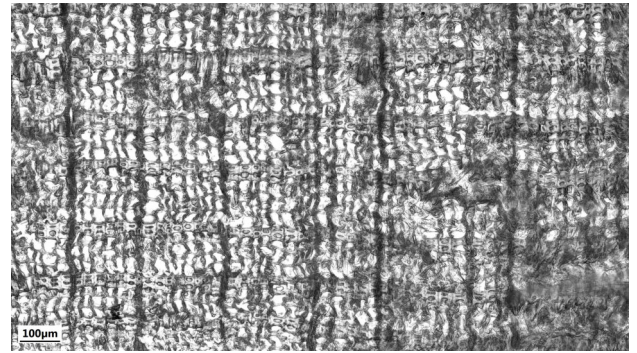
(a) Ice Patch Wood Interior



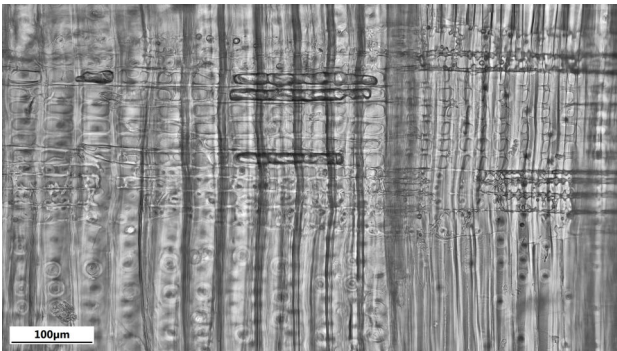
(b) Ice Patch Wood Interior



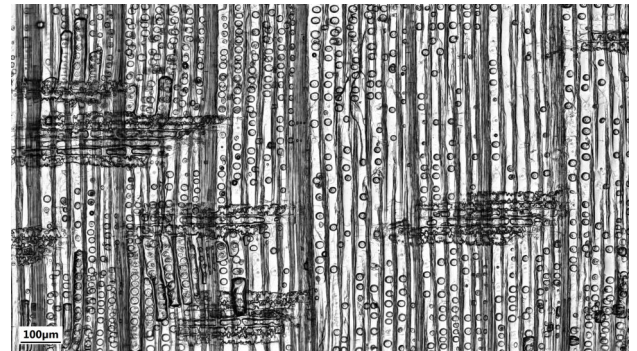
(c) Ice Patch Wood Interior



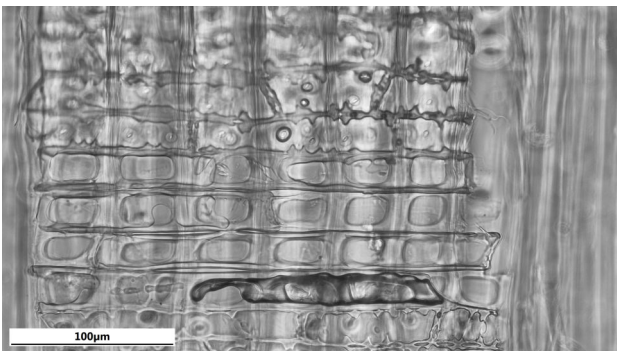
(d) Ice Patch Wood Interior



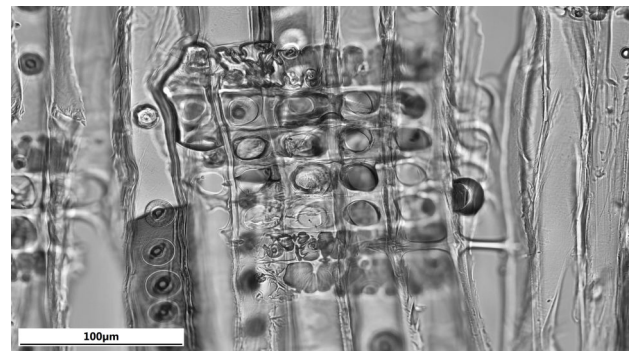
(e) Ice Patch Wood Interior



(f) Ice Patch Wood Interior



(g) Ice Patch Wood Interior

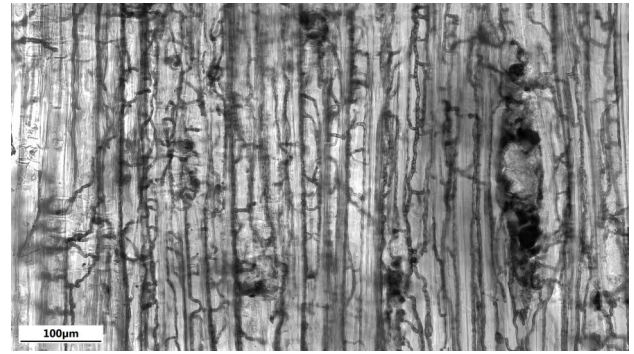


(h) Ice Patch Wood Interior

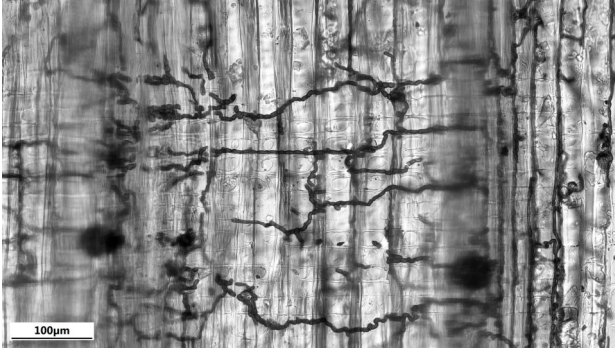
Figure 13: Light microscopy images of the interior of ice patch wood samples with anatomical characteristics used for genus-level identification. (a) C57838/4 F.238, (b) C57838/4 F.172 AG, (c) C57838/4 F.133 AG, (d) C57838/4 F.238, (e) C57838/4 F.175 AG, (f) C57838/4 F.205 AG, (g) C57838/4 F.175 AG, (h) C57838/4 F.175 AG.



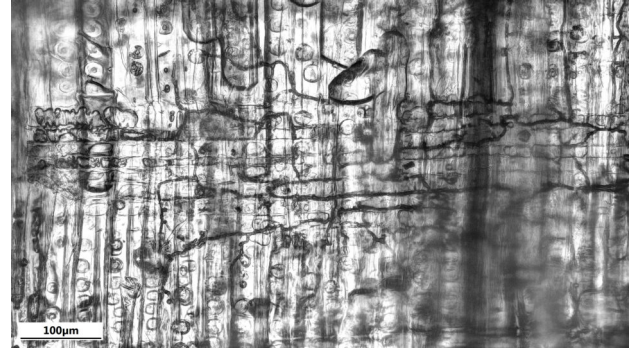
(a) Ice Patch Wood Surface with Discolorations



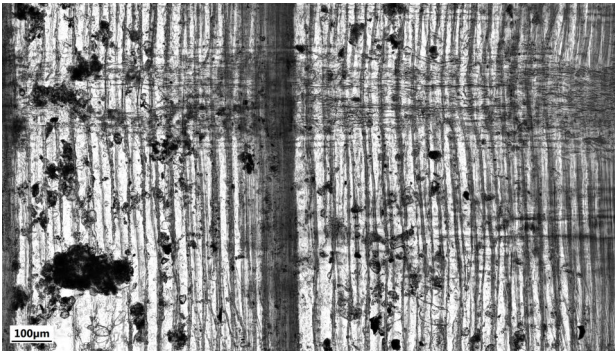
(b) Ice Patch Wood Surface with Discolorations



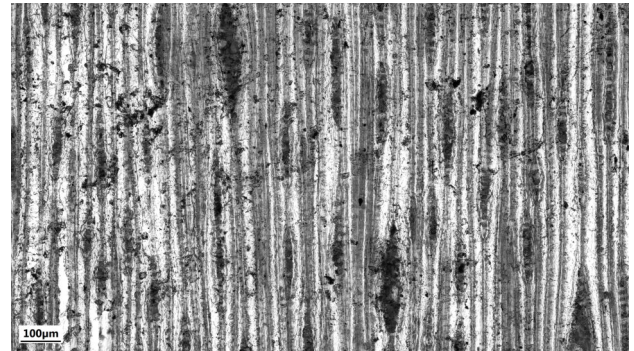
(c) Ice Patch Wood Brown Surface



(d) Ice Patch Wood White Surface



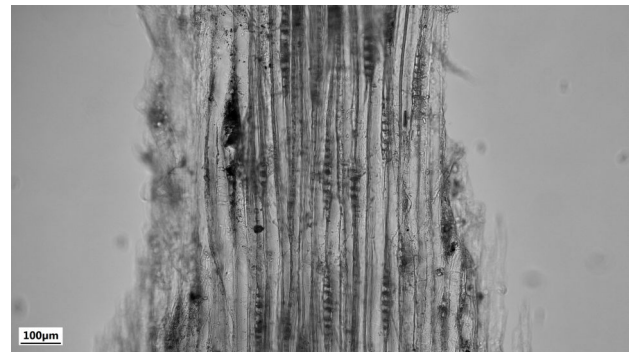
(e) Ice Patch Wood Brown Surface



(f) Ice Patch Wood Brown Surface



(g) Ice Patch Wood Brown Surface



(h) Ice Patch Wood Surface with Deposits

Figure 14: Light microscopy images of fungal hyphae (a-d) and black spots (e-h) observed on surfaces of ice patch wood samples from the *Brown Surface*, *White Surface*, *Surface with Discolorations* and *Surface with Deposits* categories. (a) C57838/4 F.172 AG, (b) C57838/4 F.172 AG, (c) C57838/4 F.133 AG, (d) C57838/4 F.133 AG, (e) C57838/4 F.230, (f) C57838/4 F.230, (g) C57838/4 F.175 AG, (h) C57838/4 F.133 AG.

5.6 ATR-FTIR Spectroscopy

5.6.1 Comparison with Reference Materials

ATR-FTIR spectroscopy is used to investigate the chemical composition of the ice patch wood samples. Figure 15 compares the spectra of ice patch wood samples with modern and archaeological pine. The surface spectra of the ice patch wood samples show two main patterns that correspond to the surface color and two representative spectra are included in Figure 15. The spectra of the interior of the ice patch wood samples show no significant differences between samples and therefore only one representative spectrum is shown. A comparison of the ATR-FTIR spectra of all ice patch wood samples is given below in Section 5.6.3.

The absorption bands in Figure 15 and their assignments are listed in Table 3. The ATR-FTIR spectra are recorded to 500 cm^{-1} , but at low wavenumbers band assignment becomes difficult due to overlapping absorption bands and limited reference data. The most characteristic absorption bands for the main wood components are located between 1800 cm^{-1} and 700 cm^{-1} and therefore this region is used for the following diagrams.

Most absorption bands in the 1800 cm^{-1} to 700 cm^{-1} region have contributions from cellulose, hemicelluloses and lignin. However, characteristic molecular structures can create absorption bands that are mainly assigned to one wood component and these bands are highlighted in Figure 15. The band at 1730 cm^{-1} is due to unconjugated C=O groups. In modern wood, this functional group is mainly present in acetyl groups in xylan and therefore this absorption band can be assigned to hemicelluloses [33]. Similarly, the band at 896 cm^{-1} from anomeric C₁-H is characteristic of cellulose [33]. Other absorption bands mainly associated with cellulose and hemicelluloses (holocellulose) are 1315 cm^{-1} , 1156 cm^{-1} , 1105 cm^{-1} and 1052 cm^{-1} [34]. As lignin is the only wood component with an aromatic ring the band from aromatic skeletal vibrations at 1509 cm^{-1} can be mainly assigned to lignin. Other bands with a large lignin contribution are 1595 cm^{-1} , 1263 cm^{-1} , 1217 cm^{-1} , 1140 cm^{-1} and 856 cm^{-1} [34].

The ATR-FTIR spectrum for modern pine shows the expected absorption bands [35]. The water-stored and freeze-dried archaeological pine have very similar spectra to each other except for an increased absorbance in the water-stored sample at 3340 cm^{-1} and 1650 cm^{-1} . These bands are assigned to water and their high intensity is caused by the high water content in the sample. In both archaeological pine samples the bands at 1730 cm^{-1} and 896 cm^{-1} are strongly decreased in intensity. Since these bands are associated with hemicelluloses and cellulose respectively, this suggests holocellulose degradation. This is also supported by the reduced intensity of all other bands with large contributions from holocellulose. In contrast, the peaks mainly associated with lignin appear more pronounced in the spectrum. This degradation pattern with is consistent with the bacterial decay of wood observed in the light microscopy images where cellulose-rich cell wall layers are degraded while the lignin is preserved [34].

The ice patch wood samples show different spectra for the surface and interior. The interior of the ice patch wood samples has a spectrum similar to modern pine with the exception that

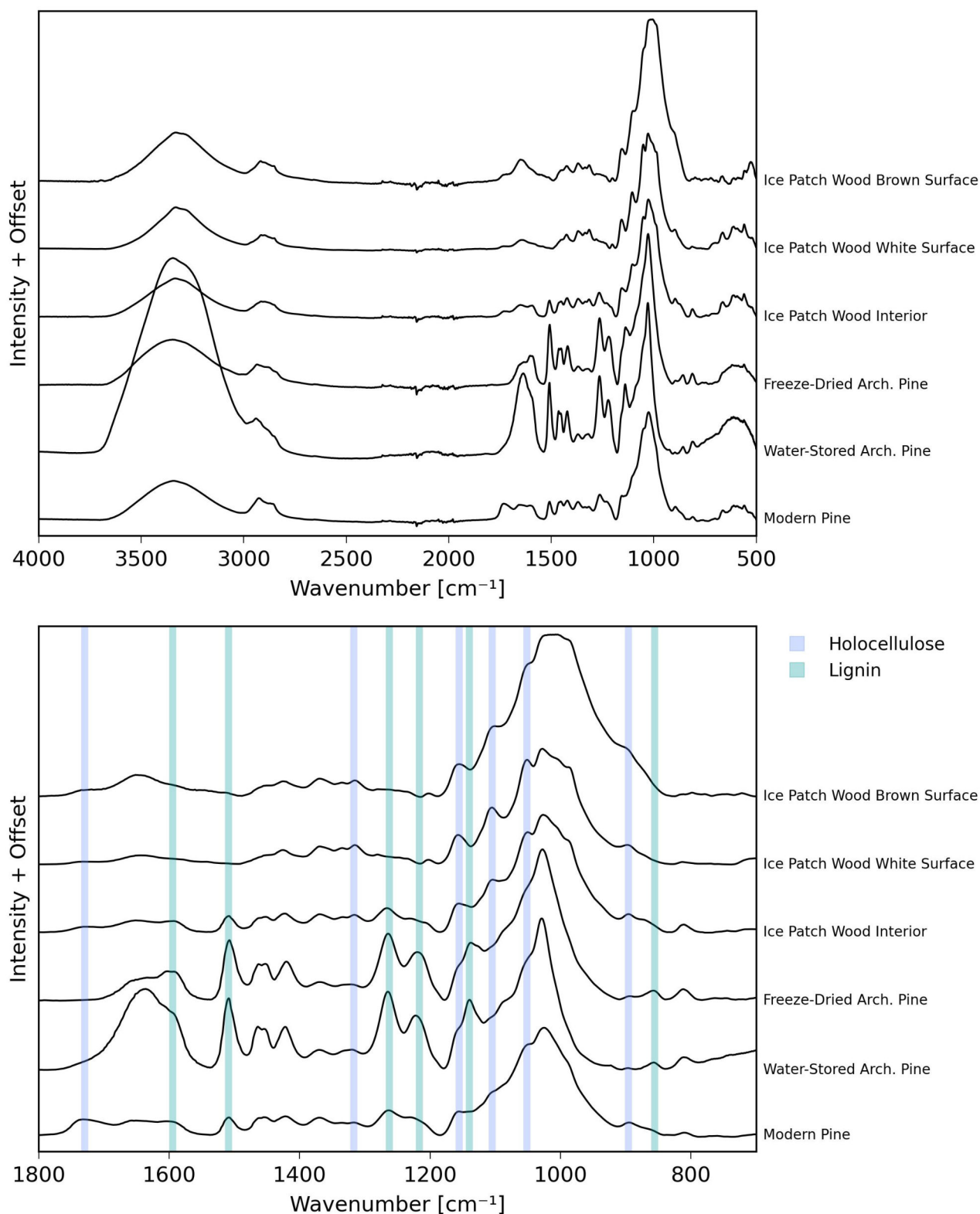


Figure 15: ATR-FTIR spectra comparing the interior and surface of ice patch wood samples with modern and archaeological pine in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below). The spectra of the ice patch wood surfaces show two main patterns. The absorption bands mainly associated with holocellulose and lignin are shown. Ice Patch Wood Interior: C57838/4 F.238, Ice Patch Wood White Surface: C57838/4 F.238, Ice Patch Wood Brown Surface: C57838/4 F.230.

Table 3: Assignment of ATR-FTIR absorption bands.

Band (cm^{-1})	Band Assignment
3340	O–H stretching and hydrogen-bonded O–H stretching and absorbed water [11, 35]
2926	C–H stretching [11, 35]
2853	C–H stretching [11, 35]
1730	C=O stretching in unconjugated ketone, carboxyl and ester groups [11]
1657	C=O stretching in conjugated p-substituted aryl ketones [11]
1650	adsorbed water [35]
1595	aromatic skeletal vibrations and C=O stretching [11]
1509	aromatic skeletal vibrations [11]
1464	asymmetric C–H vibrations in CH ₃ and CH ₂ [11]
1454	asymmetric C–H vibrations in OCH ₃ [35]
1422	aromatic skeletal vibrations and C–H in-plane vibrations [11]
1370	C–H stretching in CH ₃ , not in OCH ₃ [11]
1335	phenolic OH vibrations in condensed guaiacyl/syringyl lignin [11, 35]
1315	CH ₂ rocking [35]
1280	C–H deformation [35]
1263	C–O stretching of guaiacyl ring [36]
1231	O–H in-plane bending and C–O of acetyl in hemicellulose [35, 37]
1217	C–C and C–O and C=O stretching [11]
1204	O–H in-plane bending [35]
1156	asymmetric C–O–C stretching [35]
1140	aromatic C–H in plane vibration in guaiacyl lignin [35]
1123	asymmetric ring stretching, C–C and C–O stretching [35]
1105	asymmetric ring stretching [35]
1087	C–O deformation in secondary alcohols and aliphatic ethers [11]
1052	C–O–C ether vibrations [35]
1030	aromatic C–H in-plane bending and C–O vibration in primary alcohols and unconjugated C=O stretching [11]
987	C–O stretching [35]
896	anomeric C ₁ –H bending and ring stretching [35]
856	C–H out-of-plane vibration in position 2, 5 and 6 of guaiacyl lignin [35]
810	glucomannan [35]
700	CH ₂ rocking vibration [35]
662	C–OH out-of-plane bending [35]

the hemicellulose band at 1730 cm^{-1} is reduced in intensity. This indicates that for the interior hemicelluloses are degraded while cellulose and lignin are preserved.

The surface spectra of the ice patch wood samples show significant differences compared to modern pine. In both the *Brown Surface* and *White Surface* spectra the absorption bands associated with lignin at 1595 cm^{-1} , 1509 cm^{-1} , 1263 cm^{-1} , 1217 cm^{-1} , 1140 cm^{-1} and 856 cm^{-1} are almost completely absent while most holocellulose bands have become more pronounced. The exception is the hemicelluloses band at 1730 cm^{-1} which is decreased in intensity. This suggests that on the surface of the samples lignin and hemicelluloses are degraded while cellulose is preserved. This is opposite to the degradation pattern observed in archaeological wood, where holocellulose is preferentially degraded and lignin is preserved.

The spectra for the *Brown Surface* and *White Surface* ice patch wood samples show an overall similar shape. However, the *Brown Surface* spectrum has higher intensity and broadening of the C=O bands around 1730 cm^{-1} and 1657 cm^{-1} . Additionally, differences can be seen in the region between 1180 cm^{-1} and 850 cm^{-1} where the overlapping absorption bands are altered in shape and increased in intensity for the *Brown Surface* samples.

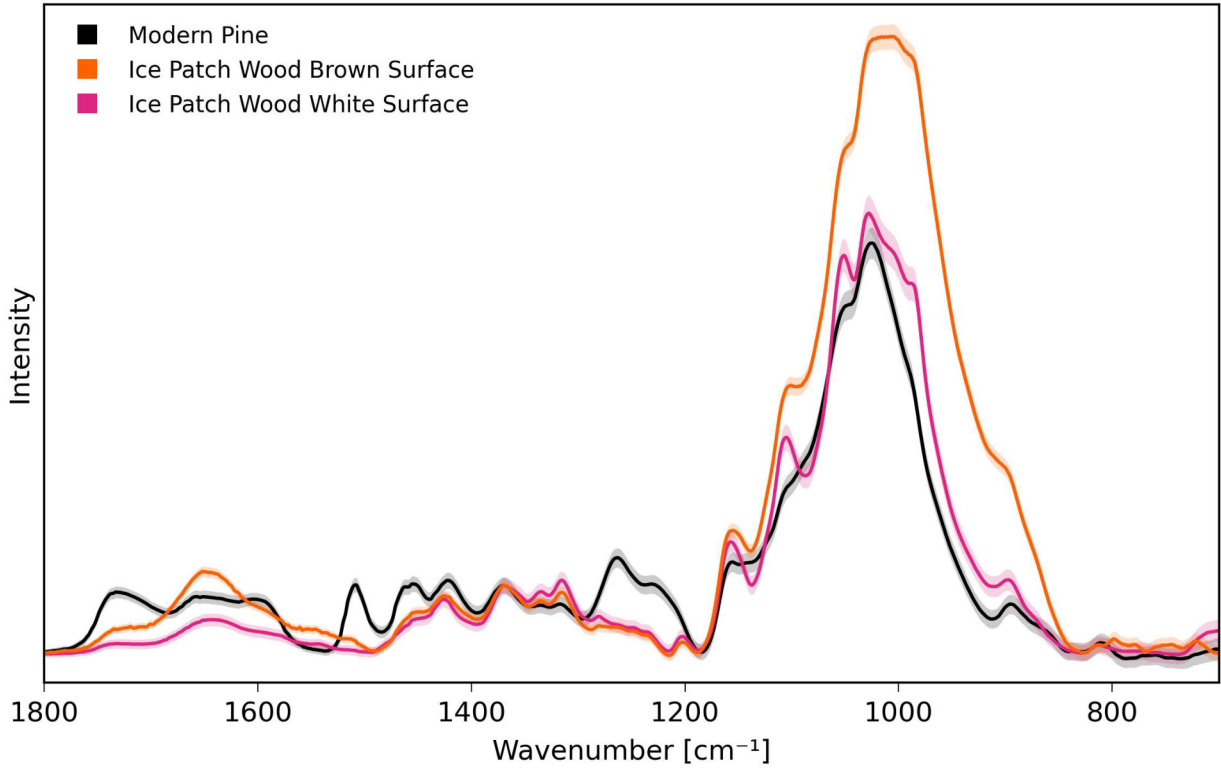


Figure 16: ATR-FTIR spectra comparing a *Brown Surface* and *White Surface* ice patch wood sample with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region. The 1σ interval is shown as a shaded area around the average spectrum. Ice Patch Wood Brown Surface: C57838/4 F.230, Ice Patch Wood White Surface: C57838/4 F.238.

5.6.2 Measurement Effects

Wood is a heterogeneous material and variations between measurements taken at different points of the same sample are expected. To account for this, each spectrum is calculated as the average of three individual measurements. Figure 16 shows the ATR-FTIR spectra of modern pine and a *Brown Surface* and *White Surface* ice patch wood sample with the standard deviation from the three measurements shows as shaded areas. The 1σ interval is relatively small compared with the observed spectral changes. The standard deviation does not change the interpretation of the results and is therefore not included in the following figures.

For the ATR-FTIR measurement, the wood chips are aligned with a reference line visible on the ATR crystal. When the longitudinal axis of the wood is parallel to this reference line, this alignment is defined as 0° . The sample is then rotated to obtain measurements for different angles between the longitudinal axis of the wood and the reference line. Figure 17 shows the spectra for modern and freeze-dried archaeological pine. The spectra do not change as the wood is rotated on the ATR crystal. For ice patch wood samples, the interior has the same behavior, as shown in Figures 18a and 18b.

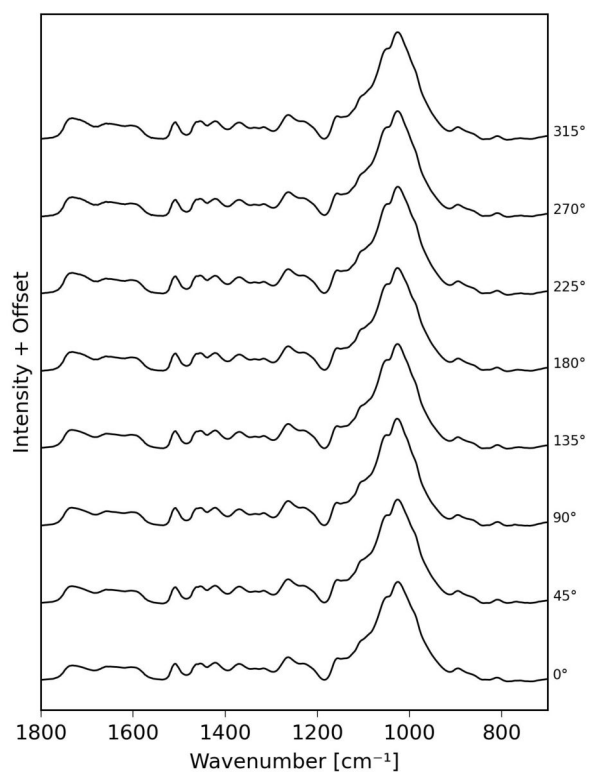
In contrast, the spectrum obtained from the surfaces of ice patch wood samples changes with the orientation of the sample on the ATR crystal. This can be seen in Figures 18c and 18d for a *Brown Surface* and *White Surface* sample. The shape and intensity of some absorption bands

changes, particularly in the region between 1180 cm^{-1} and 850 cm^{-1} . The spectra measured at angles 180° apart are identical which indicates that the spectrum depends on the sample orientation but not its direction. As the longitudinal axis of the wood can be identified from the tracheids, but not the direction of the top or bottom of the tree, 0° and 180° cannot be distinguished.

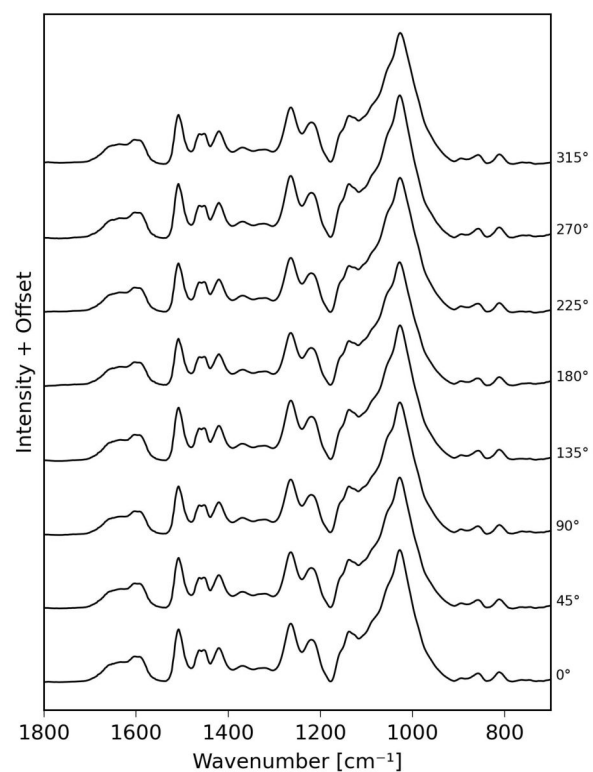
The spectrometer used in this experiment does not provide polarized infrared (IR) light. However, when the incoming unpolarized beam is reflected in the ATR crystal, the perpendicular- and parallel-polarized components relative to the plane of incidence are reflected with different Fresnel coefficients [38]. This results in different effective path lengths of the evanescent interactions with the sample which depend on the refractive indexes of the ATR crystal and the sample, the angle of incidence, the wavelength and the polarization [39]. Usually, the evanescent wave created by the parallel-polarized light has a longer effective path length than the one from perpendicular-polarized light [39]. If the sample has IR-active vibrations that are oriented in a preferential direction instead of randomly, the absorbance of these vibrations can change depending on the alignment of the dipoles with the electric field [40]. This can cause the intensity of absorption bands to vary as the sample is rotated on the ATR crystal even if the incoming IR radiation is unpolarized.

In wood, cellulose is arranged in microfibrils which can cause an anisotropic IR response [41]. On the surface of the ice patch wood samples, where hemicelluloses and lignin are strongly degraded, cellulose is the main remaining component. The observed changes in the ATR-FTIR spectra during sample rotation could be related to the IR anisotropy of cellulose. In contrast, modern and freeze-dried archaeological pine and the interior of the ice patch wood contain lignin and lignin absorption bands overlap with cellulose bands in the 1180 cm^{-1} to 850 cm^{-1} region. This might make orientation effects less visible.

The orientation-dependent effects observed in the ATR-FTIR spectra suggest an anisotropic IR response of the surfaces of the ice patch wood samples which might be caused by the high cellulose content of the surface material. Due to the complexity of the interactions on the ATR crystal this is difficult to interpret further. Further investigations using polarized IR spectroscopy could possibly provide more details on the cellulose organization in the sample [41]. For this thesis, the influence of this effect is minimized by keeping the sample orientation consistent at 0° with the wood fibers parallel to the line on the ATR crystal.

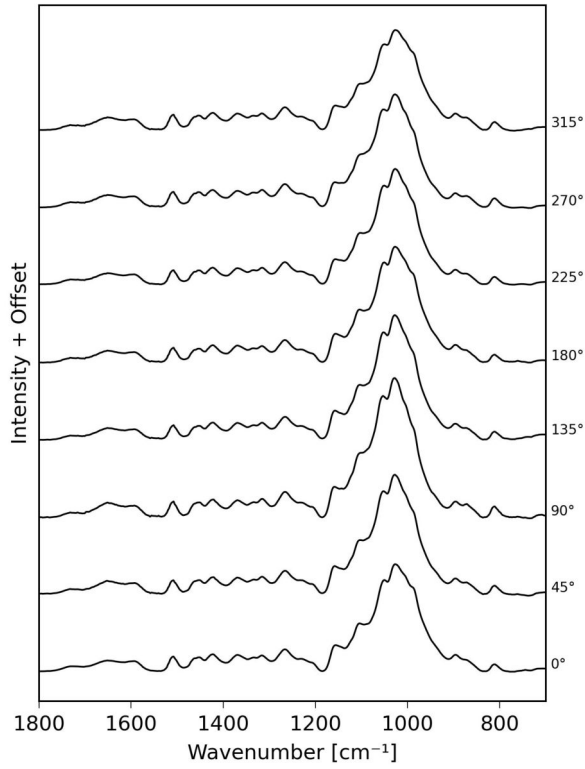


(a) Modern Pine

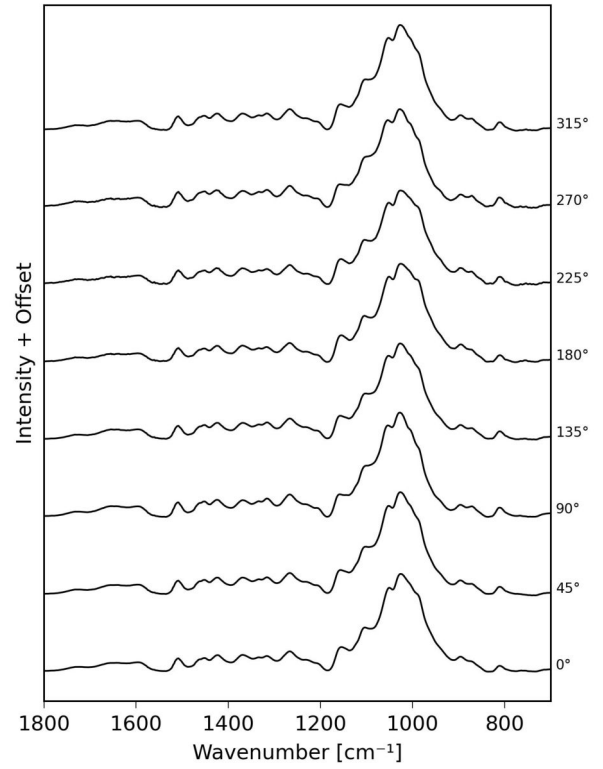


(b) Freeze-Dried Arch. Pine

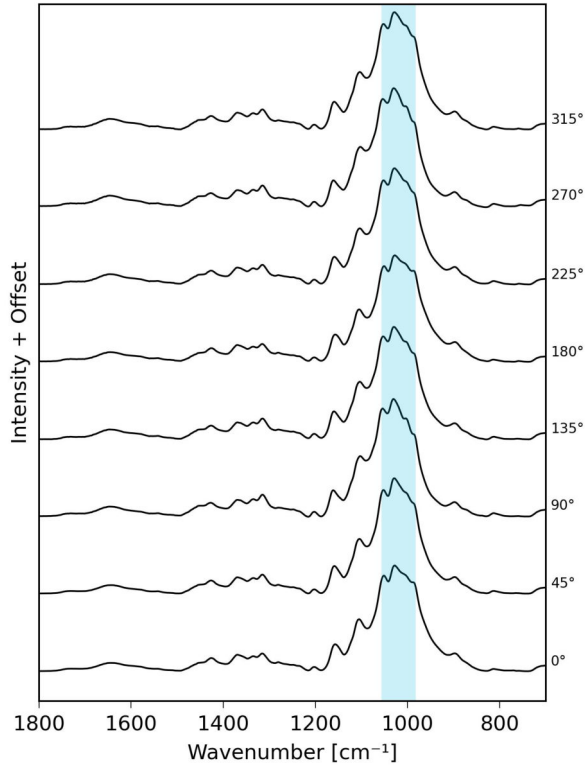
Figure 17: ATR-FTIR spectra of modern pine and freeze-dried archaeological pine at different orientations on the ATR crystal in the 1800 cm^{-1} to 700 cm^{-1} region.



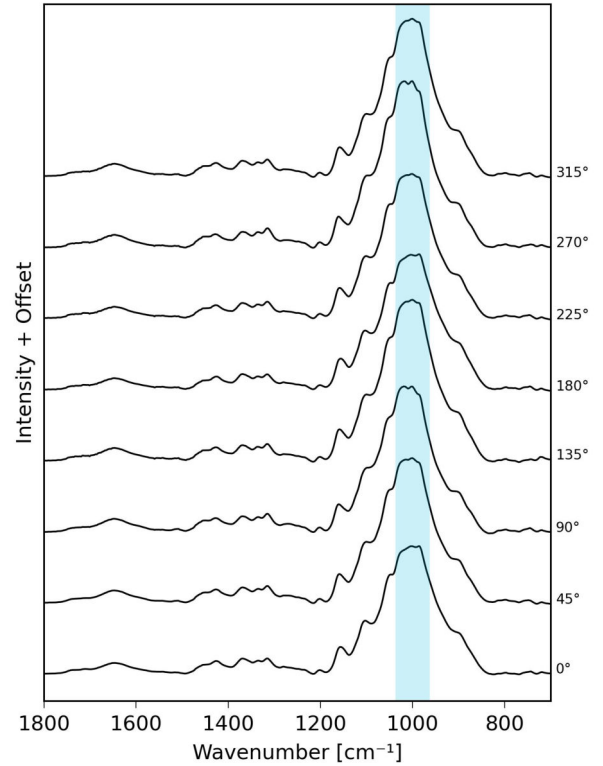
(a) Ice Patch Wood Interior



(b) Ice Patch Wood Interior



(c) Ice Patch Wood White Surface



(d) Ice Patch Wood Brown Surface

Figure 18: ATR-FTIR spectra of ice patch wood samples at different orientations on the ATR crystal in the 1800 cm^{-1} to 700 cm^{-1} region. The surfaces show an orientation-dependent effect with the shape and intensity of some absorption bands changing. The region most affected is highlighted in blue. (a) C57838/4 F.238, (b) C57838/4 F.205 AG, (c) C57838/4 F.238, (d) C57838/4 F.205 AG.

5.6.3 Variation in Ice Patch Wood

ATR-FTIR spectroscopy is used to compare the ice patch wood samples with each other, based on the surface categories defined in Table 1. Figure 19 compares the interior of ice patch wood with modern pine, with one representative sample from each surface category shown. The spectra are very similar between the samples. The surface spectra of the same ice patch wood samples are compared in Figure 20. The spectra for the *White Surface*, *Surface with Discolorations* and *Surface with Deposits* samples are very similar, while the spectrum for the *Brown Surface* sample has a significantly different shape, mainly in the region between 1180 cm^{-1} and 850 cm^{-1} .

Compared to modern pine, all ice patch wood samples show a decrease in the 1730 cm^{-1} hemicelluloses band both in the interior and at the surface. This indicates that the acetyl groups in xylan in hemicelluloses are broken down [37]. Deacetylation of hemicelluloses also reduces the intensity of the 1231 cm^{-1} band created by C–O stretching in the acetyl ester [35, 37]. Hemicellulose degradation through deacetylation is observed in Bronze Age wooden artifacts from Hallstatt, Austria, which were preserved in a salt environment without microbial activity [37] and in 150-270 year old wood used as support for paintings [42]. It may be a process of long term wood aging in absence of microbial decay and other degradation factors. At the surface, UV radiation from exposure to sunlight could be an additional contributing factor to hemicellulose degradation [14].

Surface Degradation

As the sticks are melted out of the ice when they are recovered, exposure to sunlight could have contributed to degradation on the surface. All wood components are sensitive to degradation under exposure to sunlight, but lignin is particularly sensitive as it absorbs the largest amount of ultraviolet (UV) and visible light [43, 44]. Absorbed UV photons can provide enough energy to break chemical bonds in the lignin polymer and form free radicals, which can then cause oxidation [14, 44]. One proposed mechanism involves the oxidation of lignin hydroxyl groups resulting in the formation of quinones and conjugated carbonyl structures [45]. In an FTIR spectrum, this causes an increased intensity of the unconjugated C=O stretching band around 1730 cm^{-1} and conjugated C=O band around 1657 cm^{-1} [44, 45]. UV exposure also causes demethoxylation in lignin, which can be seen in a decrease in the 1263 cm^{-1} band assigned to the C–O stretch connected to the aromatic ring and the 1217 cm^{-1} band assigned to C–O stretching [14]. The aromatic skeletal vibrations at 1595 cm^{-1} and 1509 cm^{-1} decrease as the aromatic rings in the lignin are deformed [45].

These effects from lignin photodegradation are consistent with the ATR-FTIR spectrum of the *Brown Surface* sample shown in Figure 21, which compared to modern pine has decreased intensity of the bands at 1595 cm^{-1} , 1509 cm^{-1} , 1263 cm^{-1} and 1220 cm^{-1} as well as increased intensity and band broadening of the 1657 cm^{-1} band associated with oxidation products. The

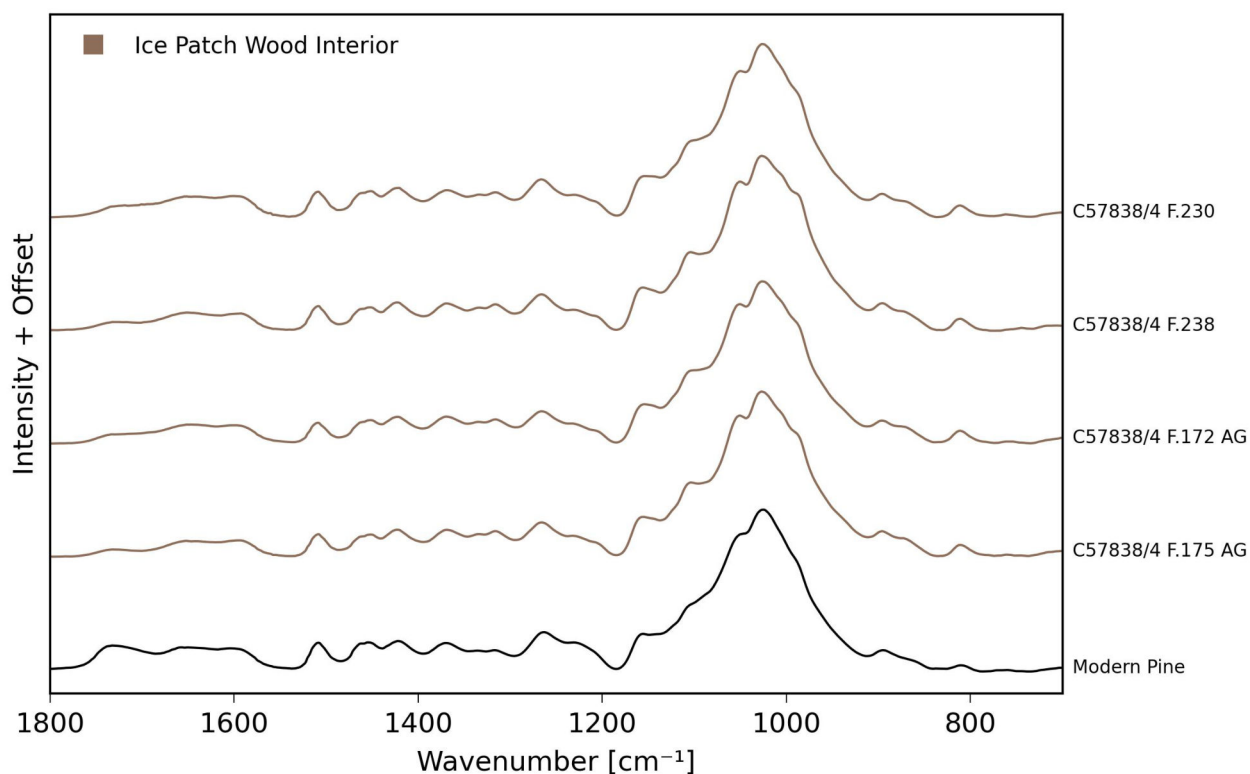


Figure 19: ATR-FTIR spectra comparing the interior of ice patch wood samples with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region.

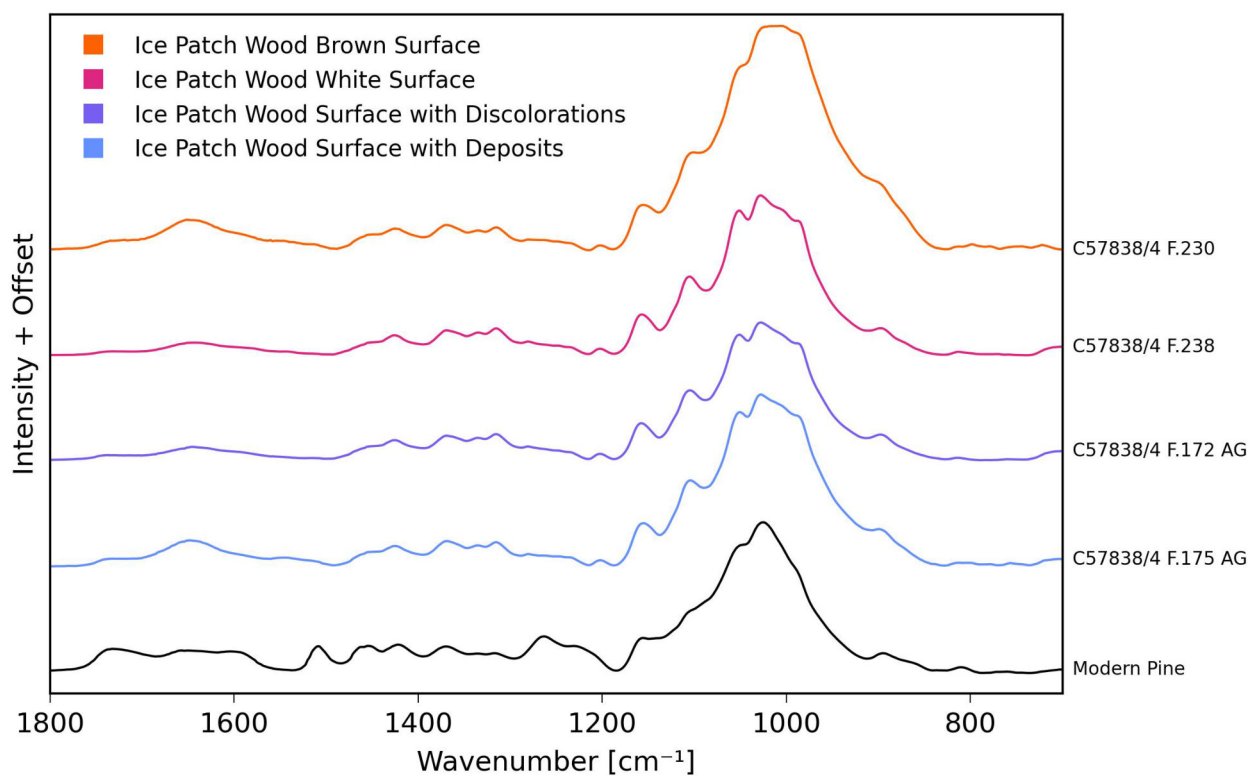


Figure 20: ATR-FTIR spectra comparing ice patch wood samples from the *Brown Surface*, *White Surface*, *Surface with Discolorations* and *Surface with Deposits* categories with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region.

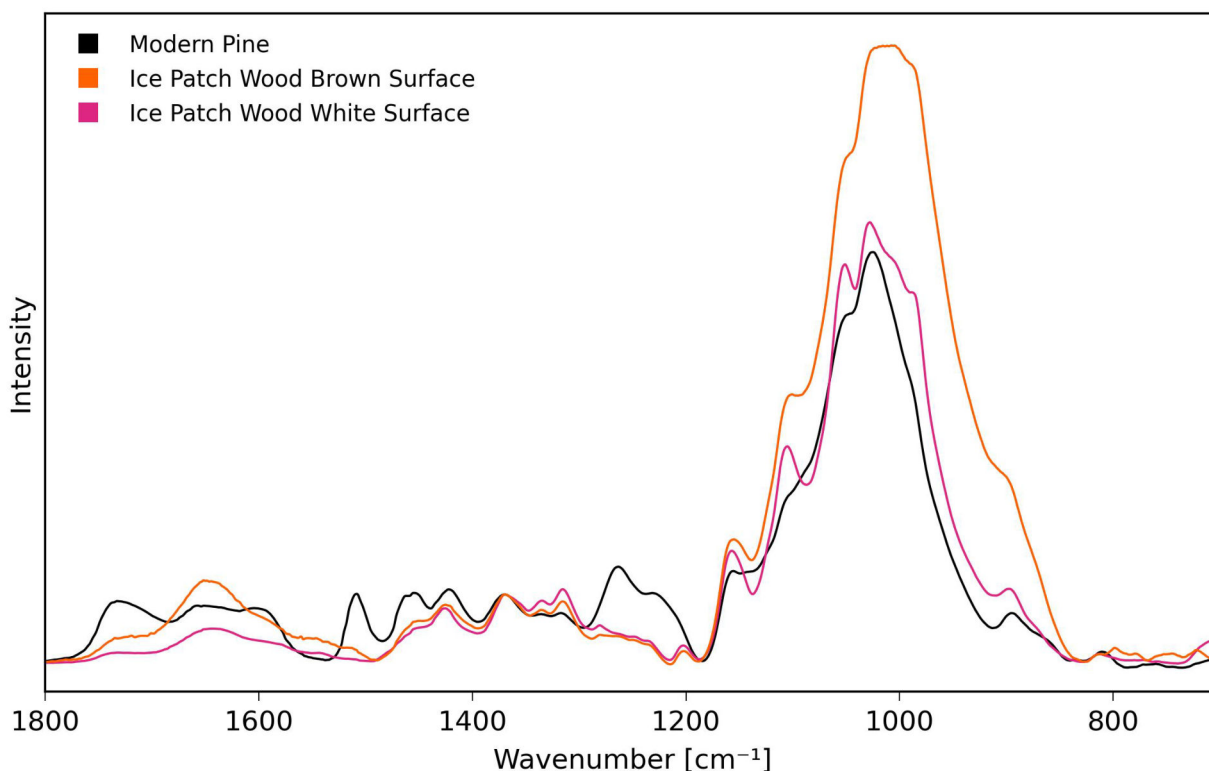


Figure 21: ATR-FTIR spectra comparing a *Brown Surface* and *White Surface* ice patch wood sample with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region. Ice Patch Wood Brown Surface: C57838/4 F.230, Ice Patch Wood White Surface: C57838/4 F.238.

1730 cm^{-1} band shows a decrease in intensity even though an increase would be expected from oxidation products [44, 45]. One possible explanation is that oxidation products are partially leached from the samples by water. Unconjugated carbonyl groups are particularly sensitive to leaching [46] and exposure to water in the ice patches is plausible. The ATR-FTIR spectrum of the *Brown Surface* sample also shows additional absorption bands between 1000 cm^{-1} and 850 cm^{-1} . This region includes $=\text{C}-\text{H}$ out-of-plane bending vibrations with the specific band positions depending on the substitution pattern and geometry [47]. The absorption could be caused by lignin degradation products containing $\text{C}=\text{C}$ bonds and the bands could be broadened because the degradation products can differ in their chemical structure and bonds can occur in different chemical environments. The higher intensity of the spectrum in the 1180 cm^{-1} to 850 cm^{-1} region could be related to normalization to the 1370 cm^{-1} band. In unnormalized spectra, the *Brown Surface* spectrum has a similar intensity in the 1180 cm^{-1} to 850 cm^{-1} region to modern pine, but has a slightly lower intensity at 1370 cm^{-1} . The dark brown surface color is also consistent with the degradation of lignin. Lignin and wood extractives are the main components contributing to wood color [14, 48]. Wood extractives are rapidly degraded during exposure to sunlight which results in a loss of color [48]. During photodegradation the chromophoric systems of lignin are broken down. Quinones and conjugated carbonyl structures formed in that process are proposed as secondary chromophoric systems with a dark brown color [14, 43, 45].

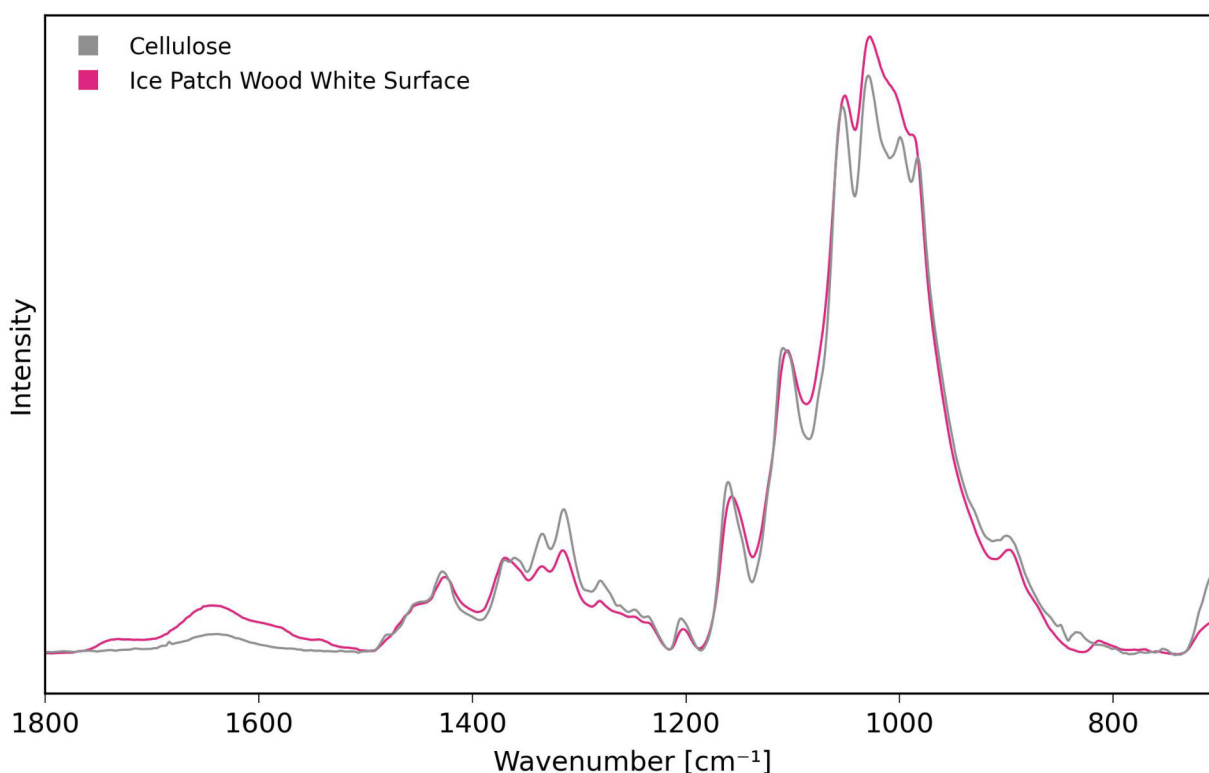


Figure 22: ATR-FTIR spectra comparing a *White Surface* ice patch wood sample with cellulose in the 1800 cm^{-1} to 700 cm^{-1} region. Ice Patch Wood White Surface: C57838/4 F.238.

An alternative explanation for the shape of the ATR-FTIR spectrum of the *Brown Surface* sample in the 1180 cm^{-1} to 850 cm^{-1} region could be the degradation of cellulose. Cellulose oxidation in wood can be linked to UV absorption by lignin, which generates free radicals and reactive oxygen species that create a local oxidative environment [44, 49]. A correlation between photodegradation of cellulose and an increase in intensity and broadening of the absorption bands in the 1180 cm^{-1} and 850 cm^{-1} region has been observed even if the specific mechanisms are unclear [50]. The oxidation products, like conjugated ketones and aldehydes, would show in an FTIR spectrum as an increase in the 1657 cm^{-1} band from conjugated C=O groups and overlap with oxidation products from lignin degradation. Some cellulose oxidation products are chromophores and absorb in the UV and/or visible light range causing discolorations [49, 51]. In cellulose-rich materials like papers this is responsible for yellowing [51]. If cellulose oxidation occurs this could contribute to the dark brown surface color.

The two possible degradation mechanisms of lignin and/or cellulose that could explain the shape of the spectrum of the *Brown Surface* sample in the 1180 cm^{-1} and 850 cm^{-1} region cannot be distinguished using only ATR-FTIR spectroscopy and require further investigation.

The ATR-FTIR spectrum of the *White Surface* sample (Figure 21) is consistent with hemi-cellulose degradation and photodegradation of lignin, which results in decreased intensity of the hemicelluloses and lignin bands and more pronounced cellulose bands compared to modern pine. The exception are the 1730 cm^{-1} and 1657 cm^{-1} bands which are expected to increase in intensity as oxidation products with unconjugated and conjugated C=O groups are formed and

which are decreased in intensity but broadened [44, 45]. It might be possible that these sticks experienced stronger leaching [46, 48]. Compared to the *Brown Surface* sample, degradation of lignin may be more advanced in the *White Surface* sample due to more exposure to sunlight, resulting in further breakdown of the secondary chromophores and change of color [14]. Figure 22 shows that the ATR-FTIR spectrum of *White Surface* sample is similar to cellulose which is exposed after the lignin is degraded. This can explain the white color and the shiny appearance, which is likely related to birefringence from cellulose [14].

The ATR-FTIR spectrum for the *Surface with Discolorations* sample in Figure 20 is very similar to the spectrum of the *White Surface* sample, which is expected as the surface is also mainly white in color. A small difference in intensity in the 1180 cm^{-1} to 850 cm^{-1} region can be seen, but similar differences are observed among many samples and are discussed below. This indicates similar degradation to the *White Surface* sample with hemicellulose degradation and lignin photodegradation which leaves cellulose as the main surface component. No differences in the chemical composition associated with the grey diffuse discolorations can be identified with ATR-FTIR spectroscopy.

The spectrum for the *Surface with Deposits* sample in Figure 20 is similar to the spectrum of the *White Surface* sample but shows an additional absorption band at 1540 cm^{-1} and an increased intensity at 1650 cm^{-1} . This could indicate the presence of an amide group. Amides have two characteristic absorption bands, Amide I and Amide II, between 1700 cm^{-1} and 1500 cm^{-1} that arise from the coupling of C=O and N-H groups [52]. This could indicate that the black deposits contain proteinaceous matter but further investigation would be required for confirmation.

Variation in Surface Categories

Figures 24-26 compare the ATR-FTIR surface spectra of the three ice patch wood samples in each surface category listed in Table 1. The spectra are generally similar within each category. The three *Brown Surface* samples show the most variation within their group which could be an indication of different degradation processes or stages. In contrast, the *White Surface*, *Surface with Discolorations* and *Surface with Deposits* samples have very similar absorption spectra with small differences in the intensity of the absorption peaks, especially in the 1180 cm^{-1} to 850 cm^{-1} region. A possible reason could be variations in the contact between the sample and the ATR crystal [12]. The same clamping strength is applied for all measurements, but surface characteristics like loose surface fibers, erosion troughs or areas with different roughness, surface structure or porosity could influence the signal.

Figure 24 also includes the ATR-FTIR spectra of the surface fibers on the *White Surface* samples which are similar to the surface spectrum with small differences in intensity in the 1180 cm^{-1} to 850 cm^{-1} region. The surface fibers of C57838/4 F.238 show absorption bands 1730 cm^{-1} and 1710 cm^{-1} . These might be due to unconjugated C=O groups in oxidation products, which are better preserved in the protruding fibers that are less affected by leaching.

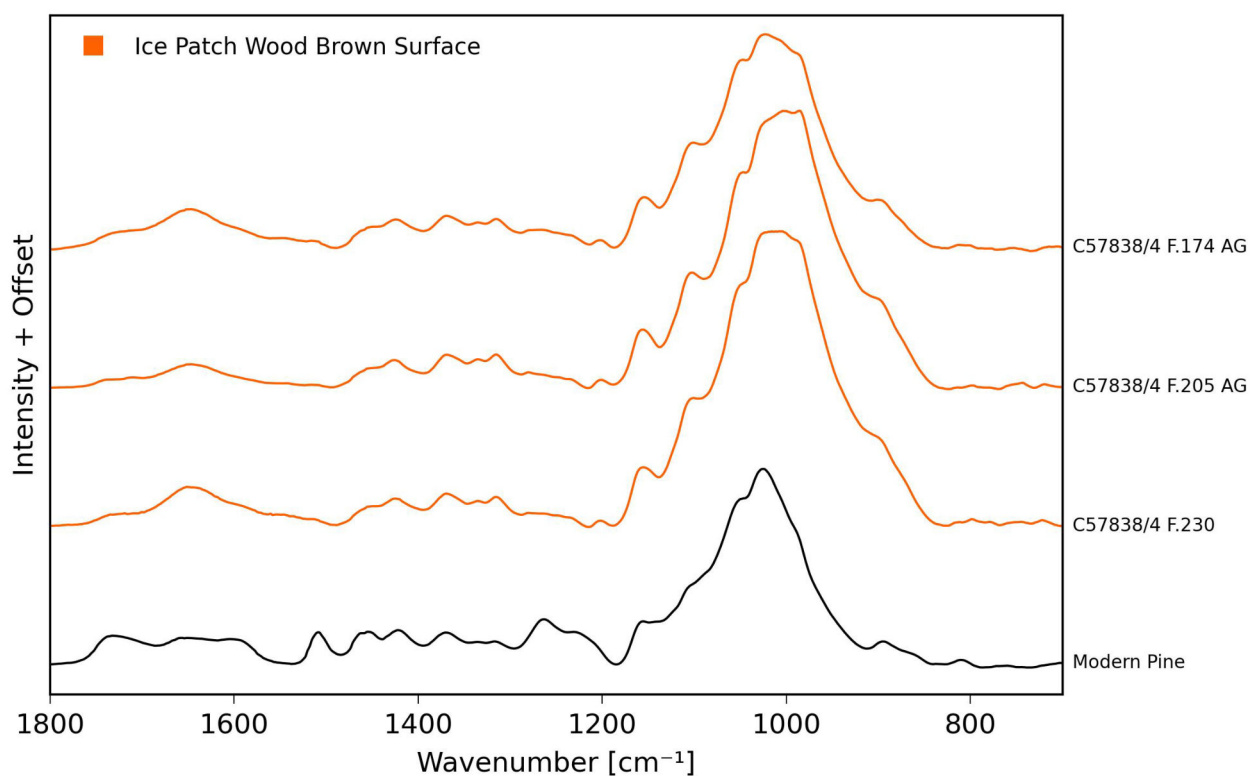


Figure 23: ATR-FTIR spectra comparing the *Brown Surface* ice patch wood samples with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region.

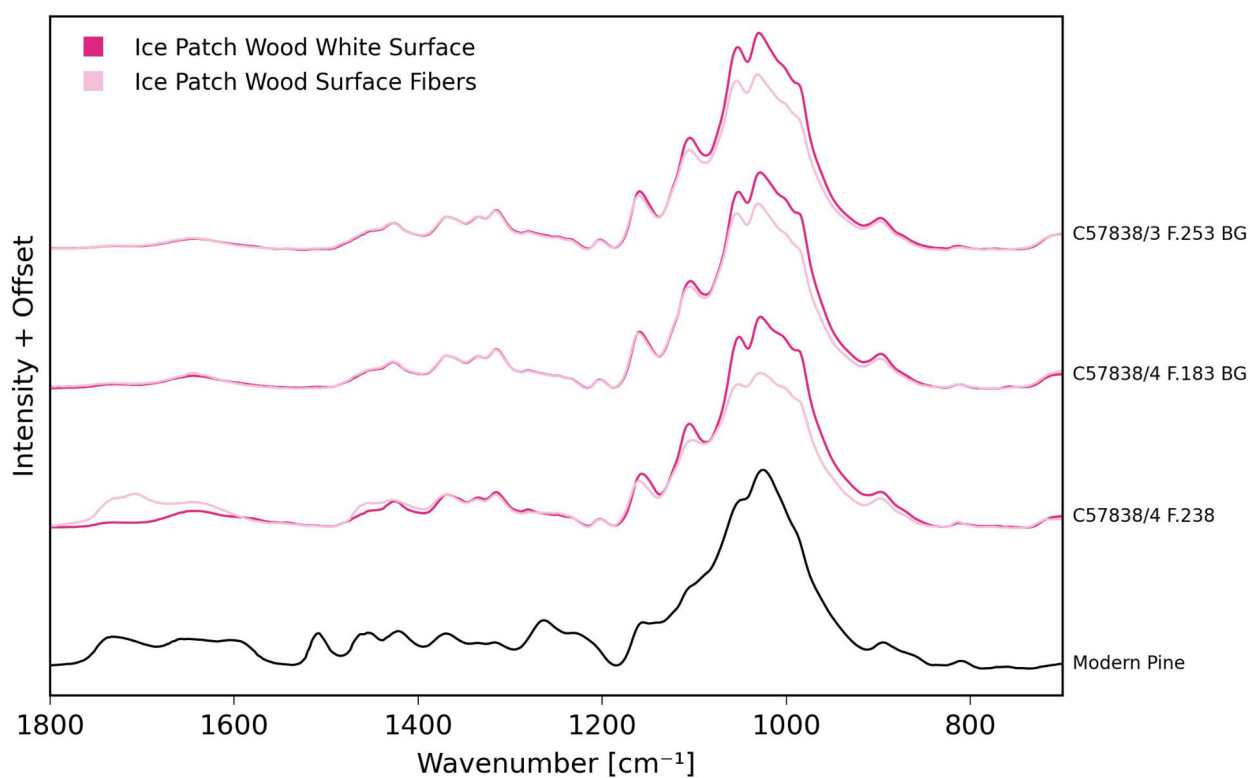


Figure 24: ATR-FTIR spectra comparing the *White Surface* ice patch wood samples with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region. The spectra of the surface fibers are shown.

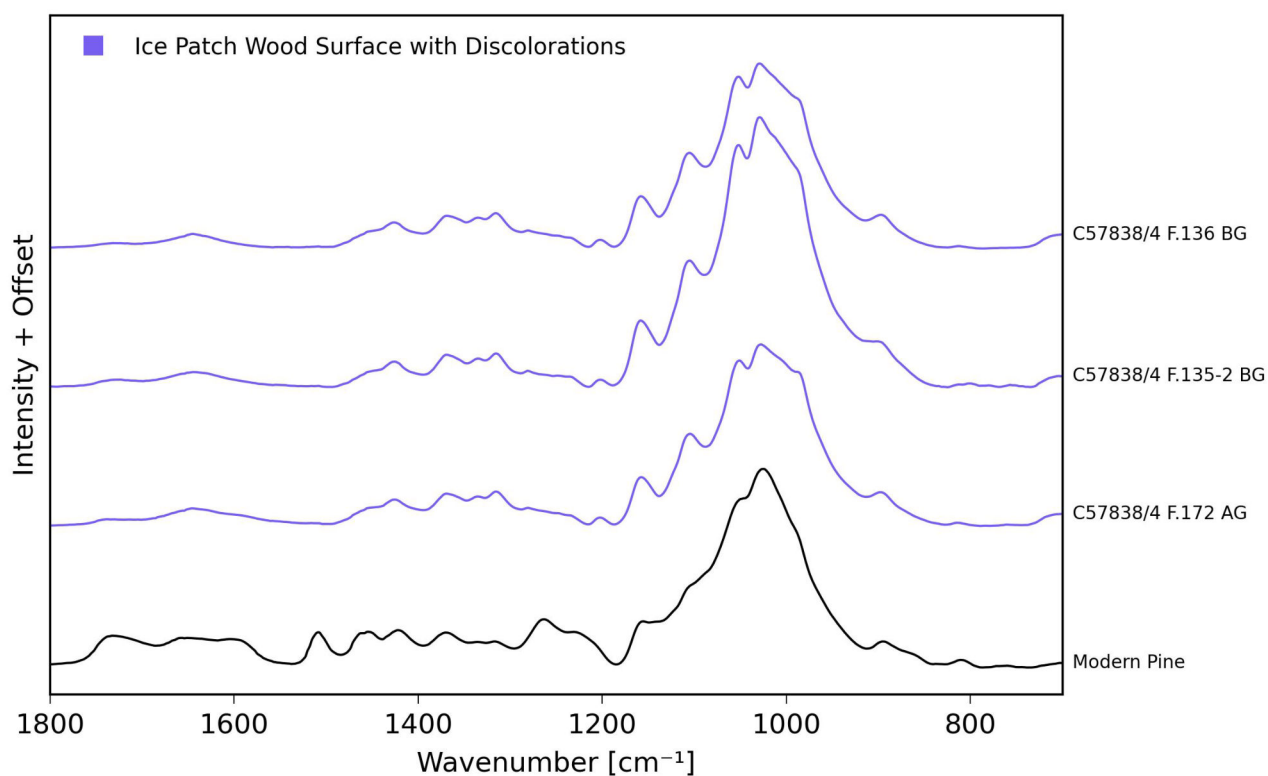


Figure 25: ATR-FTIR spectra comparing the *Surface with Discolorations* ice patch wood samples with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region.

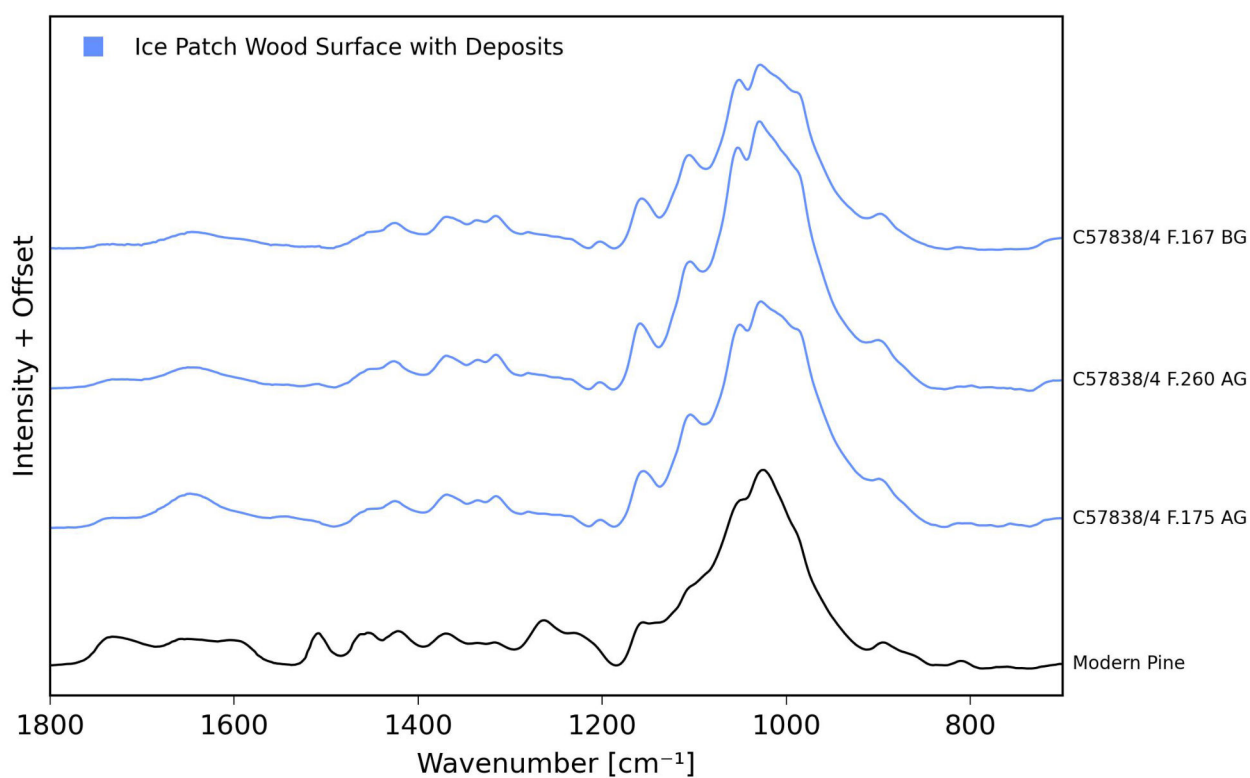


Figure 26: ATR-FTIR spectra comparing the *Surface with Deposits* ice patch wood samples with modern pine in the 1800 cm^{-1} to 700 cm^{-1} region.

Spatial Variation

For some scaring sticks two samples are taken, one from each end of the stick. Figure 27 compares the ATR-FTIR spectra for the lower section (BG) and upper section (AG) of the same stick. The lower section refers to the end that is sharpened to be driven into the ground while upper section refers to the end where a flag would be attached. The two spectra of the samples from C57838/4 F.183 and C57838/4 F.172 are similar to each other, which suggest similar degradation across the length of the stick. This is consistent with visual observations as both sticks show a similar appearance along their length. The two samples from C57838/4 F.175 have a similar spectrum, but the upper section shows higher absorption at 1657 cm^{-1} , 1540 cm^{-1} and in the 1180 cm^{-1} to 850 cm^{-1} region. Visually, the two sections have a similar appearance and these differences might be caused by differences in leaching and measurement-related effects causing different intensities as described above. The two sections from C57838/4 F.205 show two significantly different spectra, while the upper section has a spectrum similar to *Brown Surface* samples, the lower section has spectrum like *White Surface* samples. This is especially interesting as the upper section appears lighter in color than the lower section.

The scaring sticks are cut longitudinally from the original wood and the radial and tangential sections are exposed on the outer surfaces. The ATR-FTIR measurements above are all collected from the radial surfaces. Figure 28 compares the radial and tangential surfaces from the same sample. The two surfaces of C57838/4 F.238, C57838/4 F.172 AG and C57838/4 F.175 AG have similar spectra and a similar visual appearance which suggests that the samples have experienced similar degradation on both sides. In contrast, for C57838/4 F.230 the radial and tangential surfaces show different spectra. The radial section has a spectrum like the *Brown Surface* samples that matches its brown color while the lighter tangential section has a spectrum like the *White Surface* samples.

Different degradation patterns can be observed not only on opposite ends of the same stick, but also on different surfaces of the same sample which suggests that degradation is localized and depends on the specific conditions of each surface. Degradation likely depends on exposure to degradation factors like sunlight, meltwater and wind and on the interaction of these factors, for example the rate of photodegradation relative to the rate of material loss through meltwater or wind [53]. Sticks can also be displaced after melting from the ice [2]. The degradation of a surface might be the result of multiple environmental conditions at different times.

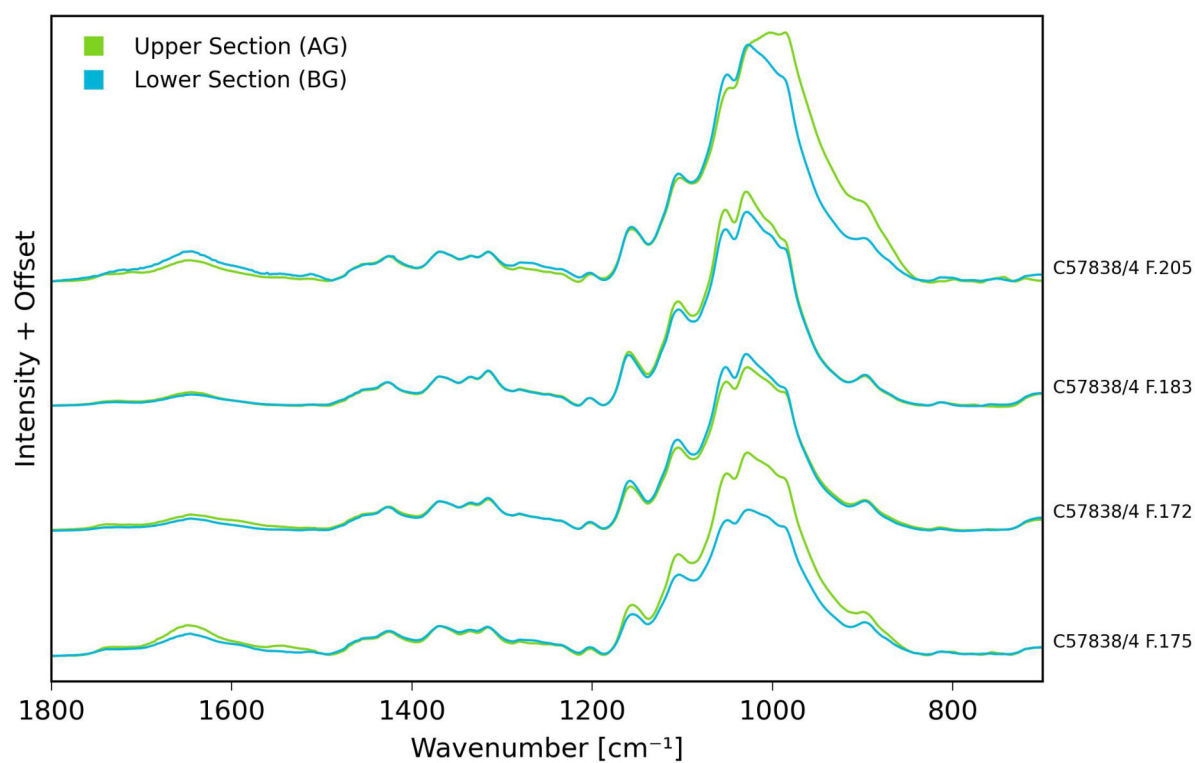


Figure 27: ATR-FTIR spectra comparing two samples from different sections of the same scarring stick in the 1800 cm^{-1} to 700 cm^{-1} region.

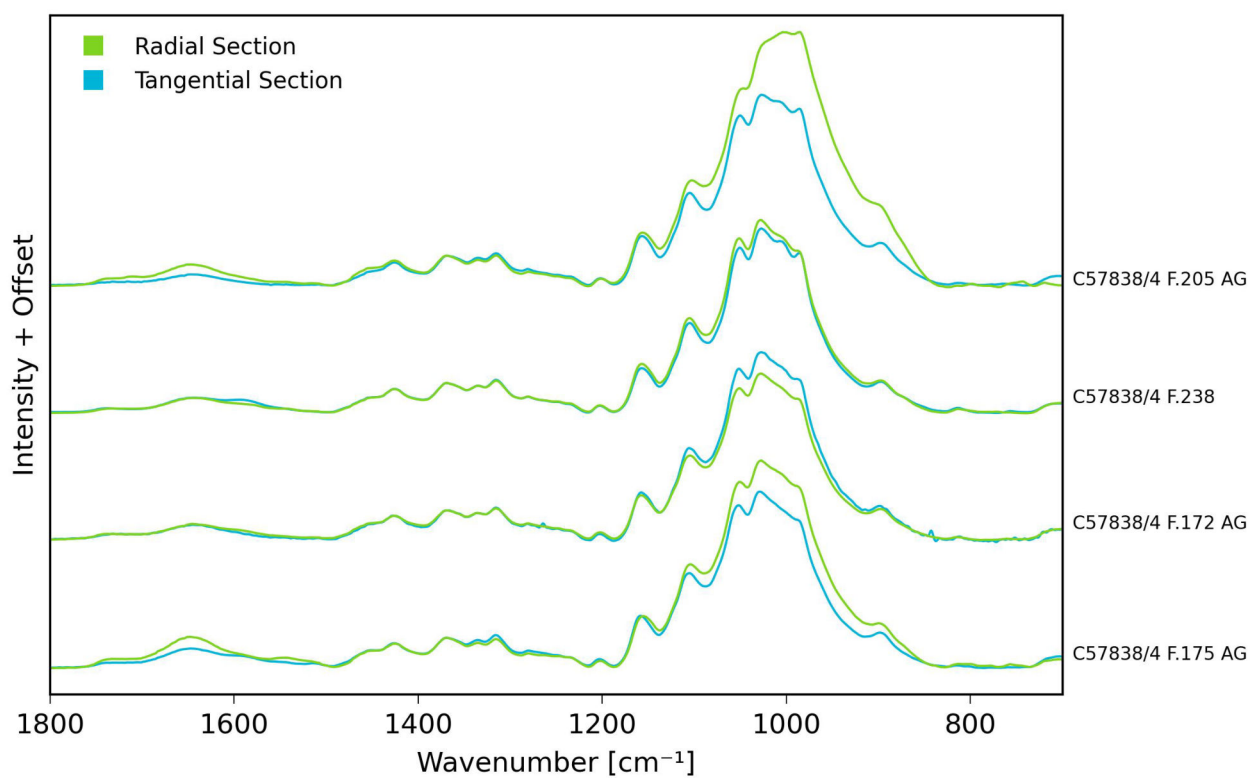


Figure 28: ATR-FTIR spectra comparing radial and tangential sections of the same ice patch wood sample in the 1800 cm^{-1} to 700 cm^{-1} region.

5.7 DVS Measurement

DVS measurement is used to investigate the moisture sorption behavior of the ice patch wood samples. Figure 29 shows the set and observed temperature and RH values during the experiment. The temperature is kept constant at 25°C with small deviations of less than ± 0.1 K. At the beginning of the measurement the samples are conditioned at 0% RH for 72 h. The lowest measured RH value is $\sim 1\%$. One of the equilibrium conditions is met for all samples at all humidity steps, except the last step at 90% RH where the instrument malfunctions before equilibrium is reached. The time to reach equilibrium is longest near 0% RH, decreases for the following humidity steps and increases from 60% RH on.

Figure 30 shows the adsorption isotherm which describes the relative mass change compared to the lowest measured mass as a function of RH at equilibrium. The lowest measured mass of all samples is at $\sim 1\%$ RH where the samples have the lowest moisture content. As this is used as a reference for all other mass measurements, the mass change in Figure 30 is equivalent to the equilibrium moisture content relative to the dry state.

All samples show an S-shaped adsorption curve which is characteristic for wood [54, 55]. As expected, the two pieces measured for each sample show similar behavior. The adsorption curve of modern pine is slightly lower than literature values [56, 57]. These deviations could be explained by natural variations, such as differences between earlywood and latewood or heartwood and sapwood, which can influence the DVS behavior [56, 57].

The adsorption curve for freeze-dried archaeological pine shows a higher equilibrium moisture content compared to modern pine at all humidity values. This is consistent with literature as degradation of cellulose and hemicelluloses through biological decay changes the chemical composition and organization of the cell wall which results in different hygroscopic behavior [54, 55].

In contrast, all ice patch wood samples have a lower equilibrium moisture content compared to modern pine at all humidity values. The relative difference to modern pine is small at low humidity, then increases with increasing RH but appears to decrease over 80% RH. In the interior of the ice patch wood samples hemicelluloses are degraded while cellulose and lignin are preserved. The removal of hemicelluloses from wood can cause a decrease in hygroscopicity as the remaining material has less hydroxyl groups that can form hydrogen bonds with water [58]. All ice patch wood samples show very similar adsorption behavior. This is expected as the hygroscopic properties are likely dominated by the interior, which is very similar between all samples as shown by the color measurement, light microscopy and ATR-FTIR investigations.

Due to instrument failure, only water adsorption is measured while typically both adsorption and desorption are recorded as they often show a hysteresis effect. Since hysteresis is often stronger for freeze-dried archaeological wood than modern wood [55], it would be interesting to investigate if ice patch wood samples show a similar behavior.

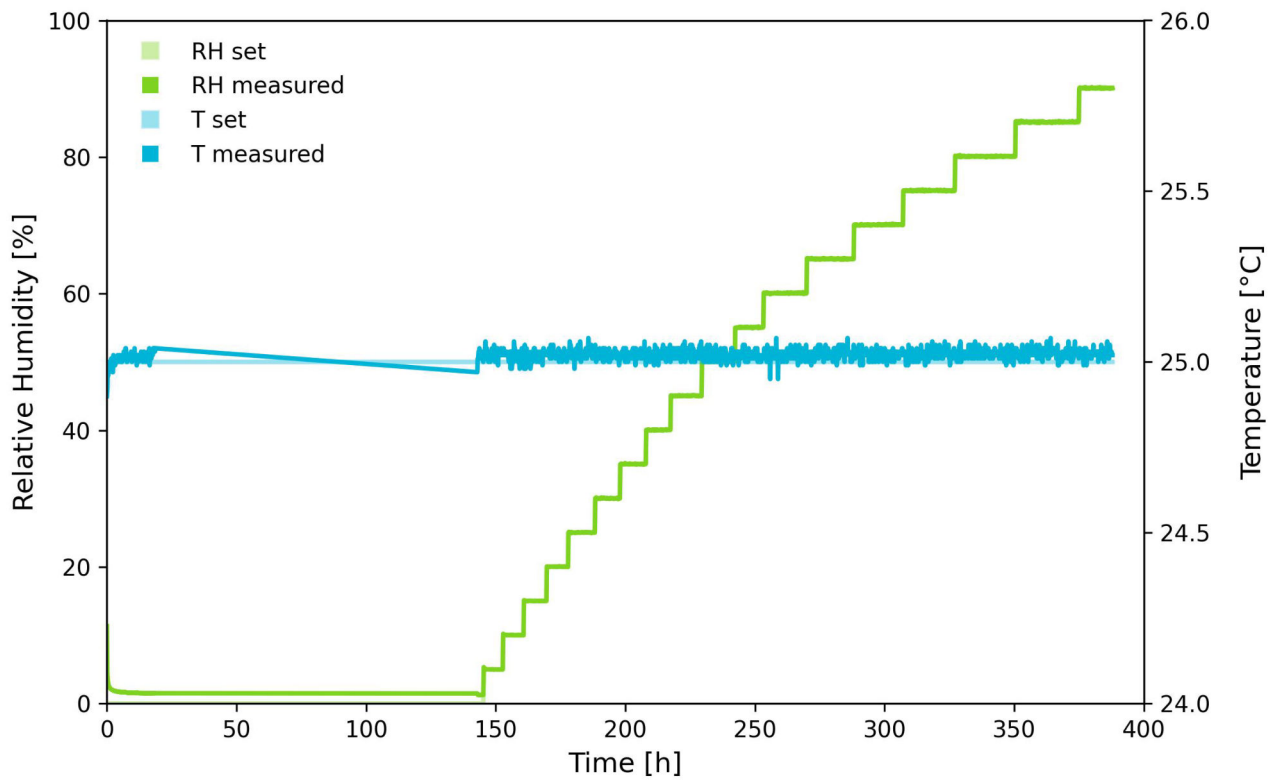


Figure 29: DVS conditions with the set and measured RH and temperature.

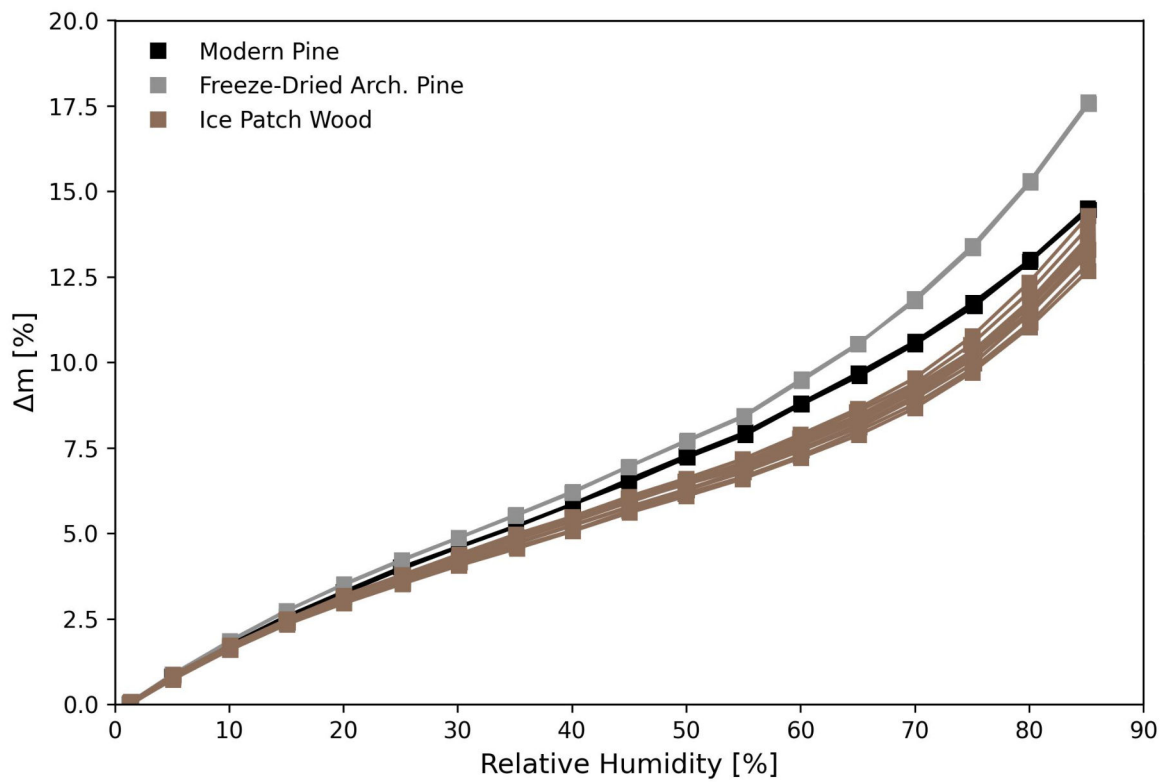


Figure 30: DVS adsorption isotherms with the mass change relative to the lowest measured mass as a function of RH at equilibrium comparing ice patch wood samples with modern and freeze-dried archaeological pine. Ice Patch Wood Samples: C57838/4 F.230, C57838/4 F.205 AG, C57838/4 F.238, C57838/4 F.172 AG, C57838/4 F.175 AG.

6 Conclusion

The wood of scaring sticks preserved in ice patches has a well-preserved interior while the surfaces show different degradation patterns. The interior of the 13 samples investigated in this thesis, which are identified as *Pinus* sp., has a similar color and density to modern pine. The cells and cell walls are well-preserved and show no evidence of biological decay, which supports that the wood was continuously frozen for most of the time it was preserved in the ice. Cellulose and lignin remain preserved in the interior, but hemicelluloses appear degraded, which may be a process of long-term wood ageing in the absence of microbial decay and other degradation factors [37, 42]. Hemicellulose degradation can also explain the lower hygroscopicity of ice patch wood compared to modern pine.

The surfaces of the ice patch wood show strong differences from the interior in color and chemical composition as well as greater variation between the sticks. Degradation is limited to the outermost layer and likely caused by environmental degradation after melting out of the ice including exposure to sunlight, which can cause photodegradation, especially of lignin [14], leaving mainly cellulose exposed at the surface. Meltwater can wash away degradation products [46], accelerate lignin degradation [46] or mechanically abrade the surface when carrying sediment particles [2]. Wind can cause movement of the sticks and contribute to surface erosion patterns [2, 21, 22]. Freeze-thaw and drying-wetting cycles can cause small cracks or splitting along the growth rings [2, 10]. Degradation is highly localized and depends on the specific environmental conditions.

The good preservation of the interior of the scaring sticks suggests that air-drying after recovery is sufficient for their conservation. However, the loss of hemicelluloses can reduce the strength of wood [59]. Investigating the mechanical properties of ice patch wood would therefore be useful for conservation, particularly for larger objects that may require different treatment.

Future research could also include additional techniques to better understand the degradation mechanisms and products on the wood surface, which cannot be identified with only ATR-FTIR spectroscopy. Some scaring sticks also show surface features that cannot be fully explained with the techniques used in this thesis. The diffuse grey discolorations on some sticks may be related to fungal activity, as light microscopy shows patterns consistent with fungal hyphae, but further investigations are needed to confirm this. ATR-FTIR spectra suggest that the black deposits on some surfaces may contain compounds with amide groups, but further analysis is needed to fully determine their composition and identify them.

Although both wood preserved in ice patches and wood preserved in waterlogged conditions are considered archaeological wood, they show different material properties, degradation patterns and require different conservation treatment. In waterlogged wood, biological decay can affect cellulose and hemicelluloses, while the lignin structure is typically preserved. In contrast, no biological decay is observed in the ice patch wood and the wood surface shows an opposite degradation pattern where lignin is degraded while cellulose remains preserved.

7 Bibliography

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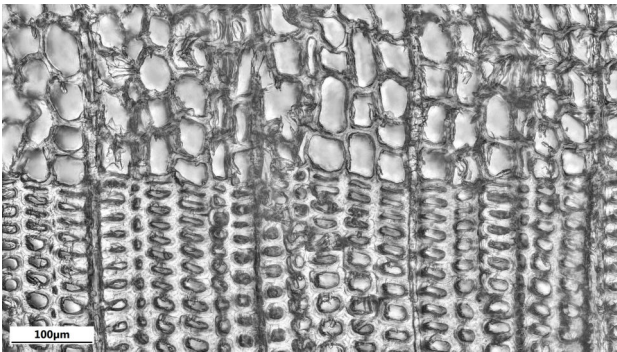
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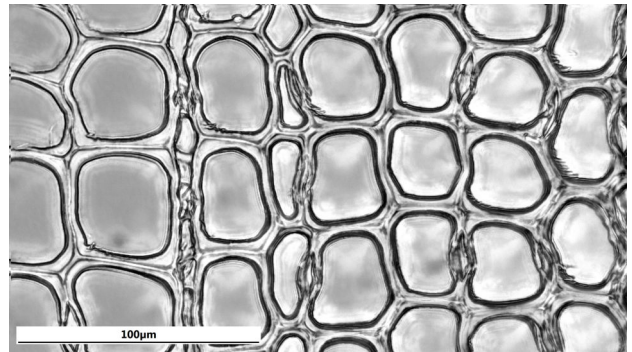
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A Appendix

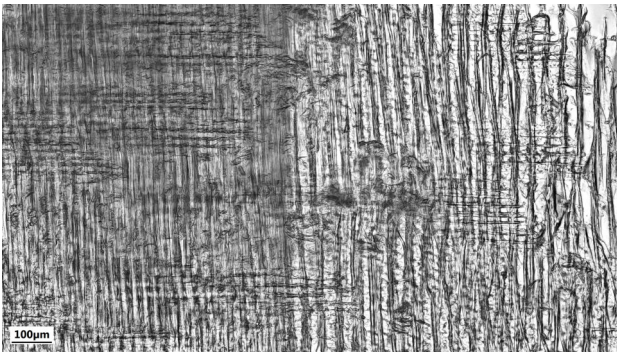
Modern Pine (Reference)



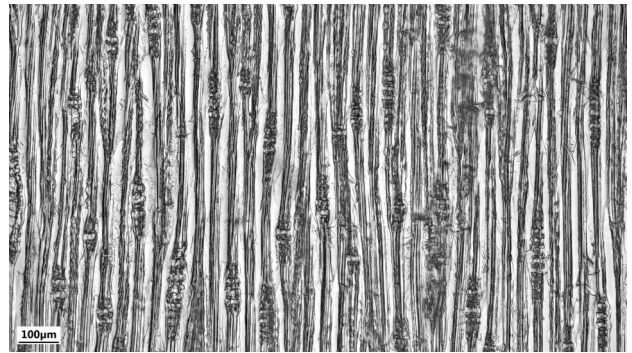
(a) Transverse Section



(b) Transverse Section



(c) Radial Section



(d) Tangential Section

Figure A.1: Light microscopy images of modern pine.

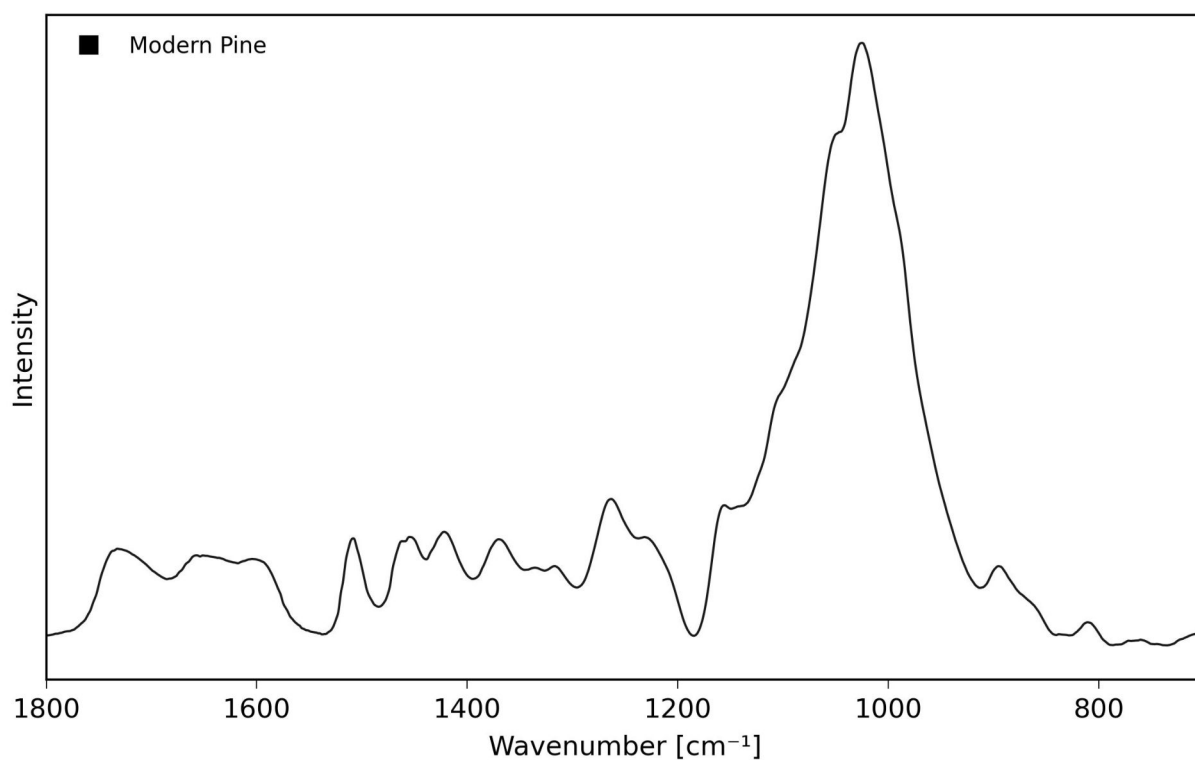
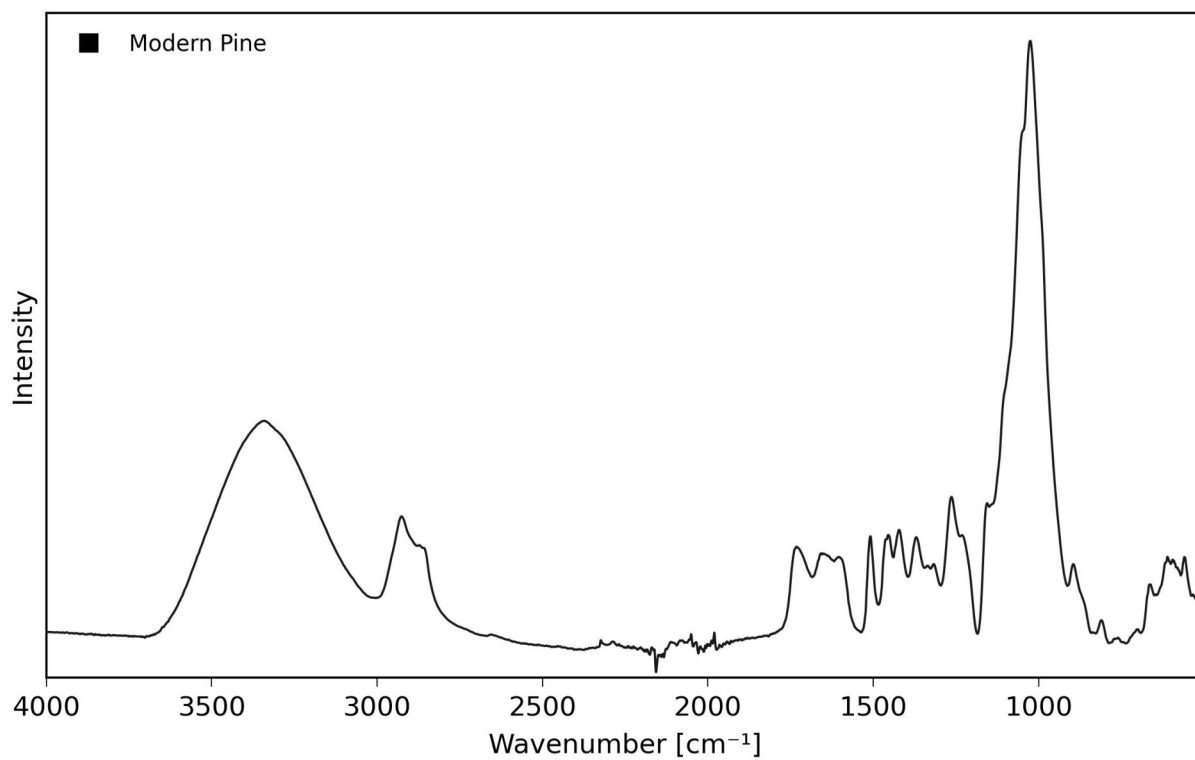
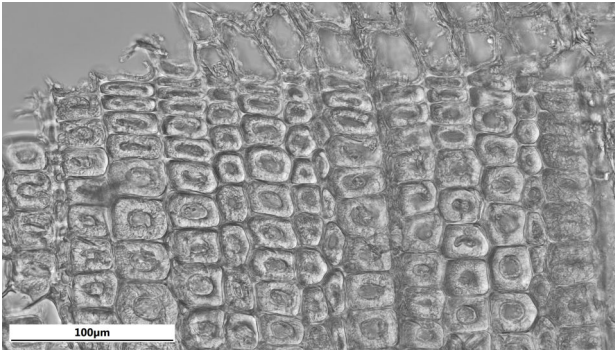
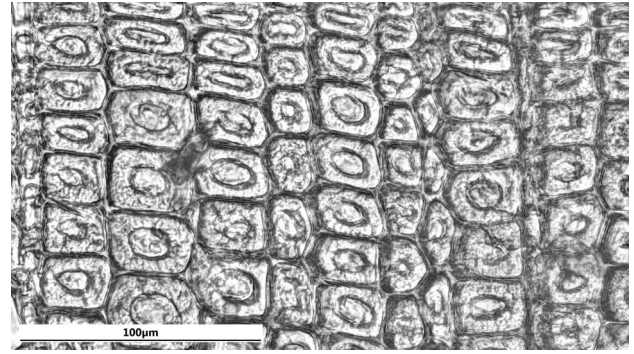


Figure A.2: ATR-FTIR spectra of modern pine in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

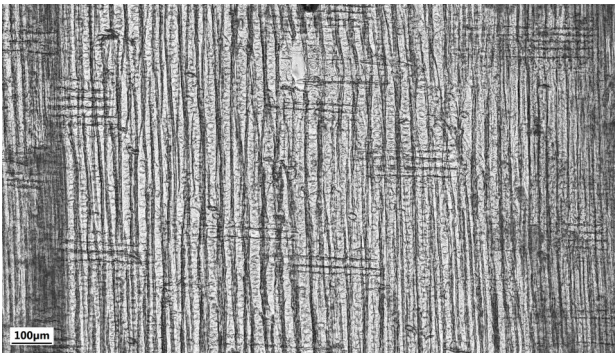
Water-Stored Archaeological Pine (Reference)



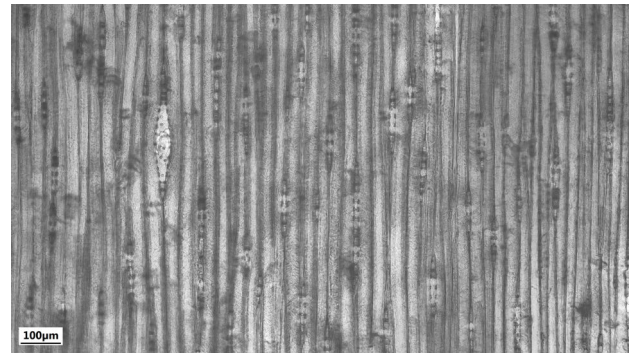
(a) Transverse Section



(b) Transverse Section



(c) Radial Section



(d) Tangential Section

Figure A.3: Light microscopy images of water-stored archaeological pine.

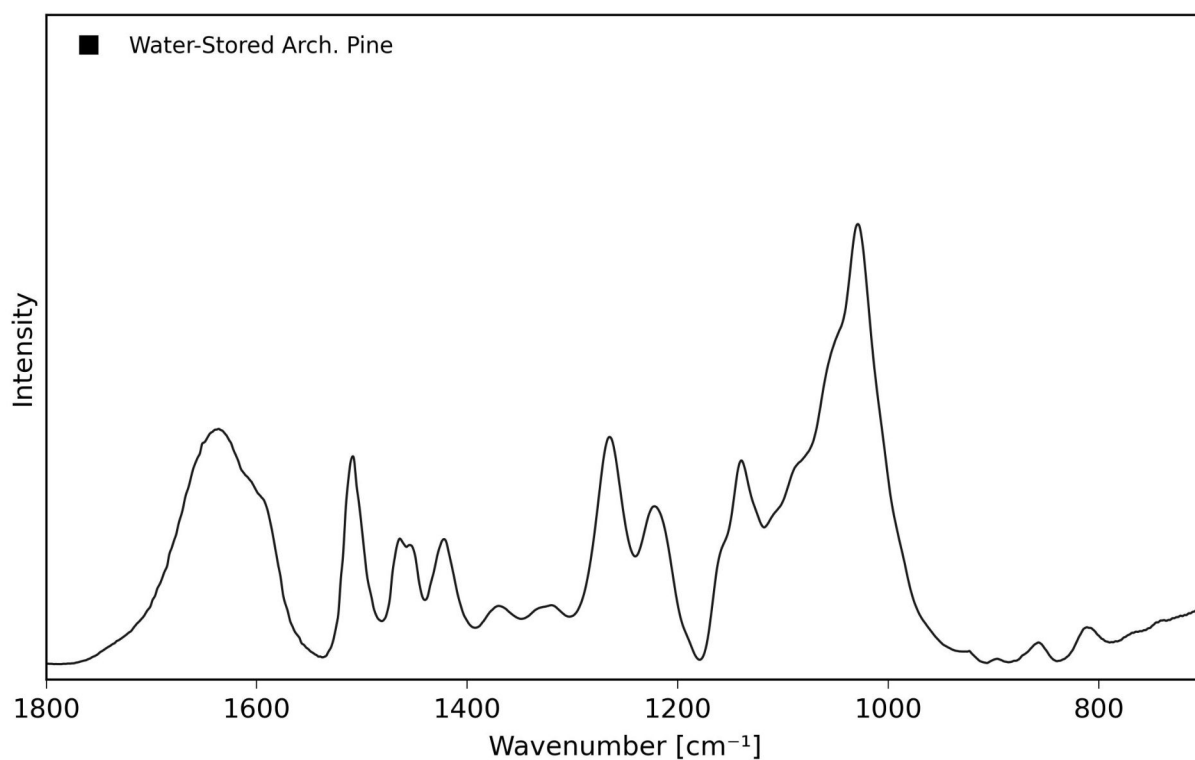
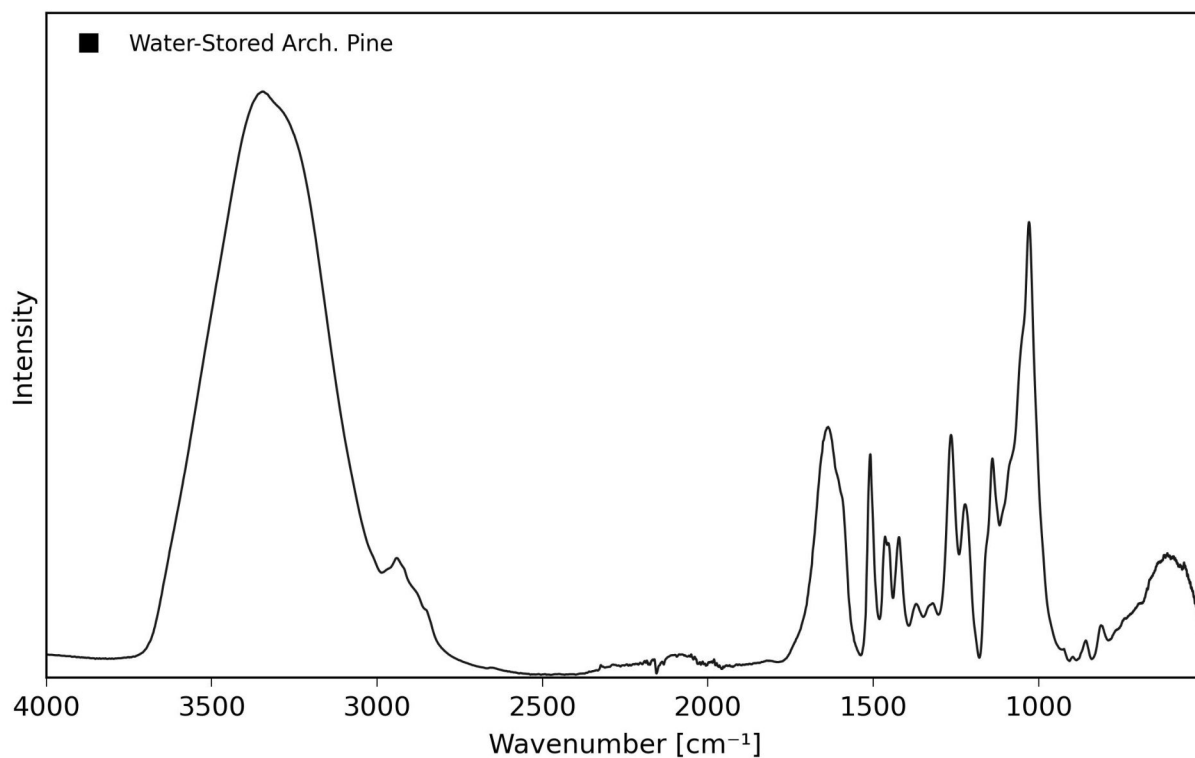
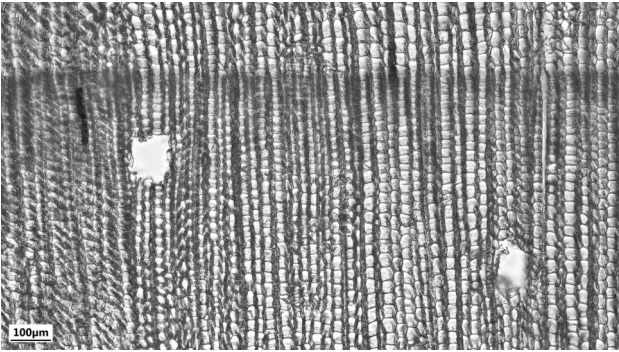
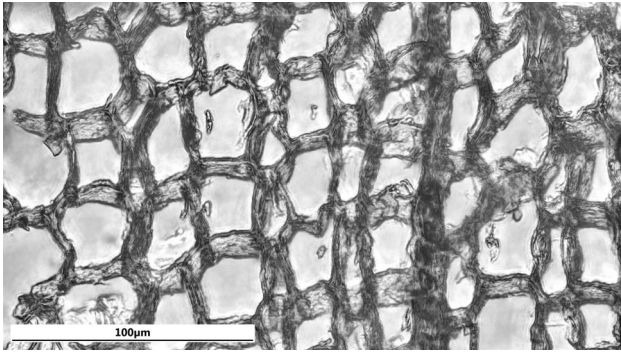


Figure A.4: ATR-FTIR spectra of water-stored archaeological pine in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

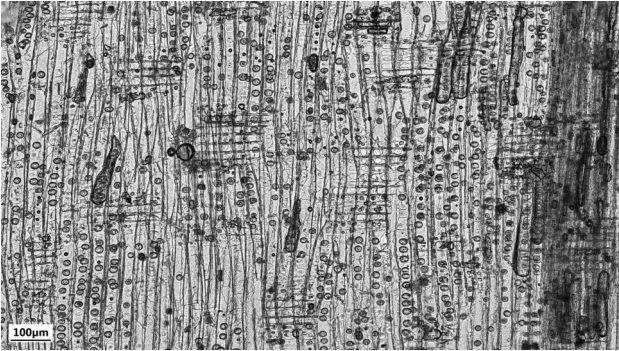
Freeze-Dried Archaeological Pine (Reference)



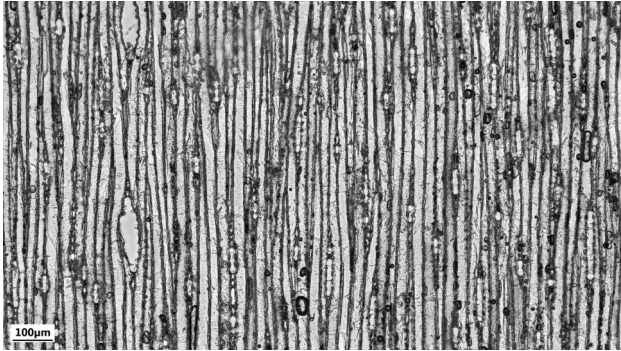
(a) Transverse Section



(b) Transverse Section



(c) Radial Section



(d) Tangential Section

Figure A.5: Light microscopy images of freeze-dried archaeological pine.

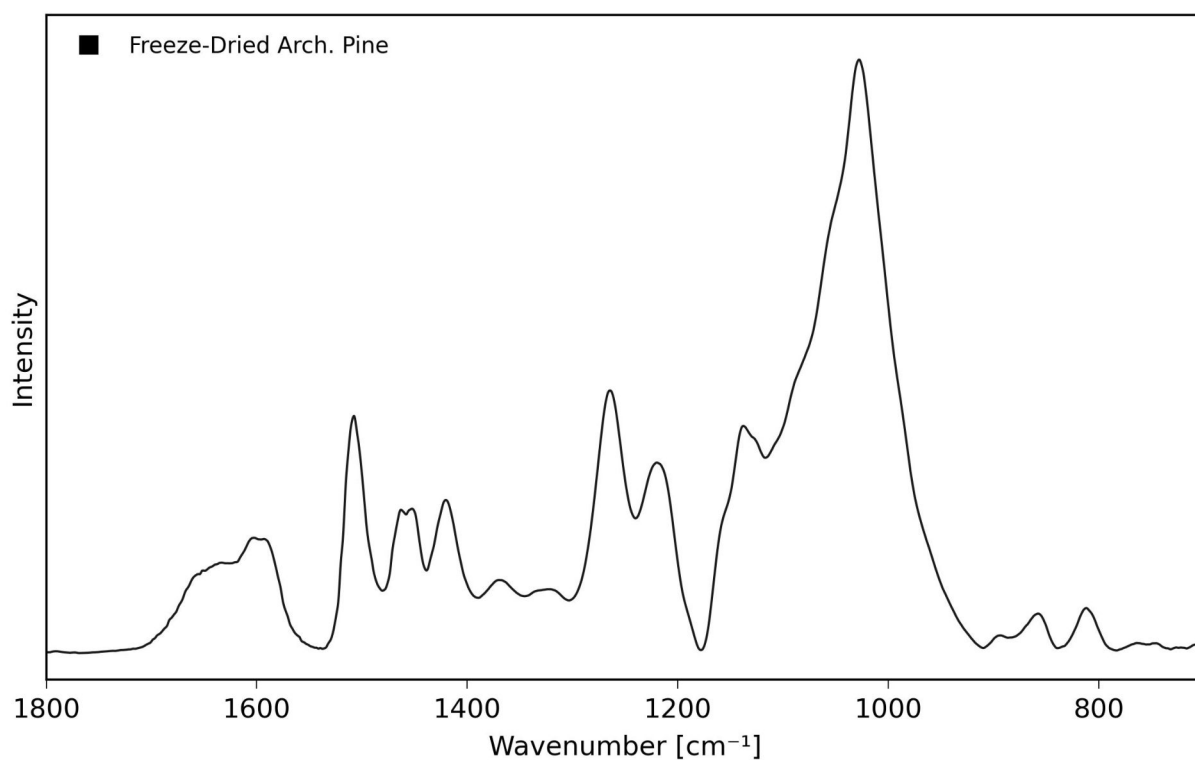
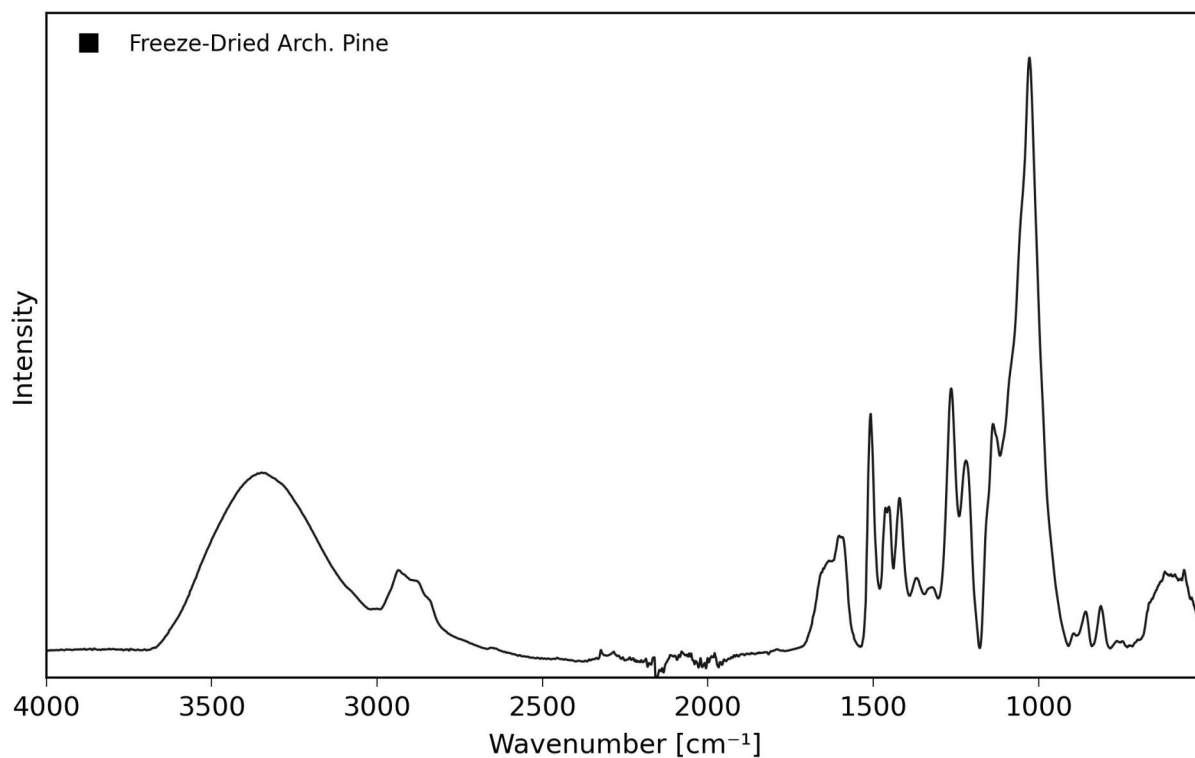


Figure A.6: ATR-FTIR spectra of freeze-dried archaeological pine in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

C57838/4 F.205 AG (Brown Surface)



Figure A.7: Photos of C57838/4 F.205. Eva Brænd, Museum of Cultural History, Norway.

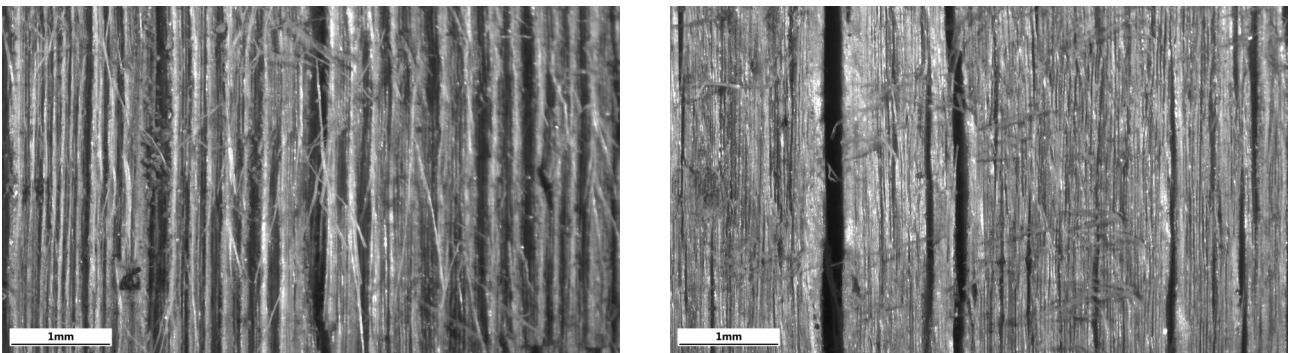
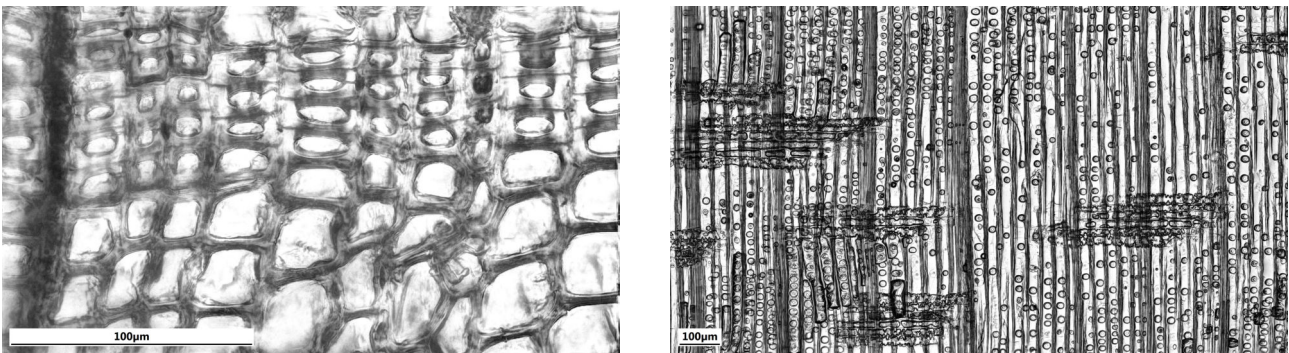
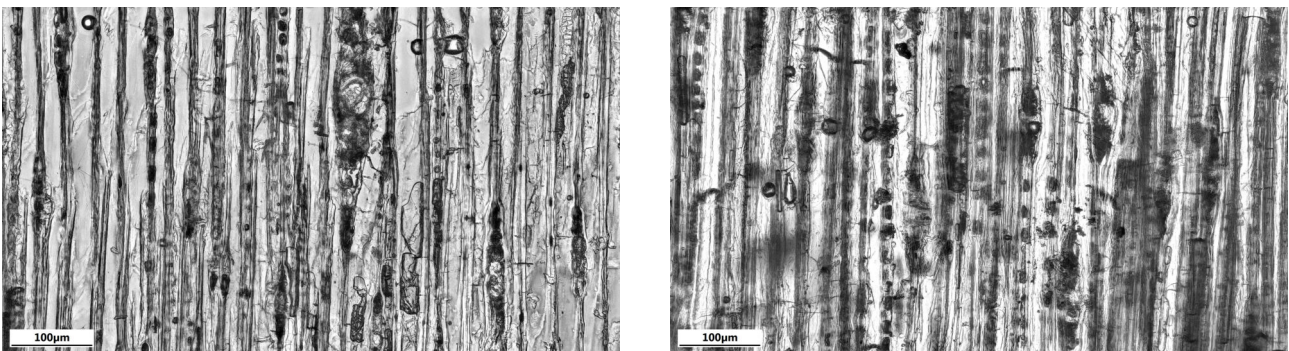


Figure A.8: Stereomicroscopy images of C57838/4 F.205 AG.



(a) Transverse Section, Interior

(b) Radial Section, Interior



(c) Tangential Section, Interior

(d) Tangential Section , Surface

Figure A.9: Light microscopy images of C57838/4 F.205 AG.

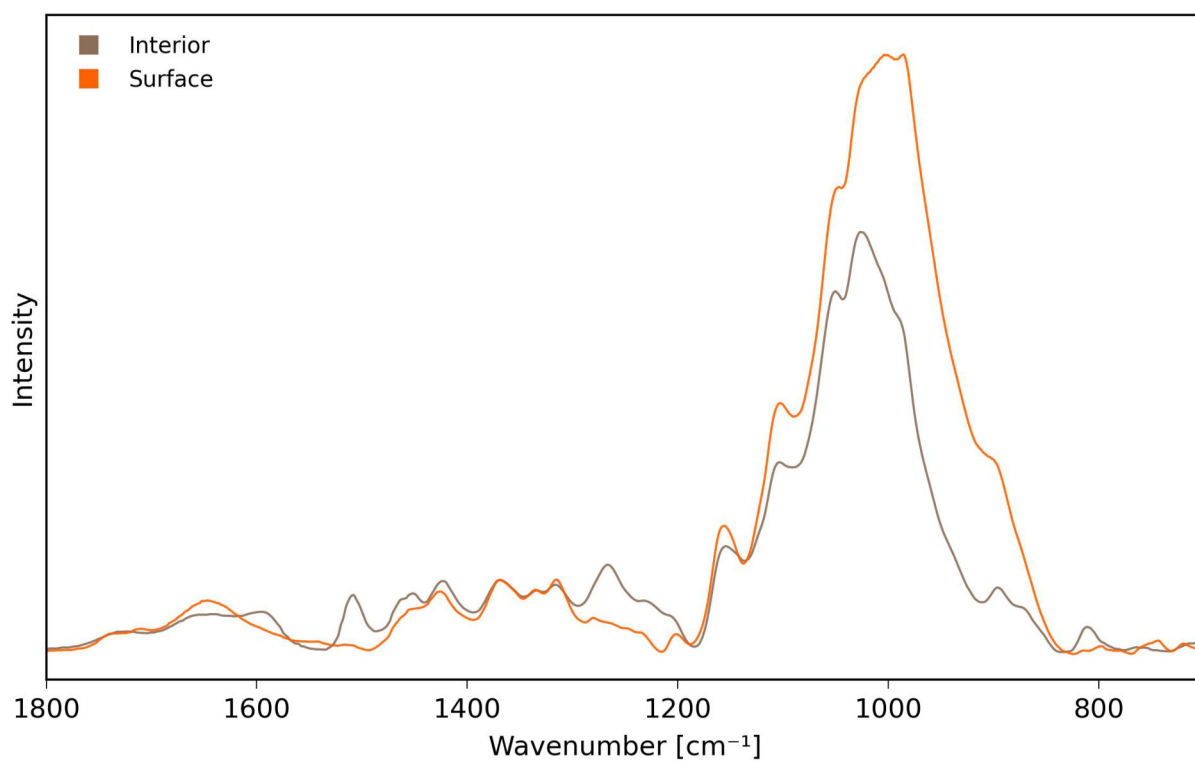
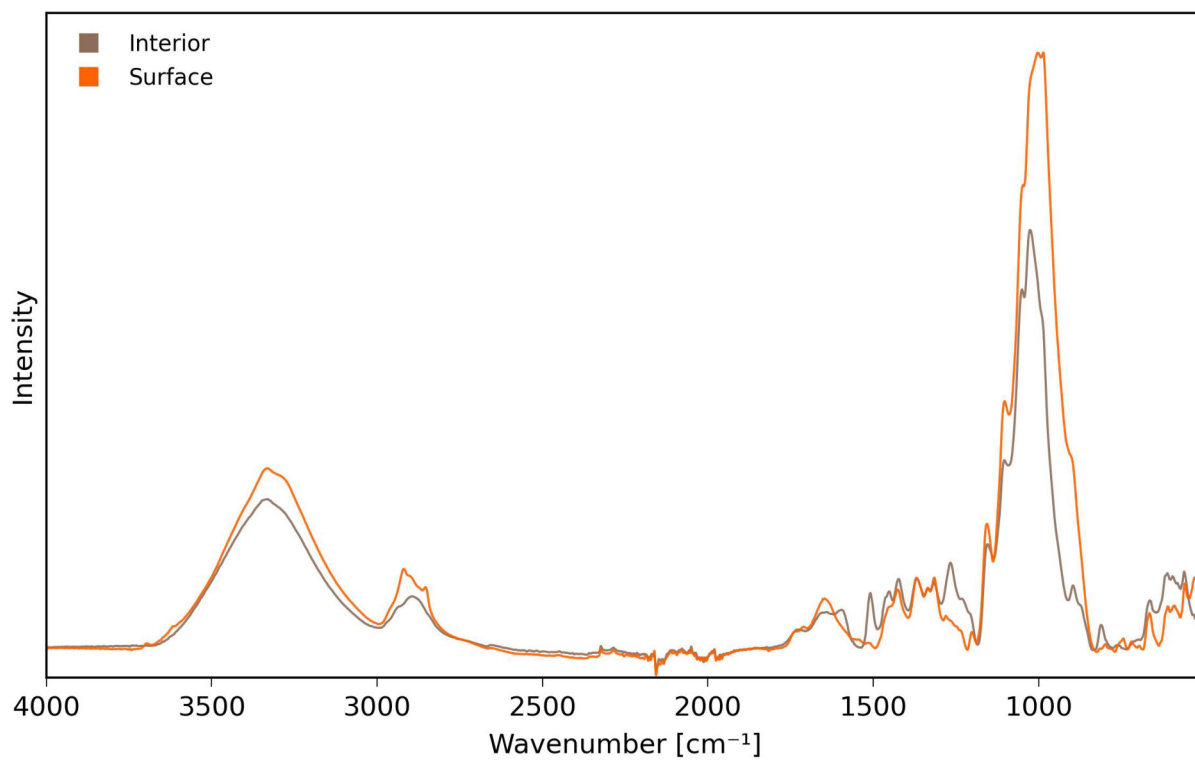


Figure A.10: ATR-FTIR spectra of C57838/4 F.205 AG in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

C57838/4 F.230 (Brown Surface)



Figure A.11: Photo of C57838/4 F.230. Eva Brænd, Museum of Cultural History, Norway.

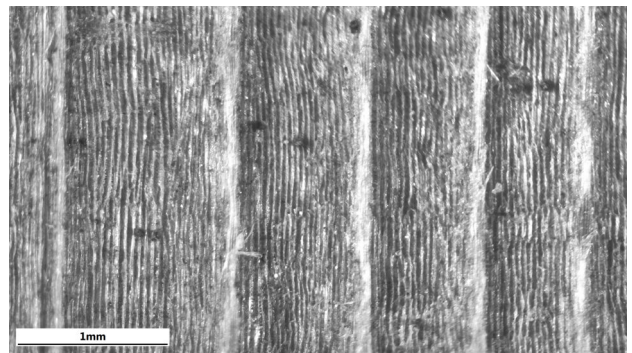
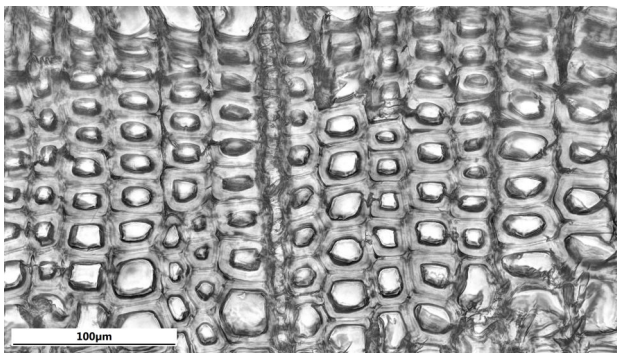
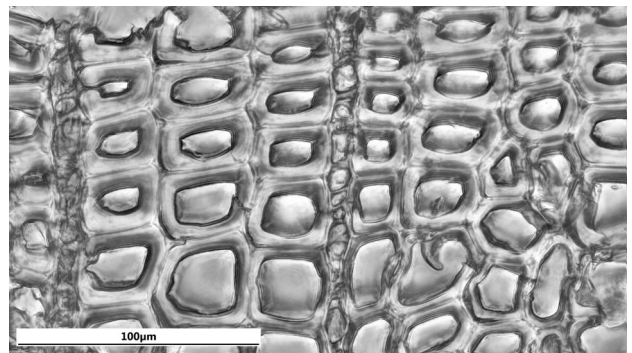


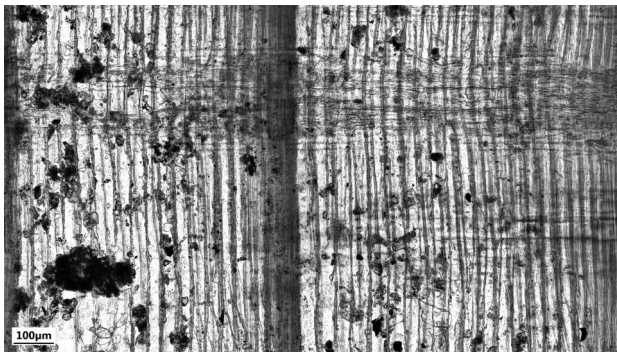
Figure A.12: Stereomicroscopy images of C57838/4 F.230.



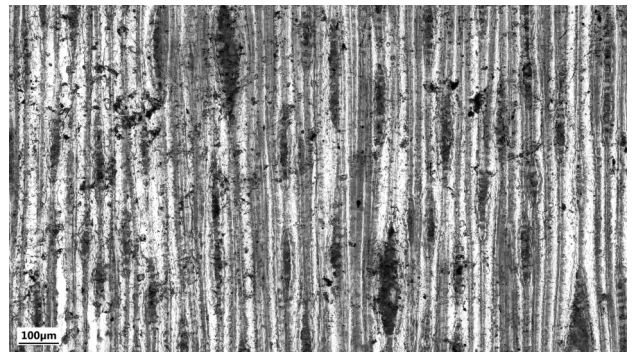
(a) Transverse Section, Interior



(b) Transverse Section, Interior



(c) Radial Section, Surface



(d) Tangential Section, Interior

Figure A.13: Light microscopy images of C57838/4 F.230.

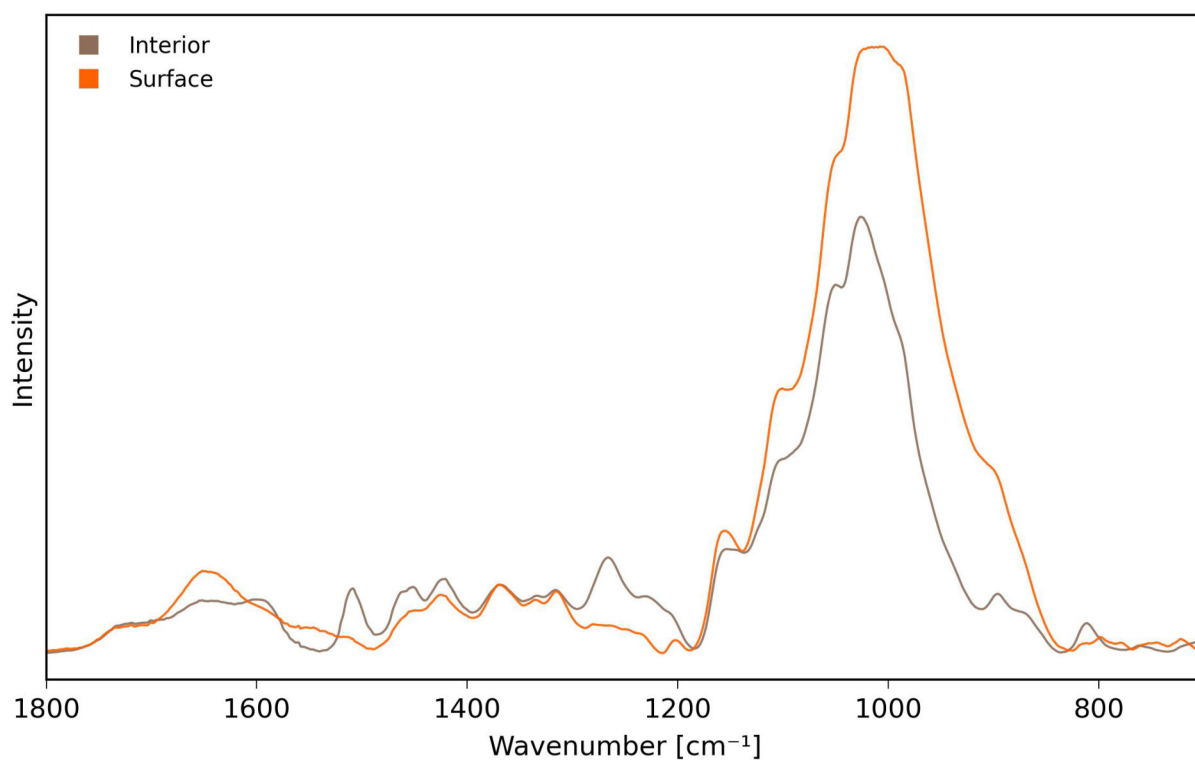
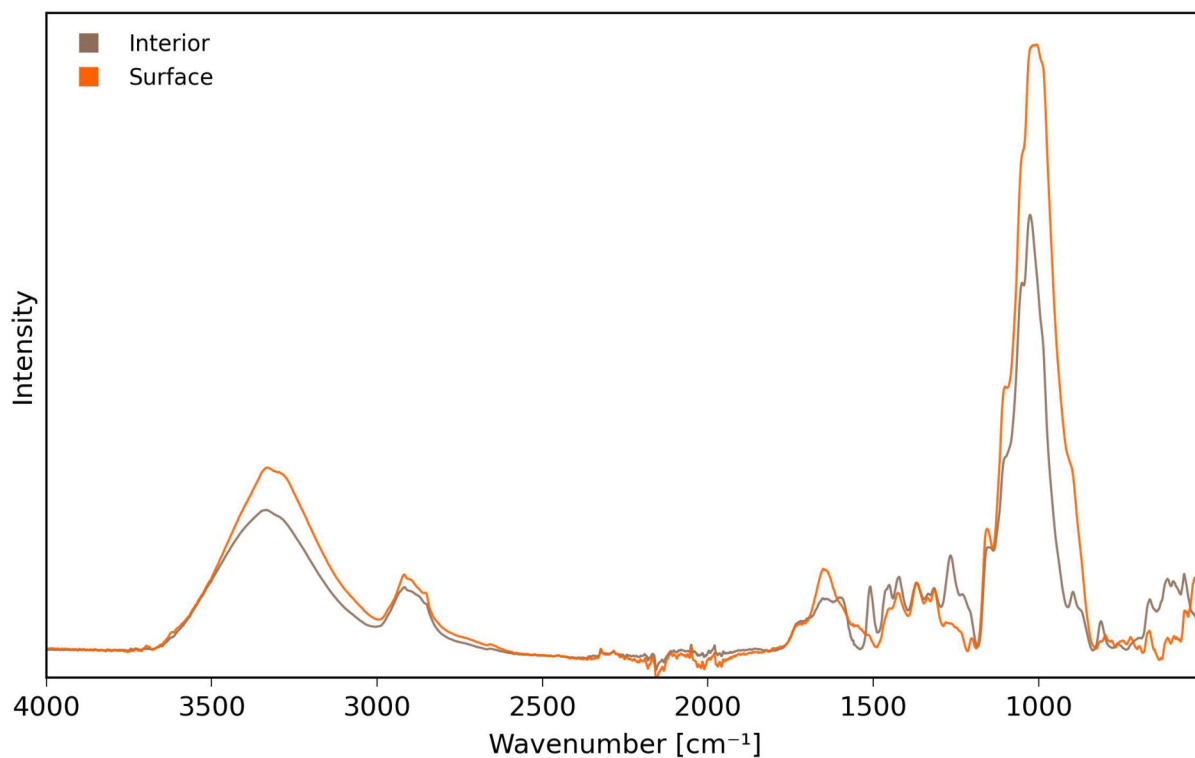


Figure A.14: ATR-FTIR spectra of C57838/4 F.230 in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

C57838/4 F.238 (White Surface)



Figure A.15: Photos of C57838/4 F.238. Eva Brænd, Museum of Cultural History, Norway.

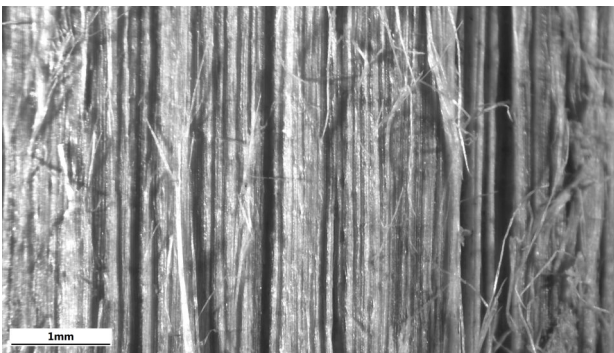
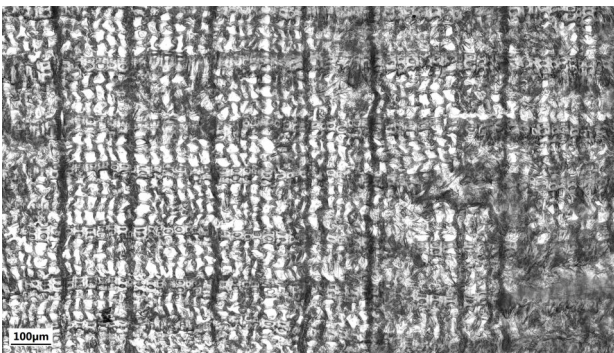
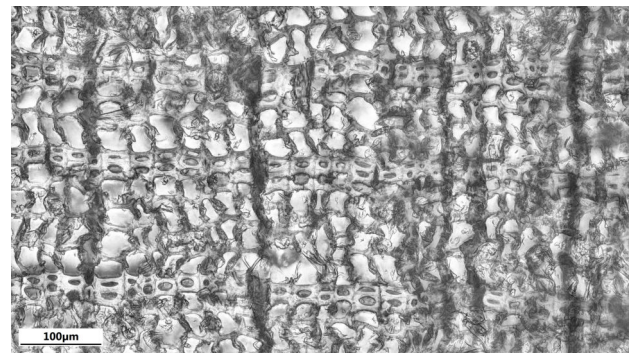


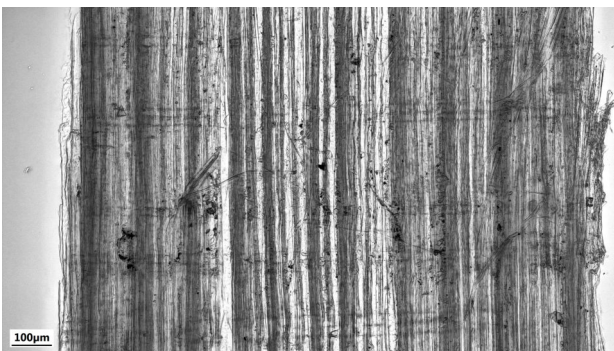
Figure A.16: Stereomicroscopy images of C57838/4 F.238.



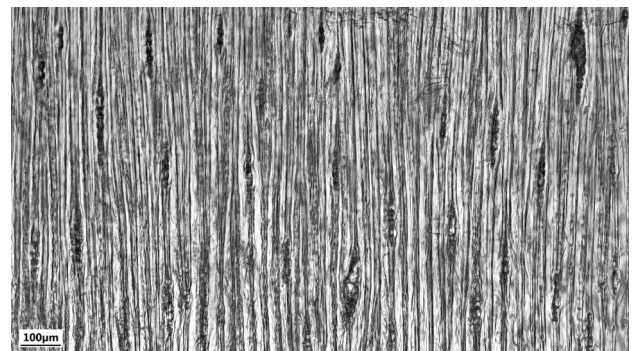
(a) Transverse Section, Interior



(b) Transverse Section, Interior



(c) Radial Section, Surface



(d) Tangential Section, Interior

Figure A.17: Light microscopy images of C57838/4 F.238.

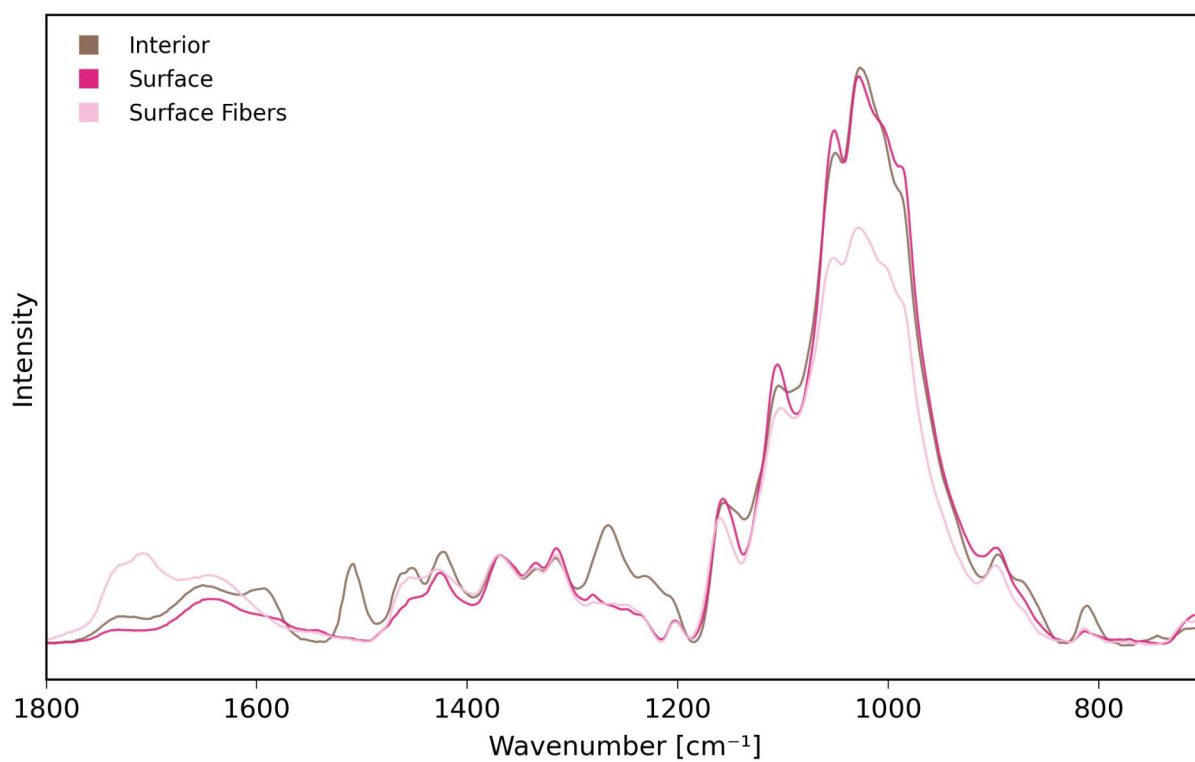
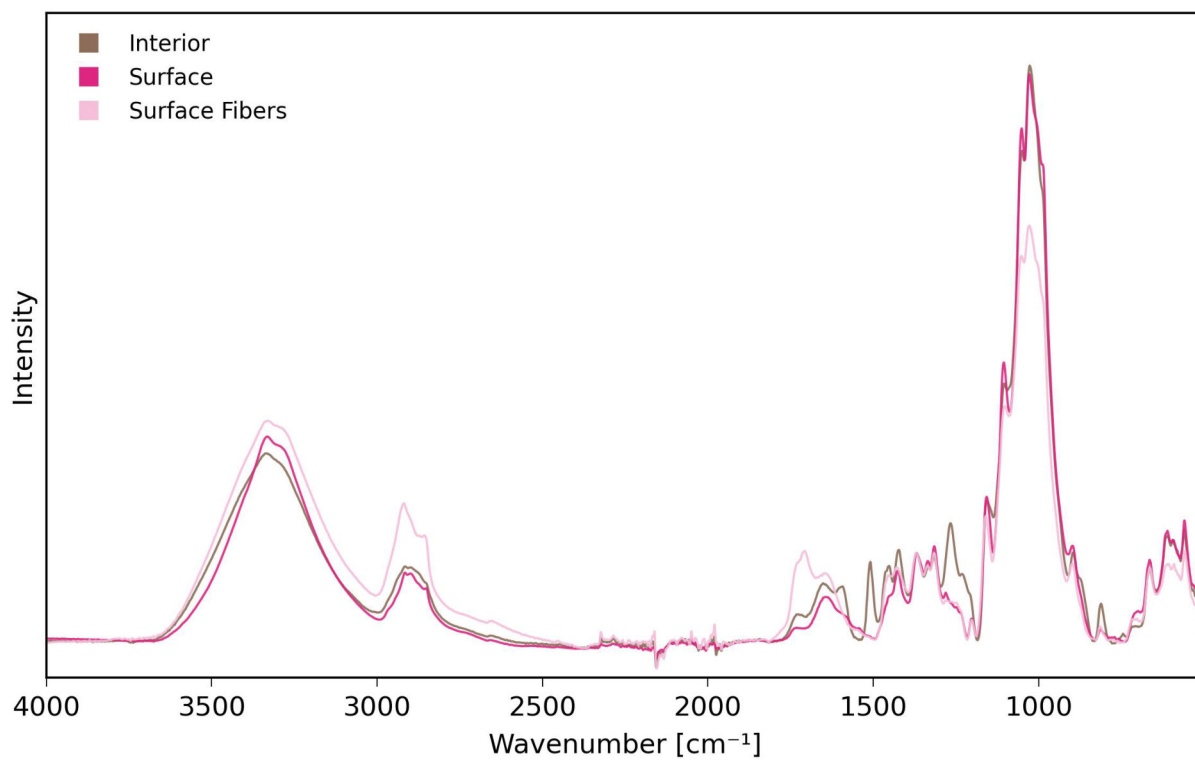


Figure A.18: ATR-FTIR spectra of C57838/4 F.238 in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

C57838/4 F.172 AG (Surface with Discolorations)



Figure A.19: Photos of C57838/4 F.172. Eva Brænd, Museum of Cultural History, Norway.

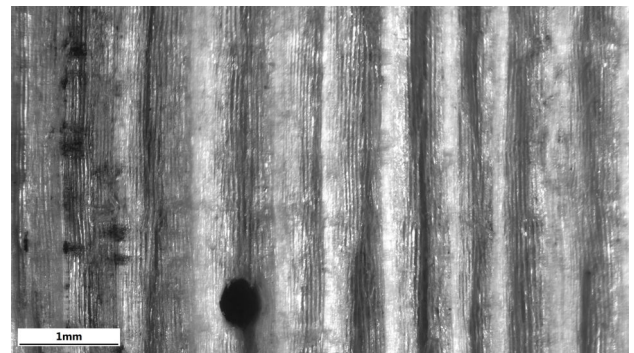
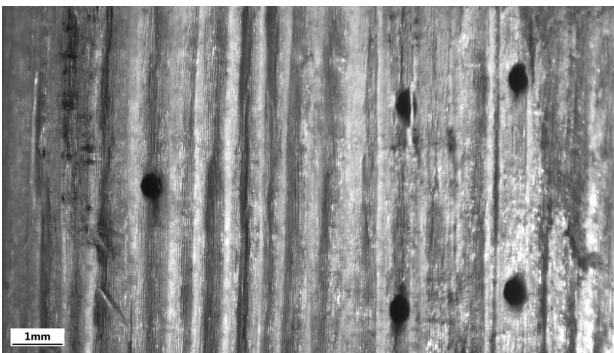
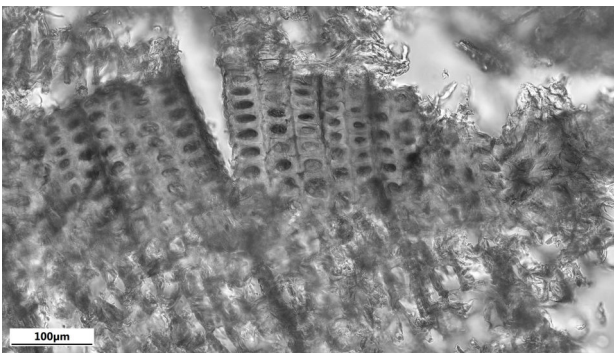
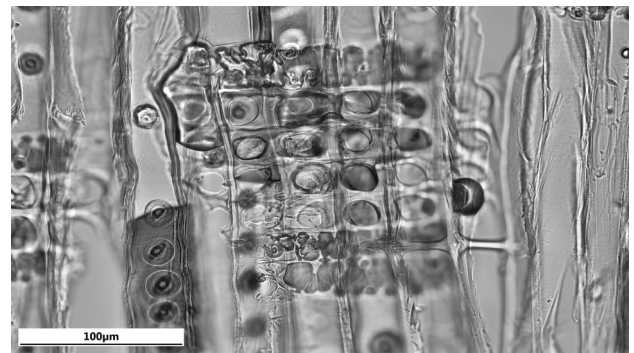


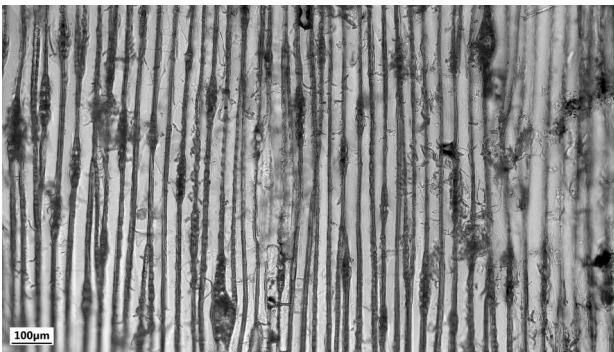
Figure A.20: Stereomicroscopy images of C57838/4 F.172 AG.



(a) Transverse Section, Interior



(b) Radial Section, Interior



(c) Tangential Section, Interior



(d) Tangential Section, Surface

Figure A.21: Light microscopy images of C57838/4 F.172 AG.

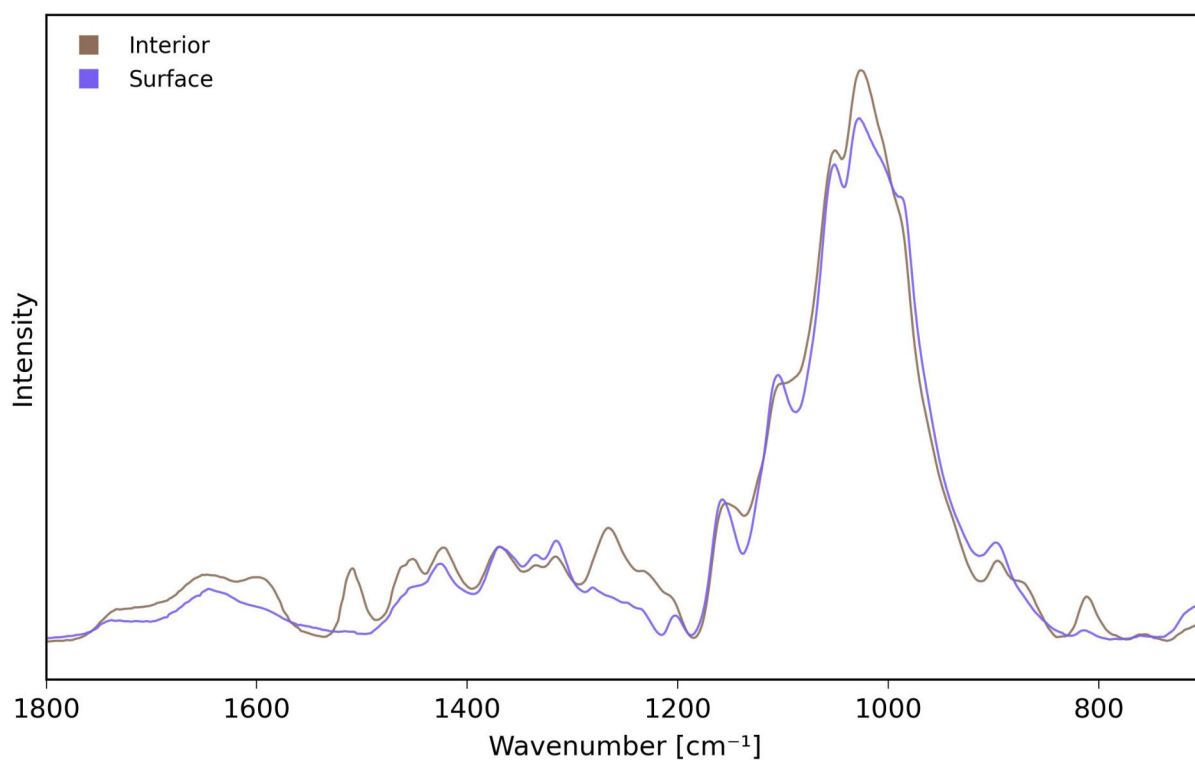
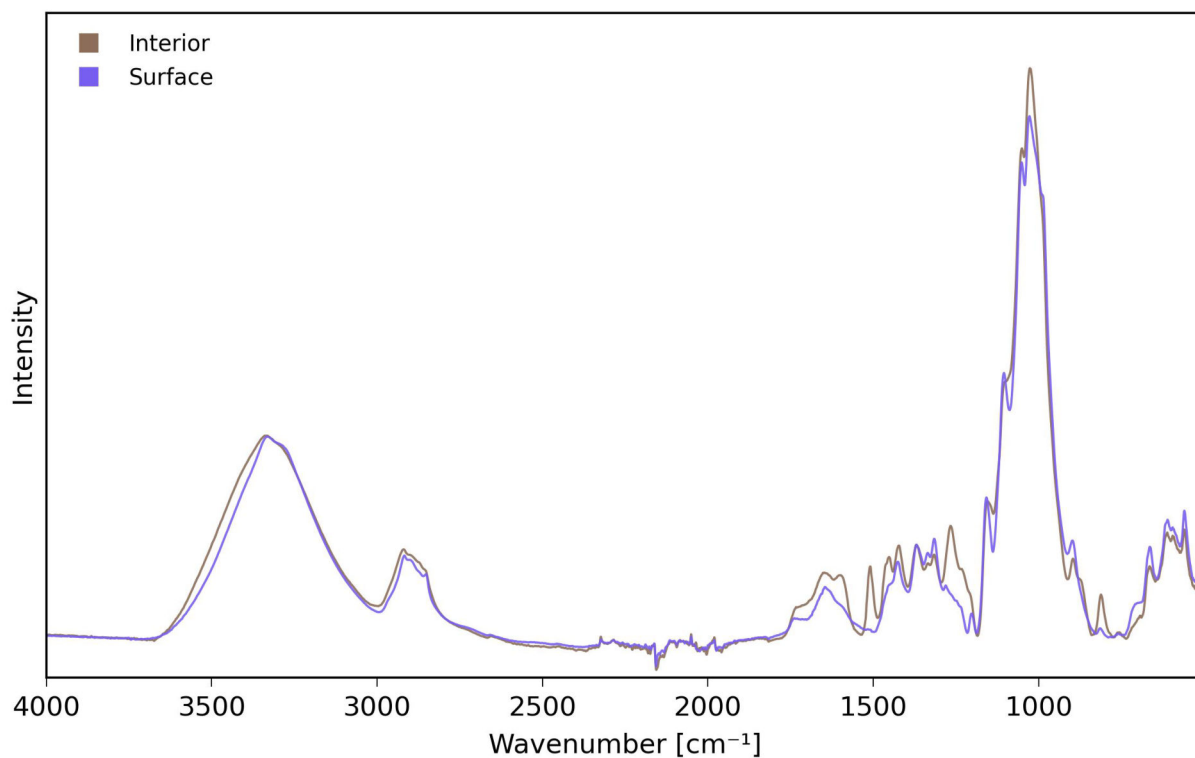


Figure A.22: ATR-FTIR spectra of C57838/4 F.172 AG in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

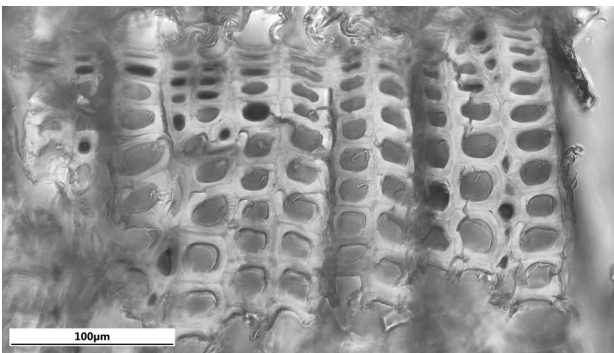
C57838/4 F.175 AG (Surface with Deposits)



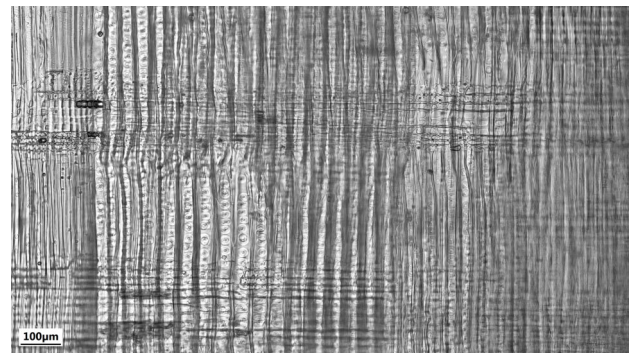
Figure A.23: Photos of C57838/4 F.175. Eva Brænd, Museum of Cultural History, Norway.



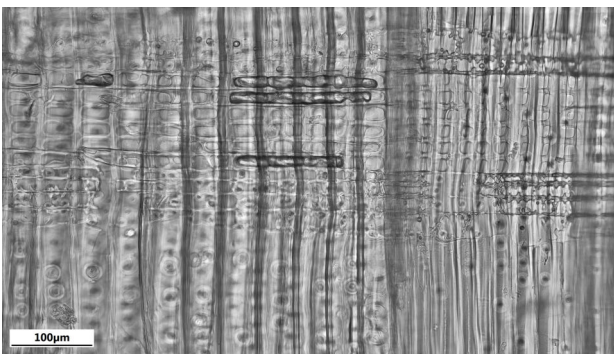
Figure A.24: Stereomicroscopy images of C57838/4 F.175 AG.



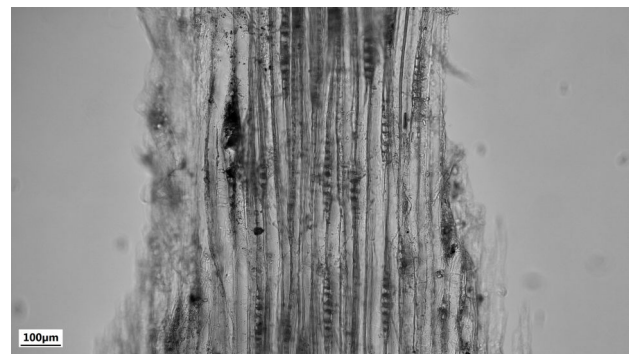
(a) Transverse Section, Interior



(b) Radial Section, Interior



(c) Radial Section, Interior



(d) Tangential Section, Surface

Figure A.25: Light microscopy images of C57838/4 F.175 AG.

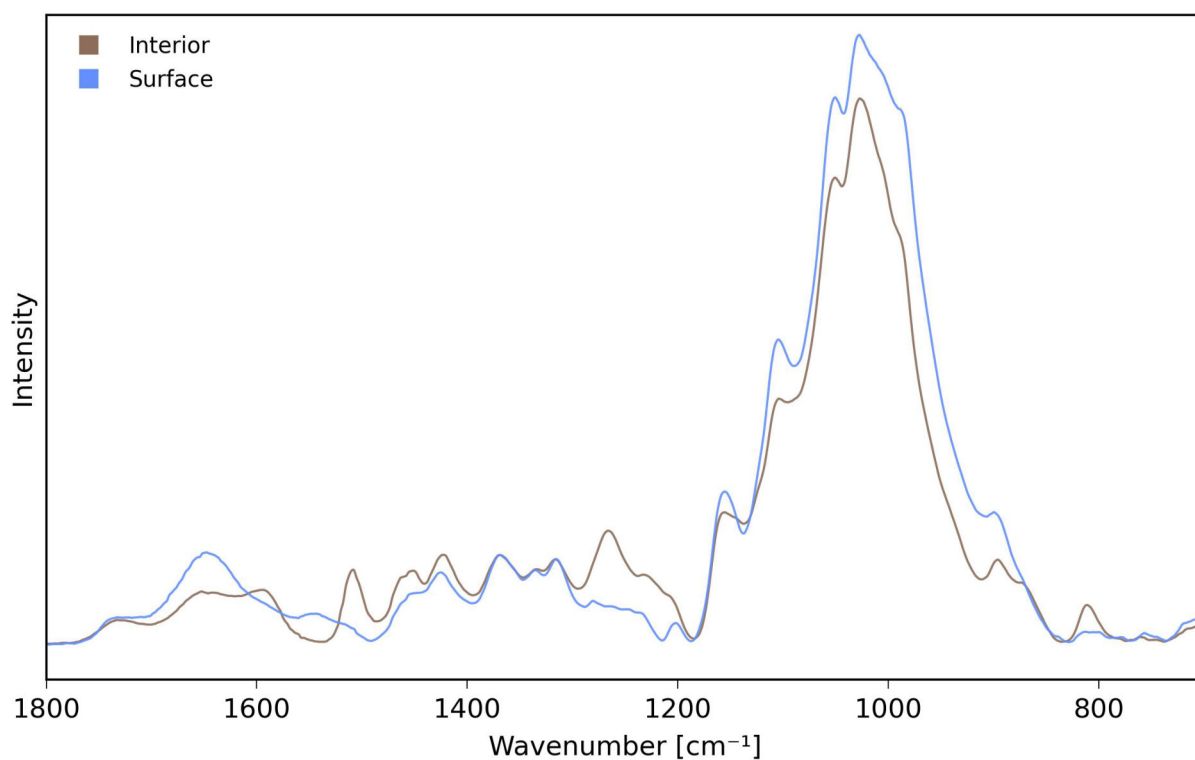
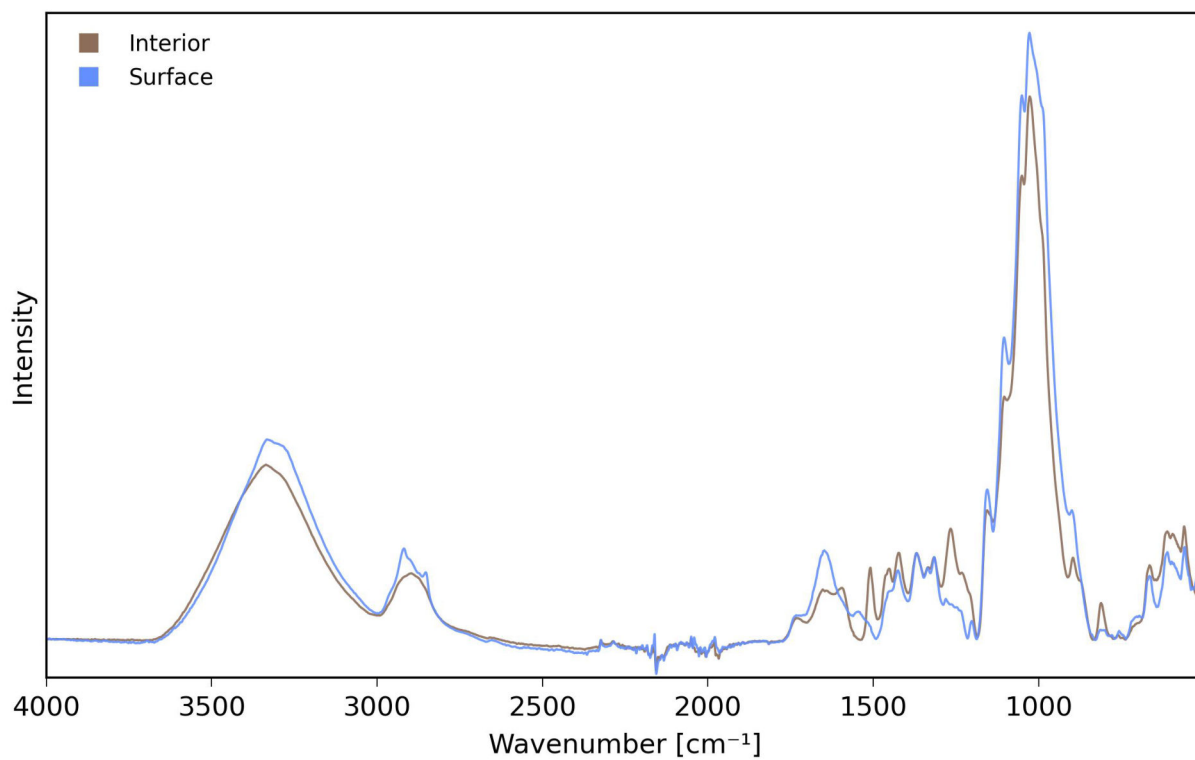


Figure A.26: ATR-FTIR spectra of C57838/4 F.175 AG in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).

C57838/4 F.133 AG (Brown and White Surfaces)



(a) Full Sample



(b) Upper Section

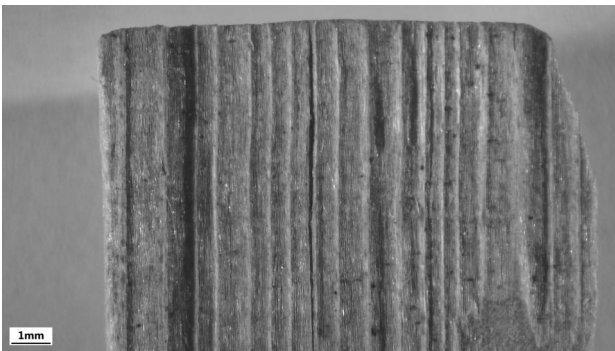


(c) Brown Surface



(d) White Surface

Figure A.27: Photos of C57838/4 F.133. Eva Brænd, Museum of Cultural History, Norway.



(a) Brown Surface



(b) White Surface

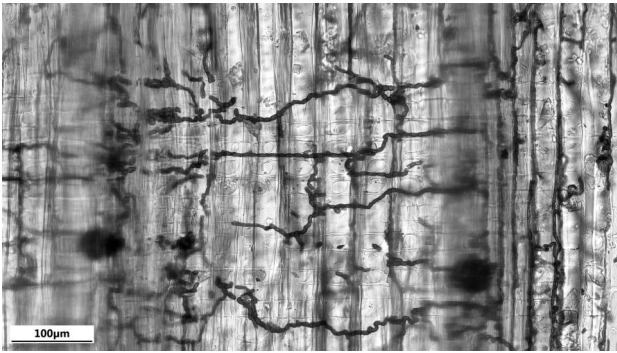


(c) Brown Surface

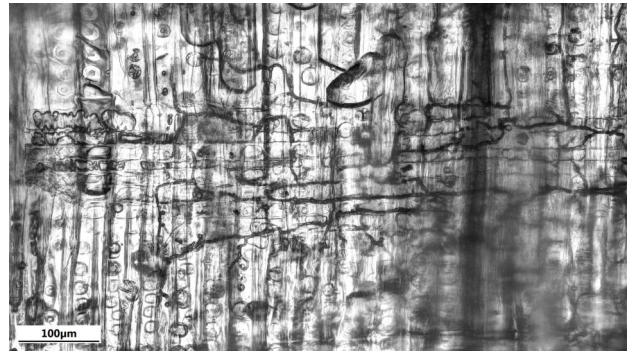


(d) White Surface

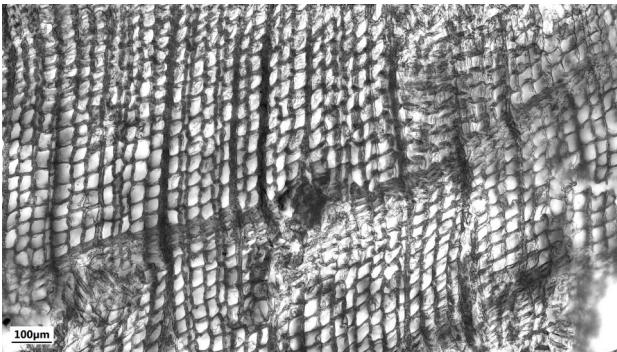
Figure A.28: Stereomicroscopy images of C57838/4 F.133 AG.



(a) Radial Section, Brown Surface



(b) Radial Section, White Surface



(c) Transverse Section, Interior



(d) Tangential Section, Interior

Figure A.29: Light microscopy images of C57838/4 F.133 AG.

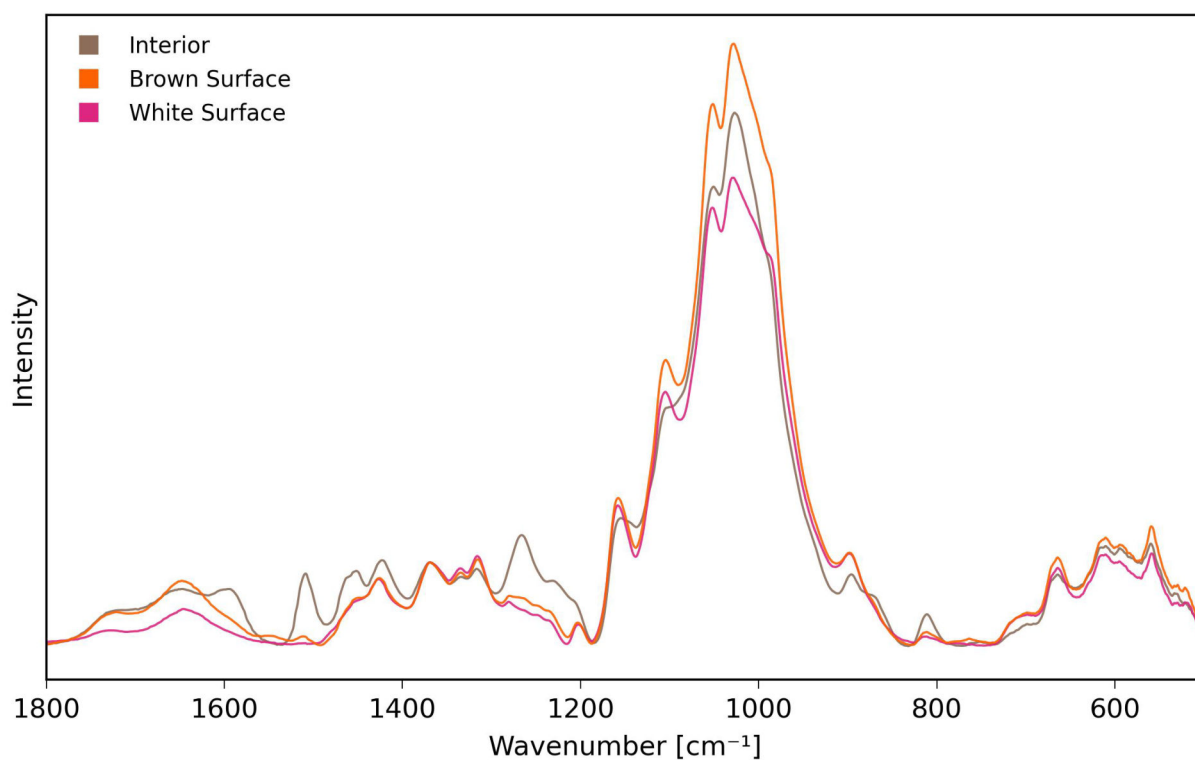
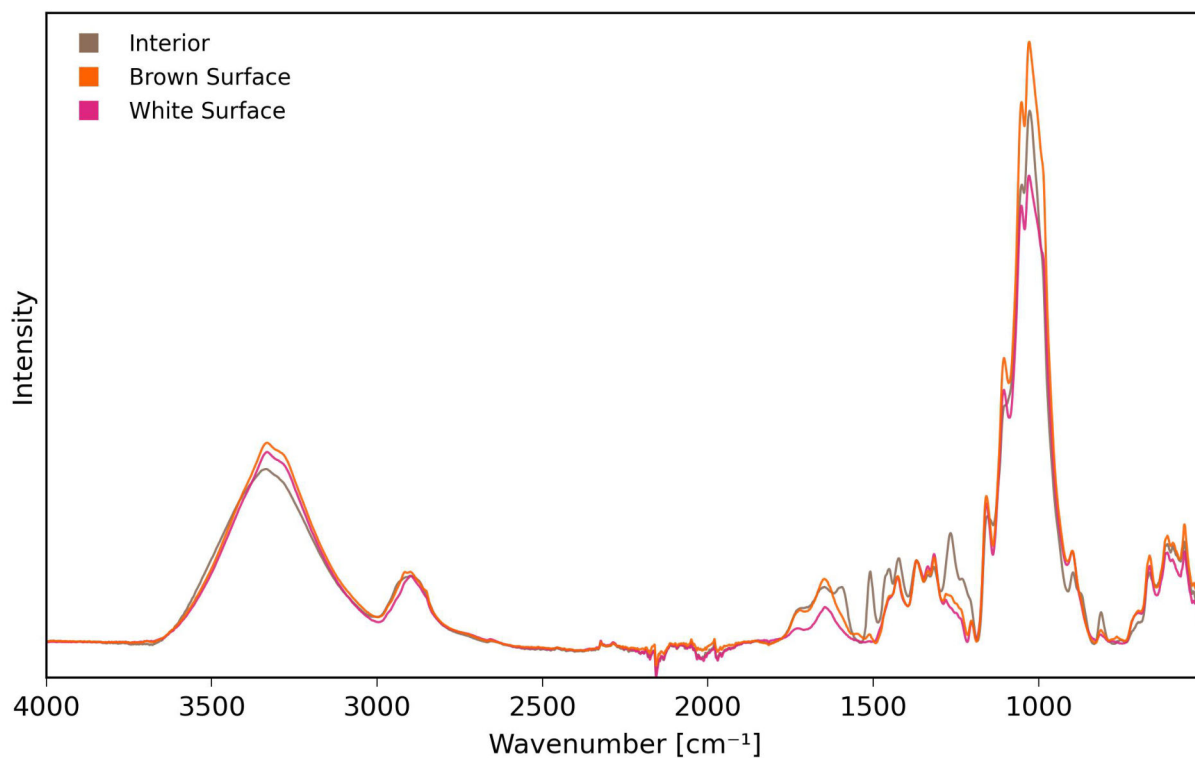


Figure A.30: ATR-FTIR spectra of C57838/4 F.133 AG in the full measured spectral range (above) and 1800 cm^{-1} to 700 cm^{-1} region (below).